Contents

Executive summary

AI-powered solutions enable retailers to optimize and automate inventory management for greater efficiency, accuracy, and real-time visibility.

As a retail decision-maker, Ahold Delhaize know the challenges of effective inventory management. In addition to meeting demand in a rapidly evolving environment, Ahold Delhaize's business has to contend with supply chain disruptions and changing customer expectations.

With traditional strategies no longer cutting it, AI inventory management is stepping in to provide the tools Ahold Delhaize need to deliver precision, efficiency, and real-time insights. The result is smarter inventory management — and a more profitable business.

By using AI for inventory management, Ahold Delhaize can significantly reduce overstocks and stock outs. Predictive analytics analyzes real-time data from sales, supply chains, and market trends to accurately forecast demand and optimize stock levels. This enables Ahold Delhaize to take a more dynamic and proactive approach to stock adjustments that leads to less wasted capital and higher sales.

Inaccurate demand forecasting

Demand for retail products is influenced by a broad range of factors, from social trends and product virility to economic uncertainty and world events. Traditionally, businesses relied on static historical data to understand future trends and predict demand. But without the ability to react in real time to changing market conditions, demand forecasting can be ineffective.

Artificial intelligence in inventory management uses predictive analytics and machine learning algorithms to analyze both historical and real-time data. This enables Ahold Delhaize's retail business to precisely predict customer demand for specific products.

Supply chain complexity

Global supply chains involve multiple stakeholders and countless moving parts. Given the level of complexity, it can be difficult — verging on impossible — to coordinate sourcing, building, and shipping of products using traditional manual processes.

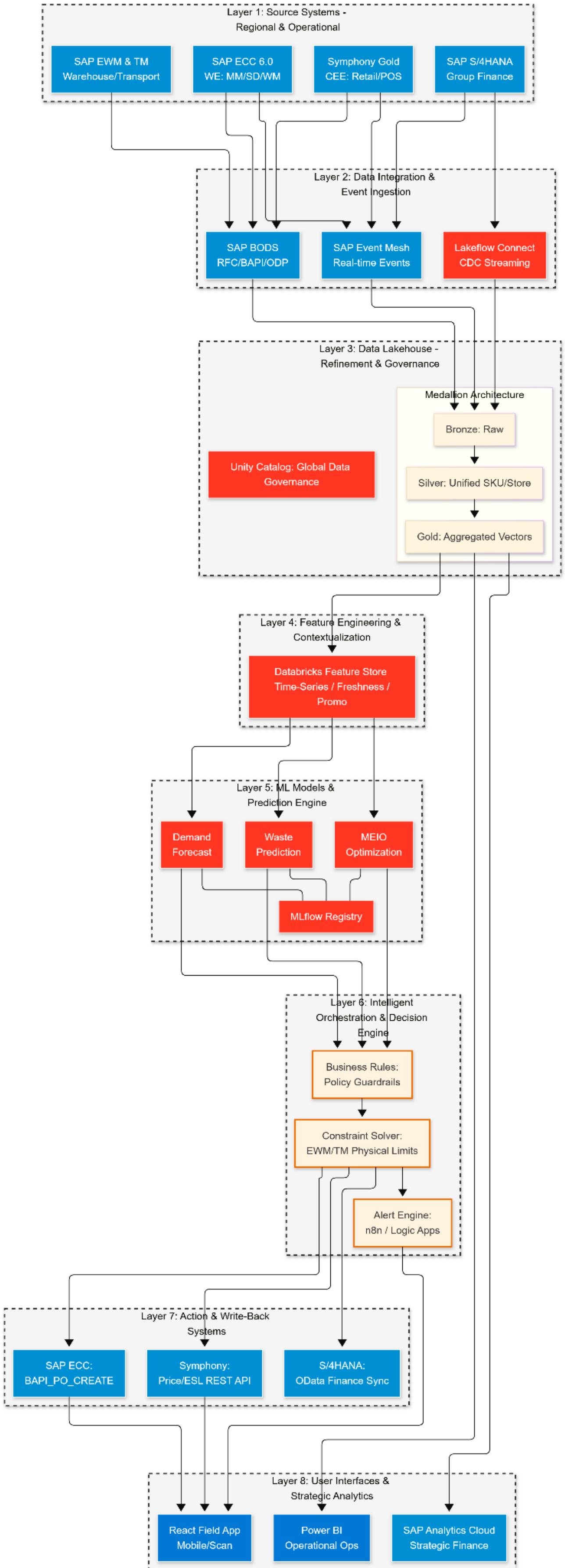
AI for inventory management reduces supply chain complexity. Using a combination of machine learning, automation, and IoT devices, AI-based systems can:

- Provide real-time visibility into supply chain operations
- Predict delays and maintenance issues
- Adjust and optimize shipping routes as issues arise

Delivered Architecture

This 8-Layer Solution Application Architecture represents the state-of-art solution for Ahold Delhaize's inventory optimization. It bridges the gap between fragmented legacy ERPs (Western Europe vs. CEE) and a centralized AI Intelligence layer, ensuring that every prediction respects physical logistics constraints (EWM/TM) and financial targets (S/4HANA).

8-Layer Solution Architecture Diagram



Layer 1: Source Systems - Regional & Operational
SAP EWM & TM Warehouse/Transport
SAP ECC 6.0 WE: MM/SD/WM
Symphony Gold CEE: Retail/POS
SAP S/4HANA Group Finance
Layer 2: Data Integration & Event Ingestion
SAP BODS RFC/BAPI/ODP
SAP Event Mesh Real-time Events
Lakeflow Connect CDC Streaming
Layer 3: Data Lakehouse - Refinement & Governance
Unity Catalog: Global Data Governance
Medallion Architecture
Bronze: Raw
Silver: Unified SKU/Store
Gold: Aggregated Vectors
Layer 4: Feature Engineering & Contextualization
Databricks Feature Store Time-Series / Freshness / Promo
Layer 5: ML Models & Prediction Engine
Demand Forecast
Waste Prediction
MEIO Optimization
MLflow Registry
Layer 6: Intelligent Orchestration & Decision Engine
Business Rules: Policy Guardrails
Constraint Solver: EWM/TM Physical Limits
Alert Engine: n8n / Logic Apps
Layer 7: Action & Write-Back Systems
SAP ECC: BAPI_PO_CREATE
Symphony: Price/ESL REST API
S/4HANA: OData Finance Sync
Layer 8: User Interfaces & Strategic Analytics
React Field App Mobile/Scan
Power BI Operational Ops
SAP Analytics Cloud Strategic Finance

Detailed Layer Descriptions

Layer 1: Source Systems (The Data Owners)

- **SAP ECC 6.0 (WE):** Manages procurement and logistics for Western Europe using legacy Oracle/DB2 databases.
- **Symphony Gold (CEE):** Manages store-level inventory and high-frequency POS transactions for Central/Eastern Europe.
- **SAP EWM & TM:** Provides the "Physical Reality"—warehouse bin capacity, labor availability, and truck load building constraints.

Layer 2: Integration & Ingestion (The Pipes)

- **SAP BODS:** Acts as the trusted proxy for ECC, extracting complex cluster tables via RFC.
- **SAP Event Mesh:** Captures real-time business events (e.g., "Truck Departed," "Item Scanned") and pushes them as web hooks to Azure Functions.
- **Lake flow Connect:** Streams Change-Data-Capture (CDC) from S/4HANA to ensure financial reporting is never more than minutes behind operations.

Layer 3: Data Lake house (The Governance Hub)

- **Unity Catalog:** Manages cross-region security. It ensures a Dutch store manager cannot see Romanian store data.
- **Medallion Architecture:** Refines data from raw (Bronze) to a **Unified Silver Layer** where SAP "Materials" and Symphony "SKUs" are merged into a single Global Product ID.

Layer 4: Feature Engineering (The Signal Generator)

- **Databricks Feature Store:** Centralizes complex signals.
  - *Freshness Feature:* Days since Goods Receipt (from EWM) + Temperature Compliance (from IoT).
  - *Promotion Feature:* Historical Lift for the specific SKU-Store combination.

Layer 5: ML Models & Prediction (The Intelligence)

- **M1 (Demand):** Uses LightGBM/LSTM to provide probabilistic forecasts (P10/P50/P90).
- **M2 (Waste):** XGBoost classifier identifies high-risk fresh items based on decay patterns.
- **M4 (MEIO):** Multi-Echelon optimization determines the best place to hold stock (DC vs. Store).

Layer 6: Intelligent Orchestration (The Decision Engine)

- **Business Rules Engine (BRE):** Applies Ahold Delhaize policies (e.g., "Always maintain 99.9% availability for Milk").
- **Constraint Solver:** The most critical component. It reconciles AI orders with **EWM Bin Capacity** and **TM Truck Volume**. It ensures orders are physically possible before they are sent.

Layer 7: Action & Write-Back (The Execution)

- **SAP ECC:** Receives automated Purchase Orders via **BAPIs**.
- **Symphony Gold:** Receives markdown instructions and price updates for **Electronic Shelf Labels (ESL)** via REST APIs.
- **S/4HANA:** Receives financial variance logs (Marketing Spend vs. Waste Savings) via **OData**.

Layer 8: UI & Analytics (The User Experience)

- **React Field App:** A mobile-first tool for store associates to handle "Exceptions" (e.g., approving a markdown or reporting a quality issue).
- **Power BI:** Daily operational dashboard for Category Managers.
- **SAP Analytics Cloud:** Executive dashboard for the C-Suite to track the €350M+ EBITDA impact and ESG/Sustainability goals.

Key Differentiators of this Architecture:

1. **Logistics-Aware AI:** Unlike standard forecasting, this system knows if a warehouse is full or a truck is unavailable (via EWM/TM integration).
2. **Hybrid Write-Back:** It handles legacy SAP (BODS/BAPI) and modern Retail AI (REST/Event Mesh) simultaneously.
3. **Human-in-the-Loop:** High-risk AI decisions are routed to the React App for manager approval, ensuring business trust.
4. **Single Version of Truth:** Unity Catalog ensures that Western and Eastern European operations are optimized using the same mathematical logic, despite being on different ERPs.

Key Architectural Decisions

1. Hybrid Automation

- **85% Automated**: Direct API calls to create orders, update prices
- **15% Human-in-Loop**: Alerts for edge cases requiring judgment
- Gradual rollout to manage risk

2. Regional Deployment

- West Europe workspace → SAP ECC countries
- Central Europe workspace → Symphony countries
- Unity Catalog for cross-workspace data sharing

3. Real-Time + Batch Hybrid

- **Batch**: Daily forecasts, weekly supplier optimization
- **Real-time**: Stockout alerts, quality issues
- **Event-driven**: Promotion launches trigger re-forecasting

Expected Business Impact

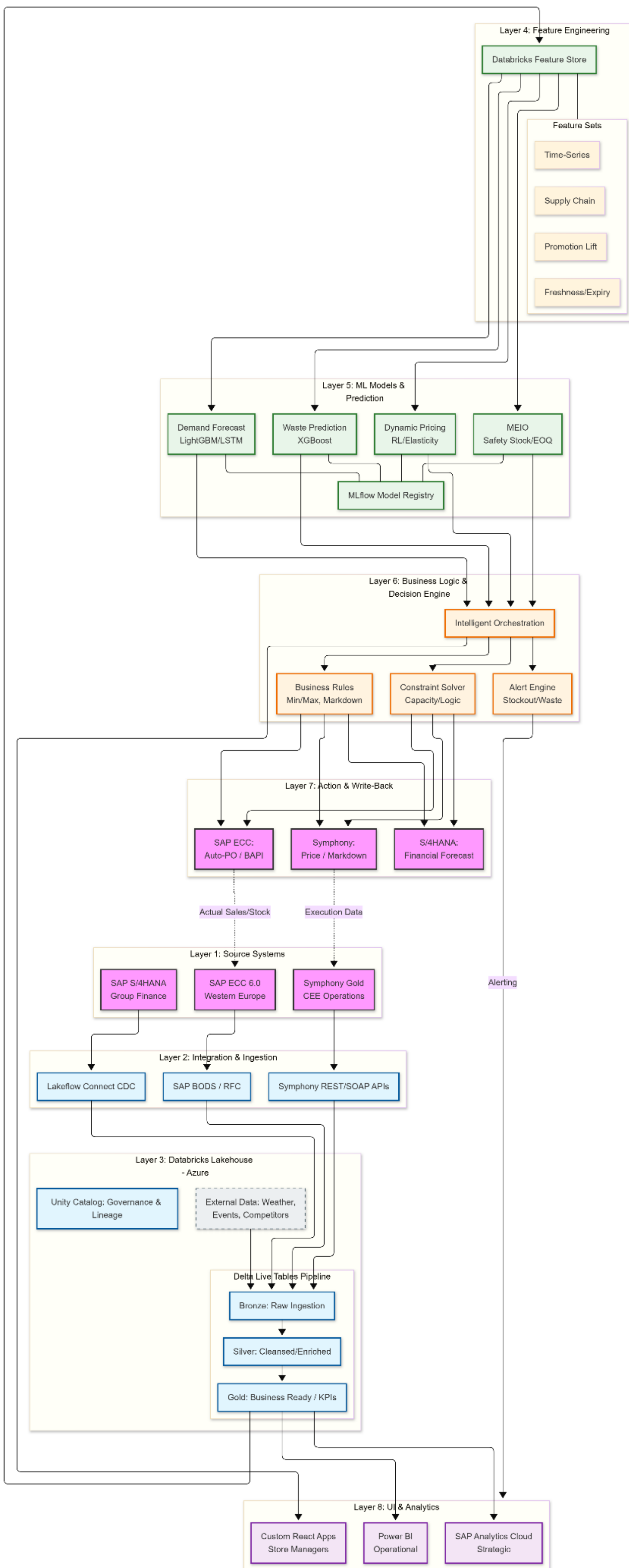
KPIs & Targets:

Metric	Current	Target	Impact
Perishable Waste	3.5%	2.0%	**€200M+** annual savings
On-Shelf Availability	94%	98%	Higher sales, better CX
Forecast Accuracy (MAPE)	25%	<15%	Better planning
Inventory Days (DIO)	28 days	24 days	**€150M** cash release
Manual Ordering Time	100%	30%	70% productivity gain

Total Annual EBITDA Impact: €350M+

Solution Component Diagram

A comprehensive **Solution Component Diagram** for Ahold Delhaize's AI-driven inventory optimization across their hybrid ERP landscape.



Layer 4: Feature Engineering
Databricks Feature Store
Feature Sets
Time-Series
Supply Chain
Promotion Lift
Freshness/Expiry
Layer 5: ML Models & Prediction
Demand Forecast LightGBM/LSTM
Waste Prediction XGBoost
Dynamic Pricing RL/Elasticity
MEIO Safety Stock/EOQ
MLflow Model Registry
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Business Rules Min/Max, Markdown
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Alert Engine Stockout/Waste
Layer 7: Action & Write-Back
SAP ECC: Auto-PO / BAPI
Symphony: Price / Markdown
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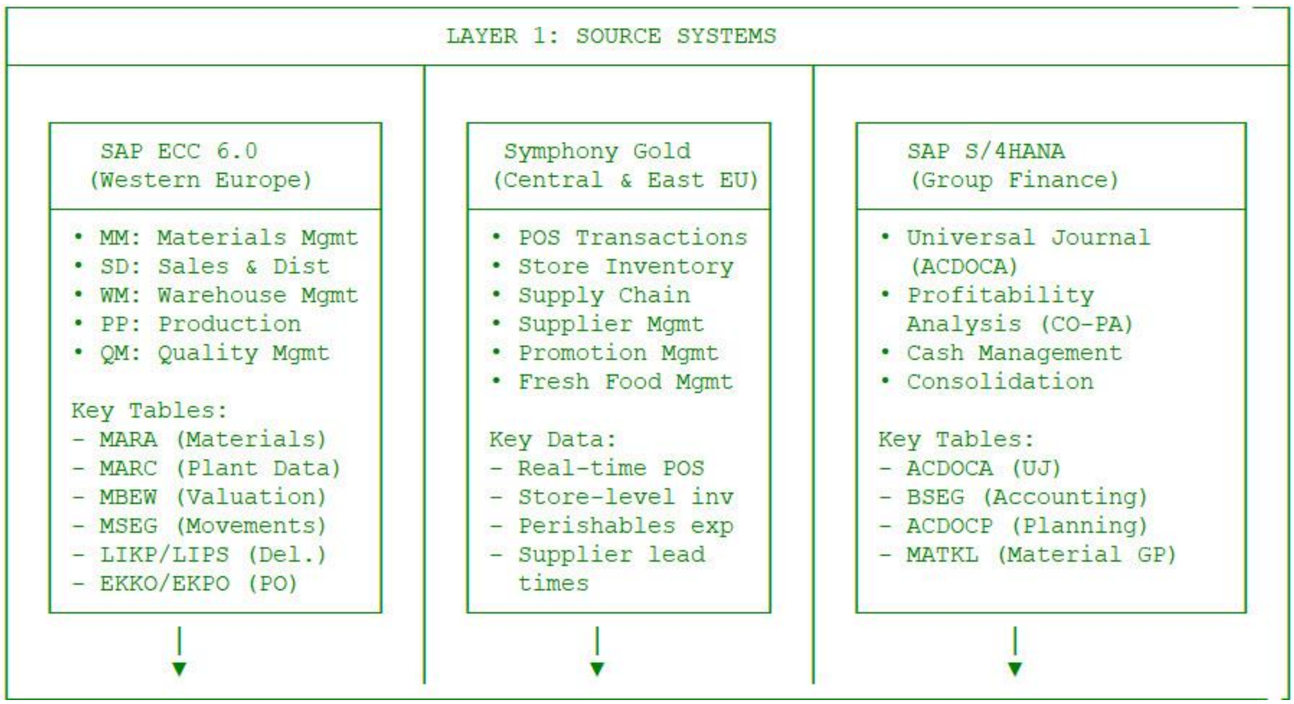
SAP S/4HANA Group Finance
SAP ECC 6.0 Western Europe
Symphony Gold CEE Operations
Layer 2: Integration & Ingestion
Lakeflow Connect CDC
SAP BODS / RFC
Symphony REST/SOAP APIs
Layer 3: Databricks Lakehouse - Azure
Unity Catalog: Governance & Lineage
External Data: Weather, Events, Competitors
Delta Live Tables Pipeline
Bronze: Raw Ingestion
Silver: Cleansed/Enriched
Gold: Business Ready / KPIs
Layer 8: UI & Analytics
Custom React Apps Store Managers
Power BI Operational
SAP Analytics Cloud Strategic

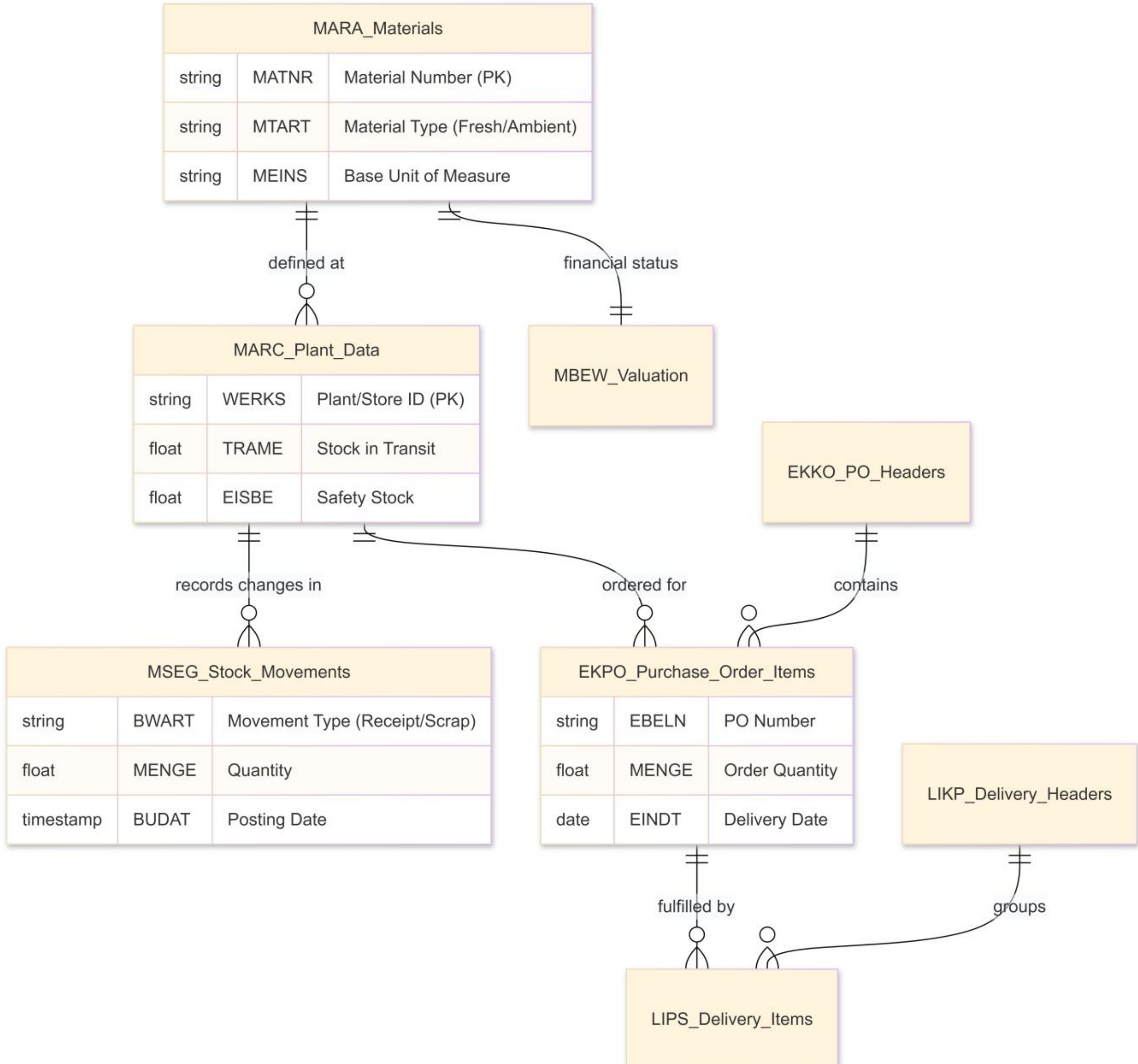


In the Ahold Delhaize landscape, the conceptual data models for the three source streams represent three different "views" of a product's life: its **operational movement** (ECC), its **retail sale and freshness** (Symphony), and its **financial value** (S/4HANA).

Below are the detailed conceptual data models for each stream.

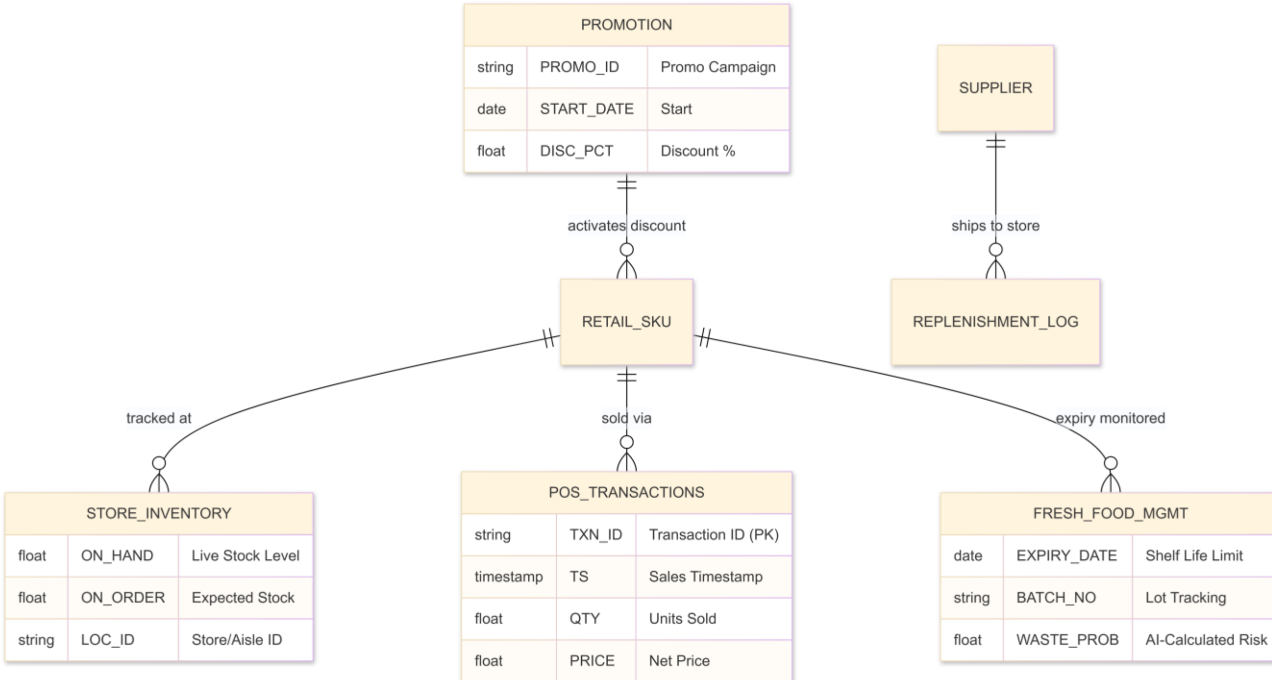
1. SAP ECC 6.0: The Operational & Supply Chain Model (Western Europe)

This model focuses on the **Logistics Lifecycle**: how a product is defined, purchased, moved into a warehouse, and shipped to a store.



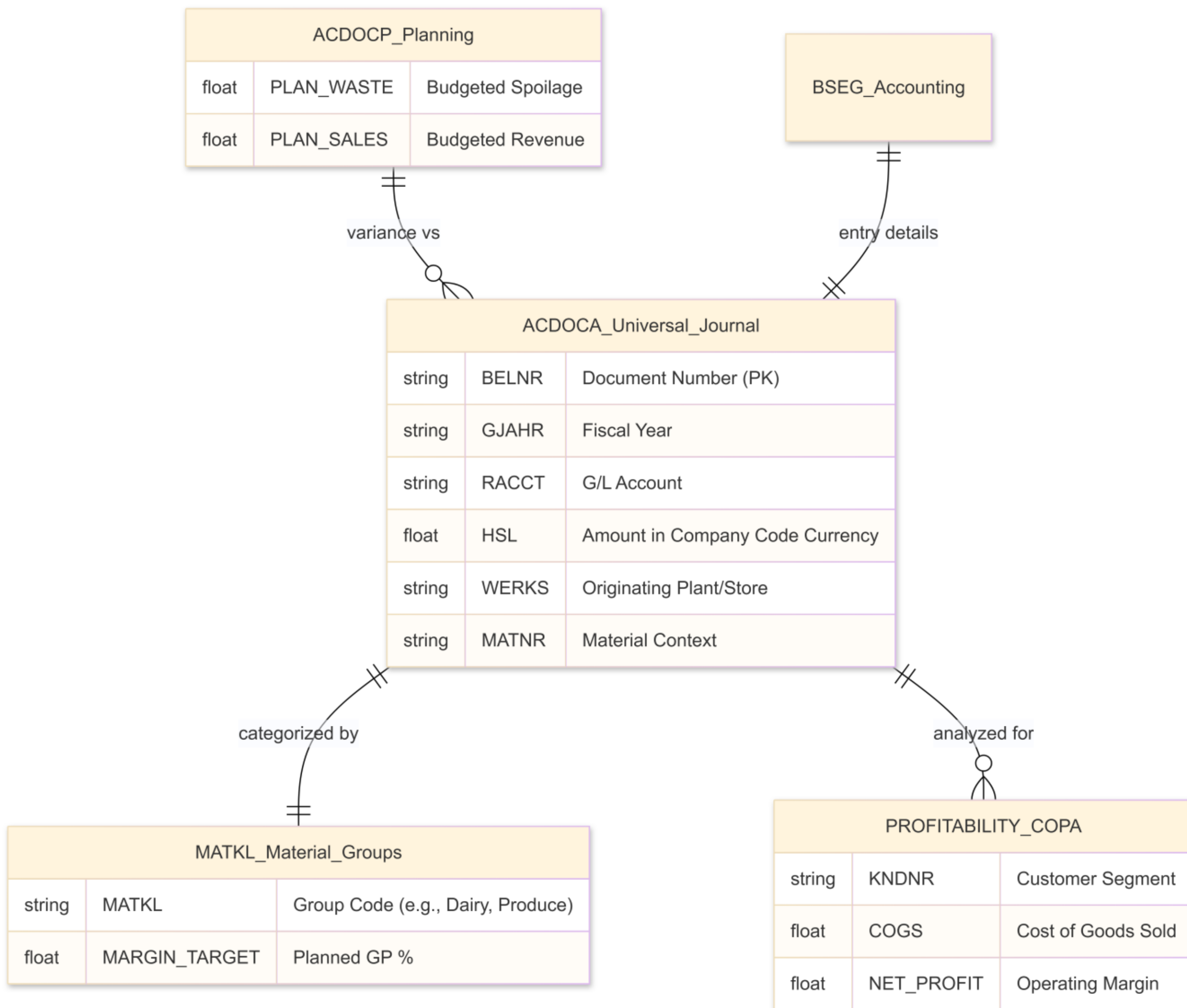
2. Symphony Gold: The Retail & Freshness Model (CEE)

This model is optimized for **Store Operations**. It prioritizes high-frequency Point-of-Sale (POS) data and the granular tracking of perishable expiry dates.



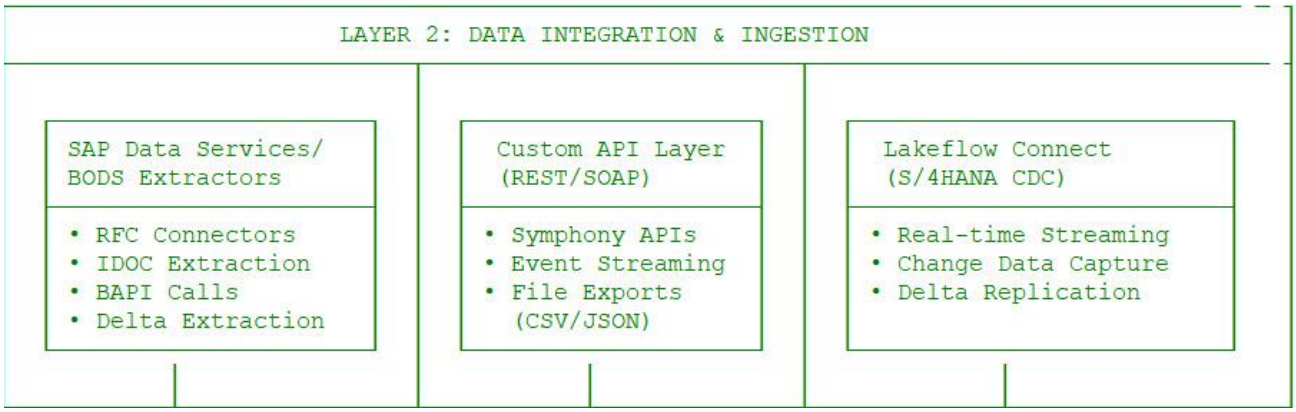
3. SAP S/4HANA: The Unified Financial Model (Group Level)

This model uses the **Universal Journal (ACDOCA)** concept, where logistics results are converted into financial truth. It is the "Single Source of Truth" for Ahold Delhaize Group Finance.



Data Model Integration Summary

1. **The Link (ID Mapping):** The MATNR (Material Number) from SAP ECC is mapped to the RETAIL_SKU in Symphony Gold via a Cross-Reference Table in the **Databricks Silver Layer**.
2. **The Transactional Flow:**
   - **ECC** provides the "How much did we buy?" (Supply).
   - **Symphony** provides the "How much did we sell?" (Demand).
   - **Databricks** calculates the "How much will we waste?" (AI Prediction).
3. **The Financial Result:** The AI-driven decisions (reduced waste, optimized stock) are posted to **S/4HANA (ACDOCA)**, specifically affecting G/L accounts for *Inventory Valuation* and *Cost of Goods Sold (COGS)*. This allows Ahold Delhaize to see the real-time financial ROI of the AI project.

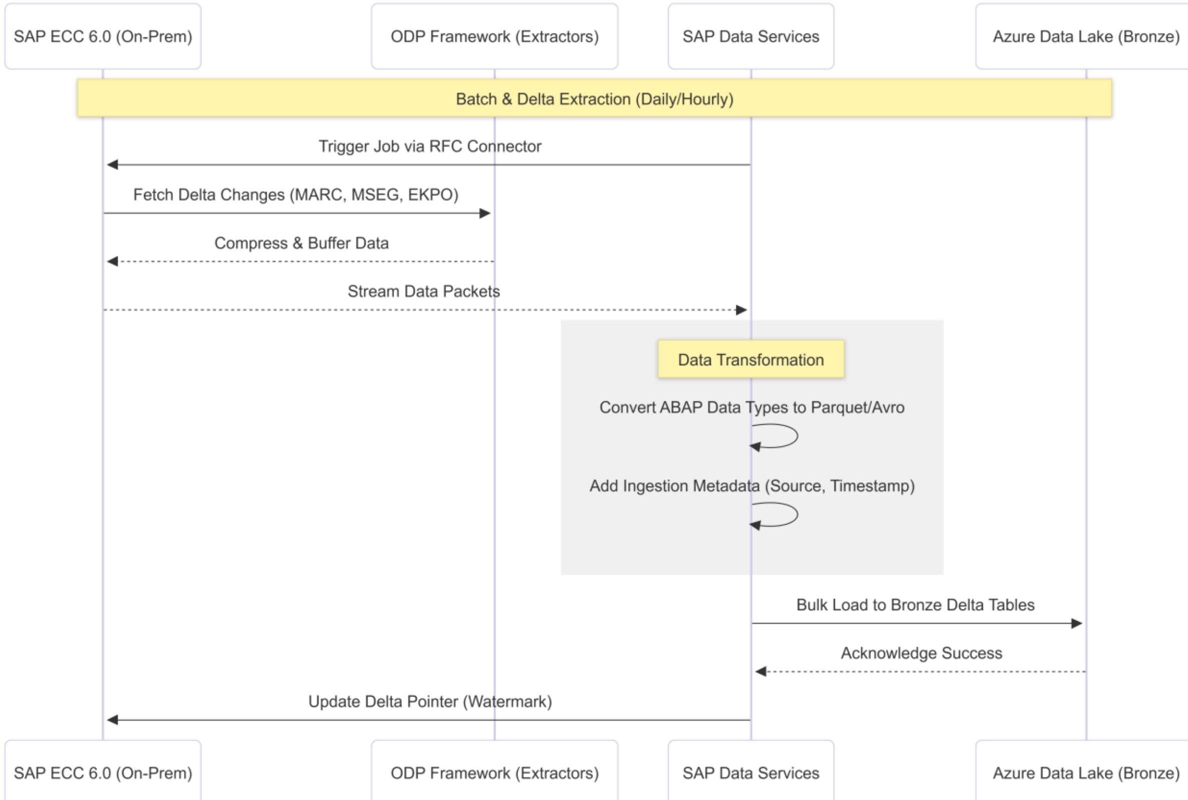


Layer 2 is the "Plumbing Layer." Its goal is to move data from the heterogenous source environments into the **Databricks Lakehouse** while maintaining data integrity and supporting both batch and real-time requirements.

Below are the detailed integration flow diagrams for each stream.

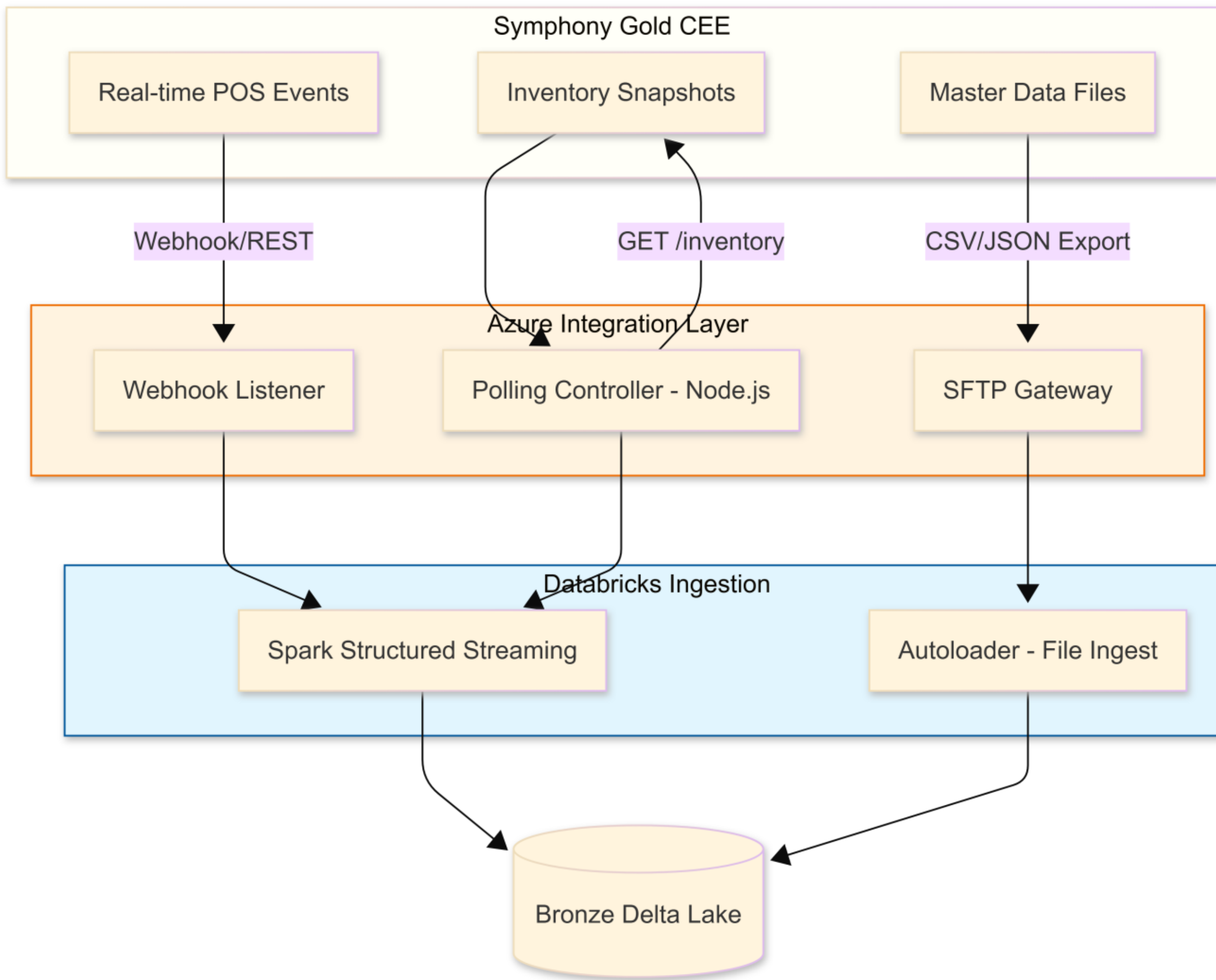
Stream 1: SAP ECC 6.0 via SAP Data Services (BODS)

This flow is optimized for the heavy, structured table architecture of legacy SAP. It uses **Operational Data Provisioning (ODP)** and **RFC** to pull delta changes (new sales, stock movements) without overloading the ECC application server.



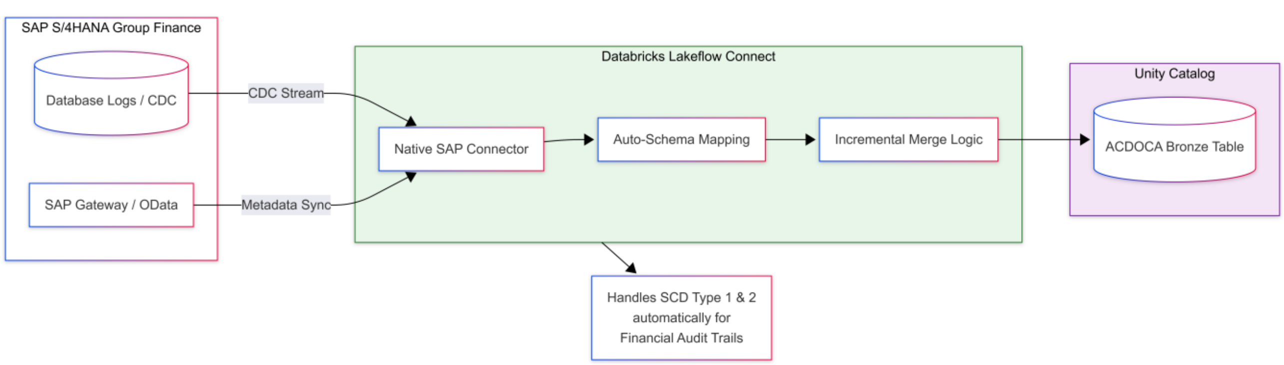
Stream 2: Symphony Gold via Custom API Layer

This flow is a **Hybrid Ingestion** model. It uses **REST APIs** for real-time POS (Point of Sale) transactions and **SFTP/File Exports** for massive master data (SKU/Supplier) synchronizations.



Stream 3: SAP S/4HANA via Lakeflow Connect

This is the most modern stream. **Lakeflow Connect** provides a "Low-Code" ingestion path that leverages **Change Data Capture (CDC)** to stream financial updates from the Universal Journal (ACDOCA) directly into Databricks with sub-minute latency.



Comparison of Integration Strategies

Feature	Stream 1 (ECC)	Stream 2 (Symphony)	Stream 3 (S/4HANA)
Primary Tool	SAP BODS	Node.js / Custom APIs	Databricks Lakeflow Connect
Data Pattern	Batch / Micro-batch	Event-driven / Hybrid	Real-time Streaming (CDC)
Protocol	RFC / NetWeaver	REST / SFTP / Webhooks	OData / gRPC
Complexity	High (Legacy Mapping)	Medium (API Handling)	Low (Native Integration)
Latency	1 Hour - 1 Day	Seconds - Minutes	< 1 Minute

Integration Logic Summary

1. **Watermarking:** For ECC and Symphony, the Integration Layer maintains a "High Watermark" (last processed ID or timestamp) to ensure no data is missed during restarts.
2. **Schema Evolution:** Lakeflow Connect (Stream 3) automatically detects if a new field is added to the S/4HANA Universal Journal and updates the Databricks table schema without breaking the pipeline.
3. **Data Quality Gate:** Every stream includes a "Check-sum" validation at Layer 2 to ensure the number of records extracted from the Source matches the number of records landed in the **Bronze Layer**.

The decision to use **SAP BODS (BusinessObjects Data Services)** specifically for **Stream 1 (SAP ECC 6.0)** is a strategic choice driven by the specific technical "debt" and architecture of legacy SAP environments.

While Lakeflow Connect is excellent for S/4HANA (Stream 3), here is why **SAP BODS** is often the superior choice for a legacy **ECC 6.0** system in a retail environment like Ahold Delhaize:

1. Handling "Cluster" and "Pooled" Tables

Legacy SAP ECC 6.0 stores some of its most critical retail data (like **BSEG** for accounting or **KONV** for pricing) in **Cluster or Pooled tables**.

- **The Challenge:** These are not standard database tables; they are compressed, proprietary SAP structures that cannot be read directly at the database level by most modern CDC (Change Data Capture) tools.
- **The BODS Advantage:** BODS is "SAP-native." It uses the SAP Application Layer to decompress and "decipher" these clusters automatically. Lakeflow Connect often requires an additional SAP metadata layer (like SAP SLT) to achieve the same result with ECC 6.0.

2. Deep Integration with ODP (Operational Data Provisioning)

SAP ECC 6.0 uses a framework called **ODP** to track changes (Deltas).

- **The Challenge:** If you pull data directly from ECC tables via a standard connector, you have to manually build "Watermarking" logic (checking AEDAT or ERDAT timestamps), which is prone to missing records if a transaction is rolled back or delayed.
- **The BODS Advantage:** BODS has a native "ODP Context" connector. It "subscribes" to SAP's internal delta queues. This ensures **100% data consistency**—if a change happens in ECC, the ODP framework catches it, and BODS pulls it without complex custom SQL logic.

3. ABAP-Layer "Push" Logic

For a large-scale retailer, pulling millions of rows of MSEG (Stock Movements) can slow down the ERP's performance.

- **The BODS Advantage:** BODS can generate and "push" **ABAP programs** directly into the ECC system. These programs run inside the SAP memory space, filter the data locally, and stream only the results to the Cloud. This is much more "Basis-team friendly" than having an external cloud tool constantly "polling" a legacy database.

4. Security and "Trusted Zone" Architecture

Legacy ECC 6.0 systems are usually the "crown jewels" of the company and sit behind very restrictive firewalls.

- **The Challenge:** Connecting a modern cloud-native tool like Lakeflow Connect directly to an on-premise ECC 6.0 production database often requires opening ports (like 1521 for Oracle or 1433 for SQL Server) that SAP Security teams usually refuse to open.
- **The BODS Advantage:** BODS typically already exists in the Ahold Delhaize landscape. It communicates with ECC via **RFC (Remote Function Call)** on standard SAP ports (33xx). It acts as a "Trusted Proxy," pulling data from the secure SAP zone and pushing it out to the Azure/Databricks zone.

5. Data "Pre-Cleaning" Before the Lake

ECC data is notoriously "messy" (e.g., German column headers, obscure 4-character codes).

- **The BODS Advantage:** You can use BODS to perform **Initial Harmonization**. For example, converting SAP "Language Keys" or "UOMs" (Units of Measure) into a standardized format *before* the data even reaches the Databricks Bronze layer. This reduces the compute load on your Databricks clusters.

Comparison Summary: Why BODS for Stream 1?

Feature	Lakeflow Connect (Ideal for S/4)	SAP BODS (Ideal for Legacy ECC)
ECC Cluster Tables	Requires SLT or complex config	**Native Support**
Delta Tracking	Database-level CDC	**Application-level (ODP)**
Security	Requires DB-port access	**Uses standard SAP RFC**
Performance	Best for HANA (In-memory)	**Best for legacy DBs (Oracle/DB2)**
Logic	"Extract then Transform"	**"Transform during Extraction"**

The "Hybrid" Recommendation

In a real-world implementation for Ahold Delhaize, we often use a **Hybrid Pipe**:

1. **BODS** performs the "heavy lifting" of extracting from the legacy ECC application layer.
2. **BODS** writes the data into a **Landing Zone** (Azure Data Lake Gen2).

3. **Databricks "Autoloader"** (part of the Lakeflow family) then instantly picks up those files and moves them into the Delta Lake.

This gives you the "Best of Both Worlds": The SAP-native reliability of BODS and the high-speed AI-ready power of Databricks.

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LAYER 3: DATABRICKS LAKEHOUSE PLATFORM (Azure)

    UNITY CATALOG (Data Governance)
    • Cross-region data lineage  • GDPR compliance  • Access control

    DELTA LIVE TABLES (DLT Pipeline)

    BRONZE LAYER            SILVER LAYER              GOLD LAYER
    (Raw Ingestion)         (Cleansed)                (Business Ready)

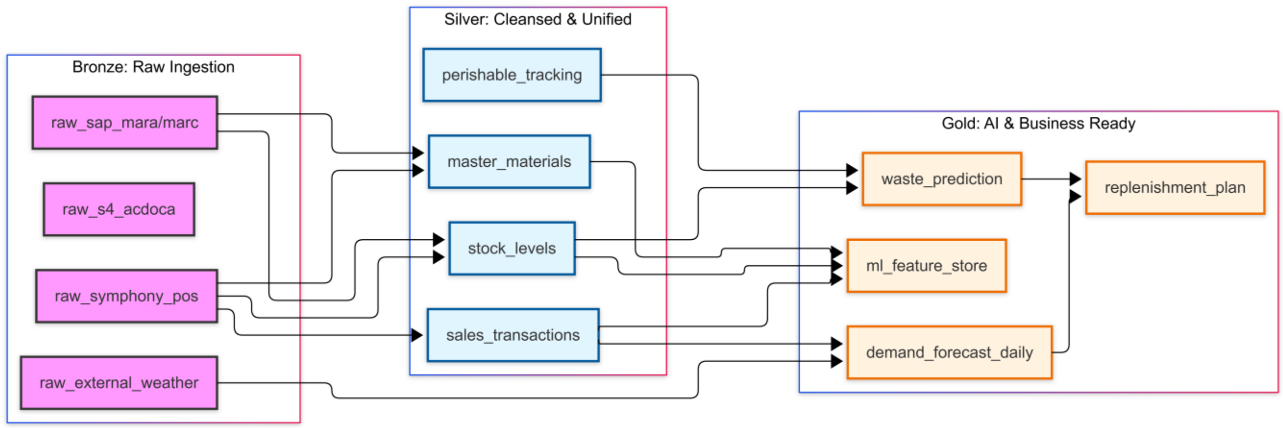
    • raw_sap_mara          • master_materials        • demand_forecast_daily
    • raw_sap_marc          • master_stores           • inventory_optimization
    • raw_sap_mseg          • master_suppliers        • replenishment_plan
    • raw_symphony_pos      • sales_transactions      • waste_prediction
    • raw_symphony_inv      • inventory_movements     • freshness_index
    • raw_s4_acdoca         • stock_levels            • promotion_impact
    • raw_s4_acdocp         • purchase_orders         • supplier_performance
    • external_weather      • delivery_performance    • anomaly_alerts
    • external_holidays     • quality_checks          • ml_feature_store
    • external_events       • perishable_tracking     • kpi_dashboards
                            • promotion_calendar
```

Designing the data model for **Delta Live Tables (DLT)** requires a focus on **Data Lineage** and **Data Quality (Expectations)**. In a DLT pipeline, the flow follows the Medallion Architecture, moving from raw, schemaless ingestion to highly optimized, project-specific aggregates.

Here is the detailed conceptual data diagram and table structure for the Ahold Delhaize Inventory Optimization pipeline.

1. DLT Medallion Flow Diagram

This diagram visualizes how raw data from SAP and Symphony is refined into AI-ready Gold tables.



2. Layer-by-Layer Data Model Detail

A. Bronze Layer (The "Append-Only" Store)

- **Purpose:** 1:1 copy of source data. No transformations.
- **Key Tables:**
  - raw_sap_mara: Material Master from ECC.
  - raw_symphony_pos: Point-of-Sale data from CEE.
  - raw_s4_acdoca: Universal Journal entries from Finance.
- **Technical Columns:** _ingest_timestamp, _source_file, _rescued_data (for schema evolution).

B. Silver Layer (The "Single Version of the Truth")

- **Purpose:** Cleaned, deduplicated, and joined. This is where we bridge the SAP-Symphony gap.
- **DLT Expectations (Data Quality):**
  - CONSTRAINT valid_sku EXPECT (sku_id IS NOT NULL)
  - CONSTRAINT positive_qty EXPECT (quantity >= 0) ON VIOLATION DROP ROW
- **Key Entity: master_materials** (Unified)
  - global_sku_id (UUID)
  - sap_id (from MARA)
  - symphony_id (from Symphony Gold)
  - category / sub_category
  - is_perishable (Boolean)
  - shelf_life_days
- **Key Entity: sales_transactions**
  - txn_id, store_id, sku_id, sales_qty, unit_price, event_timestamp.

C. Gold Layer (The "Decision Support" Store)

- **Purpose:** Aggregated metrics and ML features. Data is structured for Power BI and Model Training.

- **Table: demand_forecast_daily**
  - date, store_id, sku_id
  - predicted_qty_p50 (Median)
  - predicted_qty_p90 (Upper bound for safety stock)
  - forecast_confidence_score
- **Table: waste_prediction**
  - batch_id, sku_id, store_id
  - expiry_date
  - prob_of_waste (Calculated by Model 2)
  - markdown_recommendation (Yes/No)
- **Table: ml_feature_store**
  - Ready-to-use vectors for Model training (e.g., avg_7d_sales, is_holiday, temp_max).

3. Detailed Component Map (Integration View)

Source Table	Bronze DLT	Silver Transformation (Logic)	Gold Business Object
SAP MARA	raw_sap_mara	Unified SKU mapping, Language conversion.	master_materials
Symphony POS	raw_symphony_pos	Standardize Timezones, Currency conversion.	sales_transactions
Symphony INV	raw_symphony_inv	Calculate On-Hand vs In-Transit.	stock_levels
SAP MSEG	raw_sap_mseg	Bridge stock movements from WE warehouses.	inventory_movements
Weather API	raw_weather	Pivot by lat/long to Store_ID.	demand_forecast_daily
SAP ACDOCA	raw_s4_acdoca	Link sales margins to G/L entries.	kpi_dashboards

Strategic Advantage of this Model:

1. **Observability:** Using Databricks **Unity Catalog**, Ahold Delhaize can see exactly how a specific value in the replenishment_plan table (Gold) was derived from a raw row in raw_sap_mseg (Bronze).
2. **GDPR Compliance:** PII data (if any, like Customer Loyalty IDs in Symphony POS) can be masked at the **Silver Layer** using Unity Catalog access policies.
3. **Real-time AI:** Because DLT supports **streaming**, as soon as a sale is recorded in Symphony Gold, the demand_forecast_daily can be updated, allowing for intra-day inventory corrections.

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LAYER 4: AI/ML ORCHESTRATION & FEATURE ENGINEERING

FEATURE STORE (Databricks)

Time-Series Features:                       Categorical Features:
• 7/14/30-day sales moving averages         • Store cluster (A/B/C)
• Day-of-week/month patterns                • Product category
• Seasonality decomposition                 • Brand
• Trend indicators                          • Supplier region

Freshness Features:                         External Features:
• Days to expiry                            • Weather (temp, precip)
• Shelf-life remaining %                    • Local events/holidays
• Temperature compliance                    • Competitor pricing
• Quality score (from QM)                   • Foot traffic predictions

Promotion Features:                         Supply Chain Features:
• Active promotion flag                     • Lead time variance
• Discount depth                            • Supplier reliability score
• Promotion lift (historical)               • Transit time predictions
• Cross-promotion effects                   • Warehouse capacity utilization
```
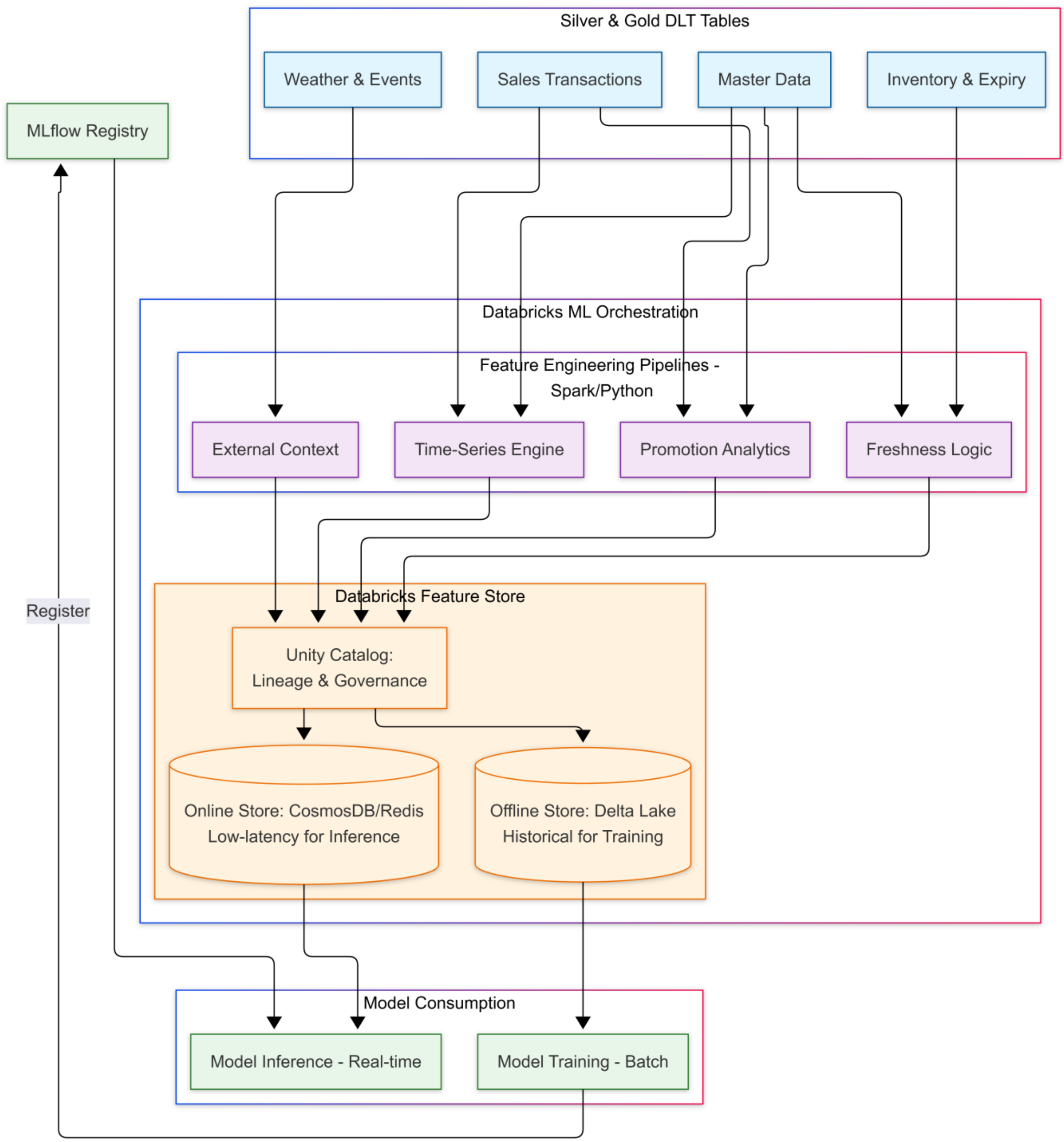

Layer 4 is the "Analytical Engine Room." It transforms cleaned data into a high-dimensional mathematical space that ML models can understand. By using the **Databricks Feature Store**, Ahold Delhaize ensures that the same feature logic used to train a model in the Lab is exactly what is used for real-time predictions in the Store (**Online/Offline Consistency**).

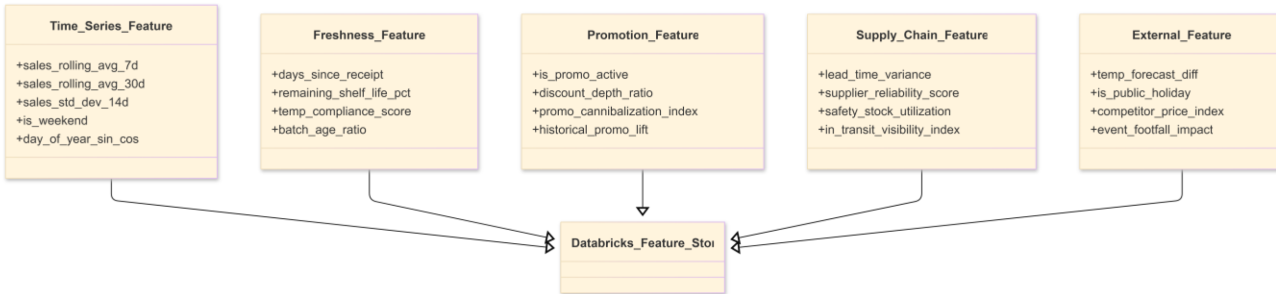
1. Feature Engineering & Orchestration Architecture

This diagram visualizes the flow from data refinement to feature persistence and model consumption.



2. Feature Group Definitions

The Feature Store is organized into logical groups. For Ahold Delhaize, these groups are calculated using **Window Functions** and **Point-in-Time Joins**.



3. Detailed Orchestration Components

A. Point-in-Time Correctness

One of the biggest challenges in Ahold Delhaize's demand forecasting is "data leakage." The Orchestrator ensures that when training a model for a past date, the model only sees features that were available *at that specific moment*. Databricks Feature Store handles this using timestamp keys.

B. The Feature Transformation Logic

- **Time-Series:** Calculated using Spark Window functions over the Gold sales tables.
  - *Logic:* avg(sales) over(partition by store_id, sku_id order by date rows between 6 preceding and current row).
- **Freshness:** Merges raw_sap_mseg (Goods Receipt) with current_date to determine the "decay" velocity of produce.
- **Promotion:** Joins the promotion_calendar (Symphony Gold) with sales to calculate the "Lift Factor."

C. Unity Catalog & Feature Lineage

Since Ahold Delhaize operates across regions, **Unity Catalog** is used to:

1. **Enforce Governance:** Only data scientists in the "CEE" group can see Symphony-based features.
2. **Audit Lineage:** If a forecast in Belgium is wrong, the system can trace the error back through the feature store directly to a specific source record in SAP ECC.

D. Online Serving (The Store App)

For the **Custom React App (Field)**, low latency is critical.

1. Features are published from the **Offline Delta Store** to an **Online Store** (like Azure CosmosDB).
2. When an associate scans a barcode, the **Model Serving Endpoint** fetches the latest "Freshness Features" (calculated minutes ago) and returns an instant prediction.

Summary of Value

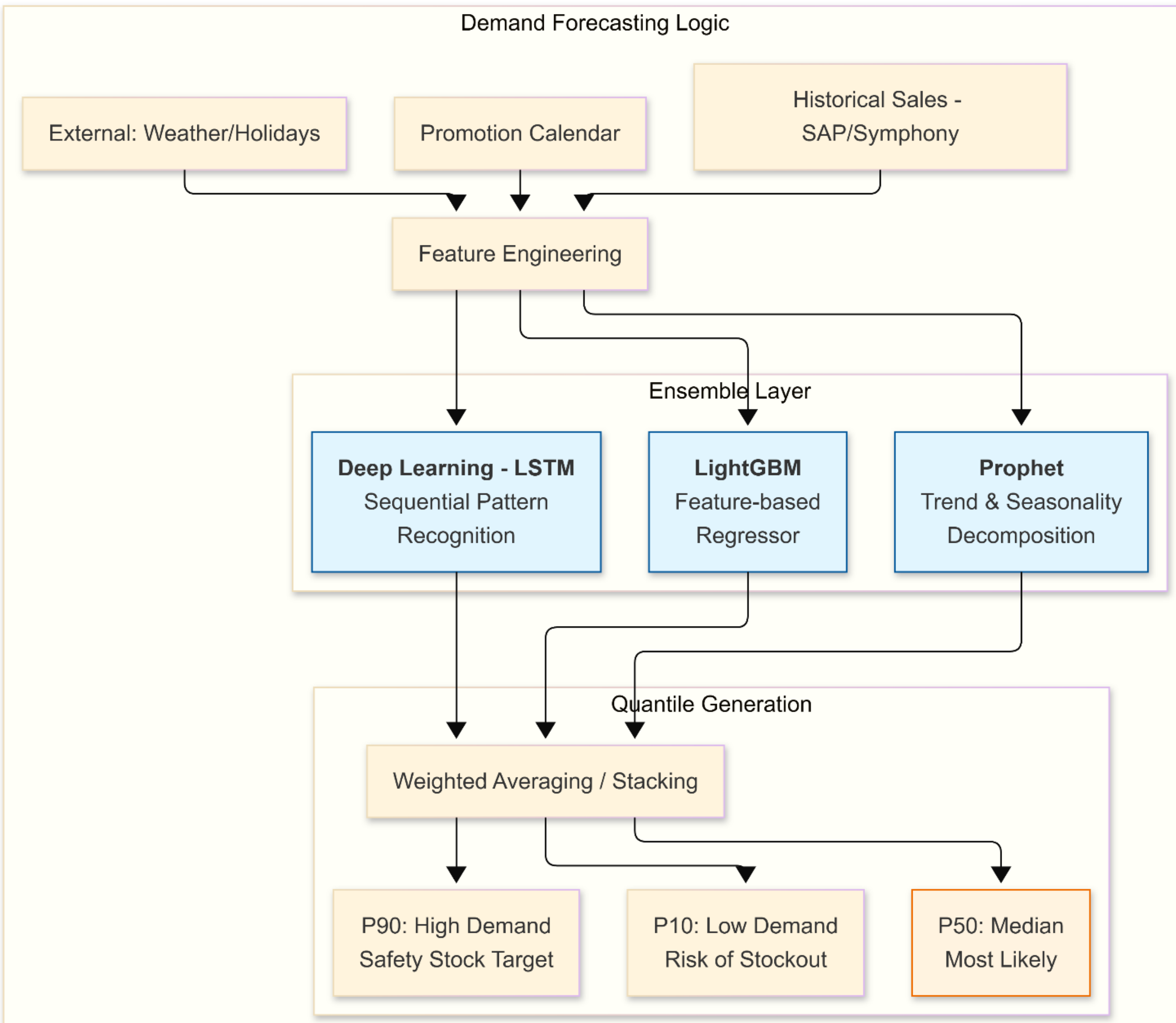
- **Reusability:** The "Lead Time Variance" feature created for Demand Forecasting can be reused by the Supplier Selection model.
- **Speed-to-Market:** Data scientists spend less time writing SQL/Python and more time tuning models because the "feature library" is already built and governed.
- **Model Accuracy:** By including "External Features" (weather/competitors), the model can explain 20-30% more variance in sales than standard ERP-based forecasting.

MODEL 1: Demand Forecasting Engine

Goal: Transition from deterministic ("Sell 10") to probabilistic ("80% chance we sell 8–12") forecasting.

In the Ahold Delhaize architecture, **Model 1 (M1: Demand Forecasting)** does not operate in a vacuum. While its primary job is to predict consumer behavior, its **accuracy and execution feasibility** are direct results of the information flow from **SAP EWM (Warehouse)** and **SAP TM (Transportation)**.

By combining AI forecasting with EWM/TM data, Ahold Delhaize moves from a "Statistical Forecast" to a **"Logistics-Aware Forecast."**

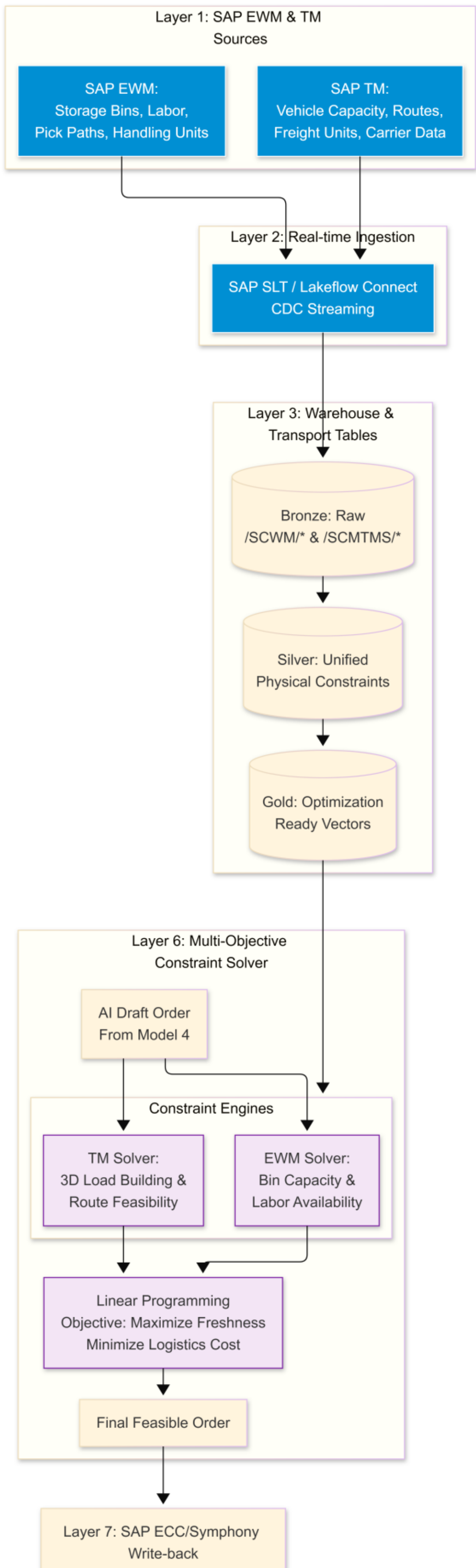


In a high-volume retail environment like Ahold Delhaize, a demand forecast is useless if the **Extended Warehouse Management (EWM)** system reports a labor strike or a blocked aisle, or if the **Transportation Management (TM)** system shows that the refrigerated truck fleet is already at 100% capacity.

To bridge this gap, we must implement a **Logistics Digital Twin** within Databricks that replicates the physical constraints of SAP EWM and TM in real-time.

1. The Logistics Constraint Integration Architecture

The solution requires adding a "Physical Execution" stream to the data model. We will use **SAP Landscape Transformation (SLT)** or **Lakeflow Connect** for real-time Change Data Capture (CDC) because warehouse and transport data change by the second.


Layer 1: SAP EWM & TM Sources
SAP EWM: Storage Bins, Labor, Pick Paths, Handling Units
SAP TM: Vehicle Capacity, Routes, Freight Units, Carrier Data
Layer 2: Real-time Ingestion
SAP SLT / Lakeflow Connect CDC Streaming
Layer 3: Warehouse & Transport Tables
Bronze: Raw /SCWM/* & /SCMTMS/*
Silver: Unified Physical Constraints
Gold: Optimization Ready Vectors
Layer 6: Multi-Objective Constraint Solver
AI Draft Order From Model 4
Constraint Engines
TM Solver: 3D Load Building & Route Feasibility
EWM Solver: Bin Capacity & Labor Availability
Linear Programming Objective: Maximize Freshness Minimize Logistics Cost
Final Feasible Order
Layer 7: SAP ECC/Symphony Write-back

2. Expanded Conceptual Data Model (Logistics Stream)

We need to ingest specific tables from EWM and TM to make the Databricks Solver "aware" of physical reality.

A. SAP EWM (Warehouse Constraints)

- **/SCWM/AQUA (Available Quantity):** To know not just total stock, but stock that is "pickable" (not stuck in quality inspection).
- **/SCWM/LAGP (Storage Bins):** To identify capacity limits per temperature zone (e.g., how many pallets of refrigerated berries can the DC actually hold?).
- **/SCWM/RSRC (Resources):** To factor in labor constraints (if only 2 forklift drivers are scheduled, the AI cannot order 50 trucks).

B. SAP TM (Transportation Constraints)

- **/SCMTMS/D_VEH_CAP (Vehicle Capacity):** Weight and Volume (m3) limits for the fleet.
- **/SCMTMS/D_TORROT (Transportation Orders):** Current truck locations and route statuses.
- **/SCMTMS/D_RESVAL (Resource Availability):** Driver and vehicle schedules.

3. The "Solver" Logic (Layer 6 Deep-Dive)

The **Constraint Solver** in Databricks uses Python libraries (like Pyomo, PuLP, or Google OR-Tools) to run a **Multi-Objective Optimization**.

The Mathematical Problem:

Maximize:

MarginMarginMargin

(from Forecast)

Subject to:

1. **Volume Constraint:**

   ∑(SKU_Volume×Qty)≤Truck_Capacity\sum (SKU_Volume \times Qty) \leq Truck_Capacity∑(SKU_Volume×Qty)≤Truck_Capacity

   (from TM)

2. **Weight Constraint:**

   ∑(SKU_Weight×Qty)≤Payload_Limit\sum (SKU_Weight \times Qty) \leq Payload_Limit∑(SKU_Weight×Qty)≤Payload_Limit

(from TM)

3. **Warehouse Constraint:**

```
∑(SKU_Pallets)≤Bin_AvailabilityZone\sum (SKU\_Pallets) \leq Bin\_Availability_{Zone}∑(SKU_Pallets)≤Bin_AvailabilityZone
```

(from EWM)

4. **Shelf-Life Constraint:**

```
Delivery_Time+Storage_Time<Total_Shelf_LifeDelivery\_Time + Storage\_Time < Total\_Shelf\_LifeDelivery_Time+Storage_Time<Total_Shelf_Life
```

Workflow Process:

1. **Draft Generation:** Model 4 proposes an order of 2,000 units based on AI demand.
2. **EWM Check:** The Solver checks DC 101. EWM data says: *"Only 1,200 units fit in the chilled zone today."*
3. **TM Check:** The Solver checks the route. TM data says: *"The morning truck is a small 10-ton vehicle; it can only take 1,000 units."*
4. **Optimization:** The Solver runs a trade-off analysis. It prioritizes high-margin/fast-expiring items (tomatoes) and cuts the order for longer-life items (potatoes) to fit the truck.
5. **Final Order:** The system outputs a **Feasible Order** of 1,000 units.

4. Integration Flow: Real-time Synchronization

Step	Component	Action
1. Trigger	**SAP EWM/TM**	A truck leaves the DC or a bin is filled. SLT captures this immediately.
2. Stream	**Lakeflow Connect**	The event is streamed into the **Silver Logistics Table** in Databricks.
3. Re-Calc	**Solver Engine**	The Constraint Solver recalculates the "Available to Order" (ATO) capacity for that DC.
4. Feedback	**Custom React App**	If a Store Manager tries to manually increase an order, the app shows a warning: *"Truck capacity reached for DC 101. Order restricted."*
5. Financials	**S/4HANA**	If the truck fill rate is consistently low (e.g., < 80%), the system sends a "Logistics Inefficiency" alert to the S/4HANA Finance dashboard.

Strategic Advantage for Ahold Delhaize

By integrating EWM and TM into Databricks:

- **Zero Rejected Orders:** You stop creating purchase orders that the warehouse cannot physically pick or the trucks cannot physically carry.
- **CO2 Reduction:** The TM solver optimizes for **Truck Fill Rate**, reducing the number of half-empty trucks on the road (ESG goal).

- **Freshness Preservation:** If EWM reports a delay in the refrigerated zone, the AI automatically shifts the next day's orders to a different DC to prevent spoilage.

MODEL 2: Waste Prediction Engine

Goal: Identify perishable items that will likely expire before being sold, using classification and survival analysis.

In the Ahold Delhaize architecture, **Model 2 (M2)** is the specialized "Risk Monitor" for perishables. While Model 1 predicts what customers *will* buy, M2 predicts what *won't* sell before it spoils.

Here is a detailed breakdown of what this component means technically and operationally:

1. The Algorithm: Why XGBoost?

XGBoost (Extreme Gradient Boosting) is a powerful Machine Learning algorithm that uses a "forest" of decision trees to make predictions.

- **Why a "Classifier"?** A classifier doesn't predict a quantity (like "5 apples"); it predicts a **Probability/Category**. In this case: *"What is the probability (0% to 100%) that this specific batch will end up in the bin?"*
- **Handling Non-Linear Data:** Food waste isn't linear. For example, a tomato's quality might stay perfect for 4 days and then collapse in 6 hours if the temperature rises. XGBoost is excellent at spotting these complex, "tipping point" patterns that traditional statistics miss.

2. What are "Decay Patterns"?

"Decay" in this context refers to the **shelf-life velocity** of a product. The model analyzes historical data to learn how different factors accelerate or decelerate spoilage:

- **Time Since Receipt:** The age of the product recorded in **SAP ECC** (Goods Receipt date).
- **Biological Decay:** Historical data on how fast "Leafy Greens" lose value vs. "Citrus."
- **Temperature Context:** IoT data from warehouses or trucks (via SAP EWM/TM). If a pallet sat on a loading dock for 2 hours in 25°C heat, the "Decay Pattern" shifts to high risk.
- **Sales Velocity Correlation:** If **Model 1** predicts low sales for tomorrow, but **Symphony Gold** shows high stock of an item with only 2 days of life left, the "Decay Pattern" signals a high-risk collision.

3. Identifying "High-Risk" Items

The model assigns a **Risk Score** to every SKU-Store combination.

- **Low Risk:** Stock is fresh, sales are high. Action: *None.*
- **Medium Risk:** Stock is aging, sales are average. Action: *Suggest "Feature" placement (move to front of aisle).*
- **High Risk:** Probability of waste > 70%. Action: *Trigger the **Markdown Workflow** in the React App.*

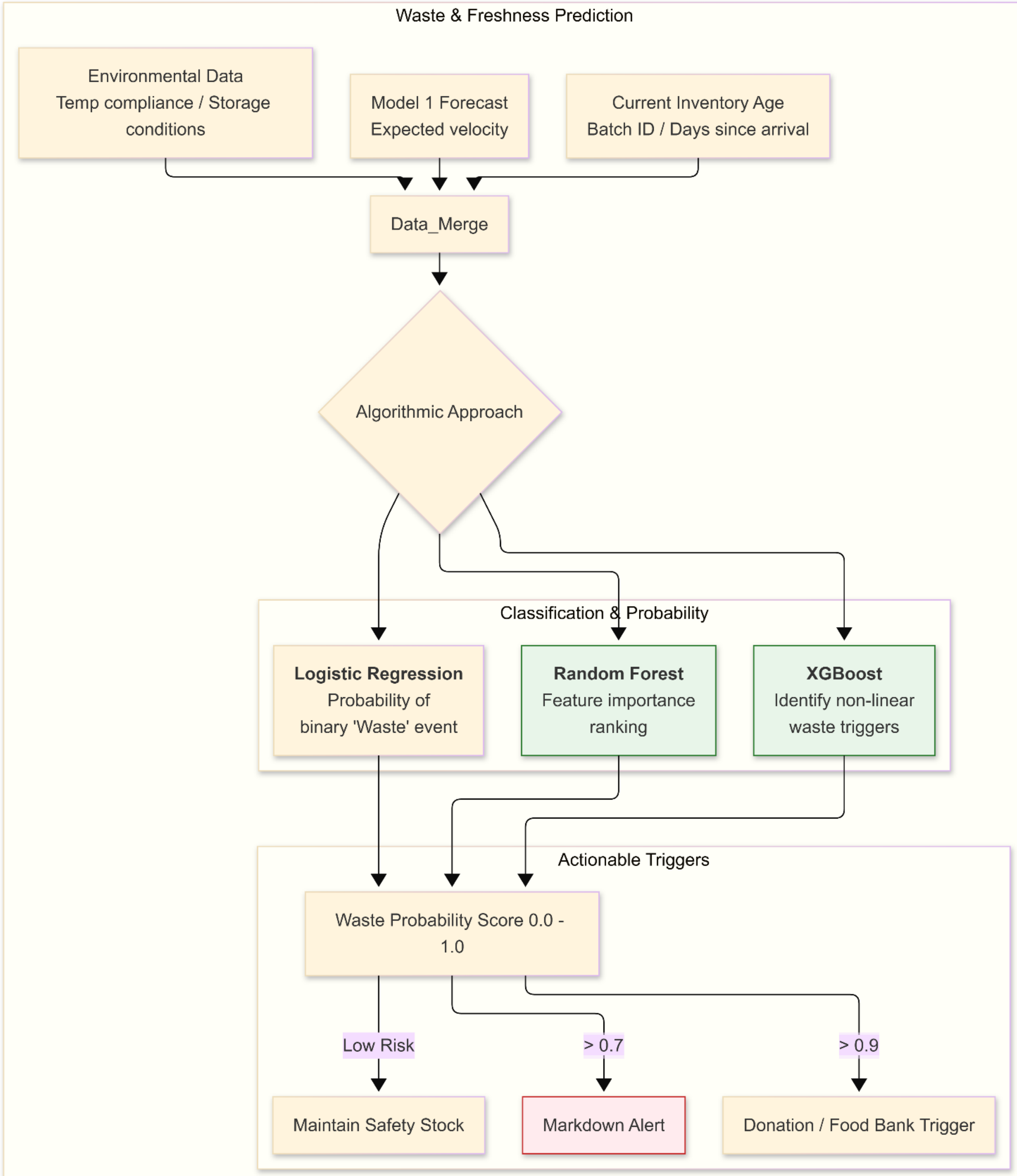
4. Operational Example: The "Strawberry Scenario"

1. **Data Input: SAP ECC** shows a shipment of 500 crates of strawberries arrived at the DC 3 days ago. **Symphony Gold** shows 100 crates are still in the Amsterdam store.
2. **External Context:** The weather forecast (External API) shows a sudden heatwave.
3. **The Analysis (M2):** The XGBoost model looks at the age (3 days), the heatwave, and the current sales speed. It identifies a **Decay Pattern** that suggests these strawberries will be unsellable by tomorrow afternoon.
4. **The Output:** It flags these 100 crates as **"High Risk."**
5. **The Action:**
    - The **React Field App** sends a notification to the Store Manager.
    - **Model 3 (Pricing)** suggests a 40% discount immediately.
    - **Model 4 (Replenishment)** automatically "throttles" (reduces) the next shipment from the DC to prevent adding fresh fruit to a failing stock.

5. Synergy with the 8-Layer Architecture

- **Layer 4 (Feature Store):** Provides the "Freshness Features" (Days since receipt, remaining shelf-life %).
- **Layer 5 (The Model):** XGBoost runs the math.
- **Layer 6 (Decision Engine):** Checks if a markdown is allowed by corporate policy.
- **Layer 7 (Write-back):** Sends the new price to the **Electronic Shelf Labels (ESL)**.

Summary: M2 is the "early warning system" that ensures Ahold Delhaize moves from **Reactive Scrapping** (throwing away rotten food) to **Proactive Selling** (discounting food while it is still high quality). This is the primary driver for the **€200M annual waste reduction goal.**



MODEL 3: Dynamic Pricing & Markdown Engine

Goal: Use Reinforcement Learning to find the optimal price point that clears stock while protecting margins.

In the Ahold Delhaize architecture, **Model 3 (M3)** is the "Profitability Protector." While traditional retail uses fixed markdown schedules (e.g., 25% off on the last day), M3 uses **Reinforcement**

Learning (RL) to treat pricing as a dynamic game where the goal is to "win" by selling the item at the highest possible price before it hits the bin.

Here is the detailed breakdown of how Model 3 functions within the Databricks/Azure environment.

1. The Core Logic: Why Reinforcement Learning?

Traditional price elasticity models are **static** (they tell you what might happen *now*). RL is **sequential** (it plans for the *future*).

- **The Problem:** If I discount strawberries by 20% today, I lose margin but sell faster. If I wait until tomorrow, I might have to discount them by 50% or throw them away.
- **The RL Solution:** The "Agent" (the model) learns through thousands of simulations which "path" of price changes leads to the highest cumulative reward (Total Margin - Waste Cost).

2. The RL Framework: State, Action, Reward

Model 3 operates in a continuous loop, receiving data from the Feature Store (Layer 4) and sending actions to Symphony Gold (Layer 7).

A. The State (

sss

) - "What is happening now?"

The agent looks at a multi-dimensional vector of data:

- **Inventory Age:** Days remaining until expiry (from Model 2).
- **Stock Pressure:** Current units on hand vs. Model 1's demand forecast.
- **Time Context:** Day of the week, time of day (e.g., foot traffic peaks at 5:00 PM).
- **External Context:** Competitor pricing and weather (e.g., high heat reduces appetite for heavy meals but spikes demand for salads).
- **Price History:** How the product has responded to previous discounts.

B. The Action (

AAA

) - "What should we do?"

The agent chooses from a set of discrete actions to maintain shelf stability:

- Action 0: Maintain current price.
- Action 1: 15% Discount (Initial nudge).
- Action 2: 30% Discount (Aggressive clearing).
- Action 3: 50% Discount (Flash sale to avoid total waste).
- Action 4: Bundle offer (e.g., "Buy 2 for X" logic).

C. The Reward Function (
RRR

) - "The Mathematical Goal"

The agent is programmed to maximize a specific score:

```
        Reward=(Sold_Qty×Net_Margin)−(Wasted_Qty×COGS)−PenaltyBrandReward =
(Sold\_Qty \times Net\_Margin) - (Wasted\_Qty \times COGS) -
Penalty_{Brand}Reward=(Sold_Qty×Net_Margin)−(Wasted_Qty×COGS)−PenaltyBrand
```

- **Net Margin:** Selling price minus the cost of the item.
- **Waste Cost:** A massive negative penalty if the item hits the expiry date without being sold.
- **Brand Penalty:** A small penalty for excessive discounting to prevent "training" customers to only buy on sale.

3. Training: Simulation-to-Reality (Offline RL)

Ahold Delhaize cannot let an AI "experiment" with real prices in a store because it might set prices to €0.01 just to learn. Instead, we use **Offline Reinforcement Learning**:

1. **Market Simulator:** Using 3 years of Symphony Gold POS data, we build a simulator that predicts how customers react to price changes.
2. **Training in the "Gym":** The RL agent (using **PPO - Proximal Policy Optimization**) plays millions of games in the simulator.
3. **Deployment:** Once the agent consistently beats the "Manual Markdown Policy," it is registered in **MLflow** and deployed.

4. Integration with the 8-Layer Architecture

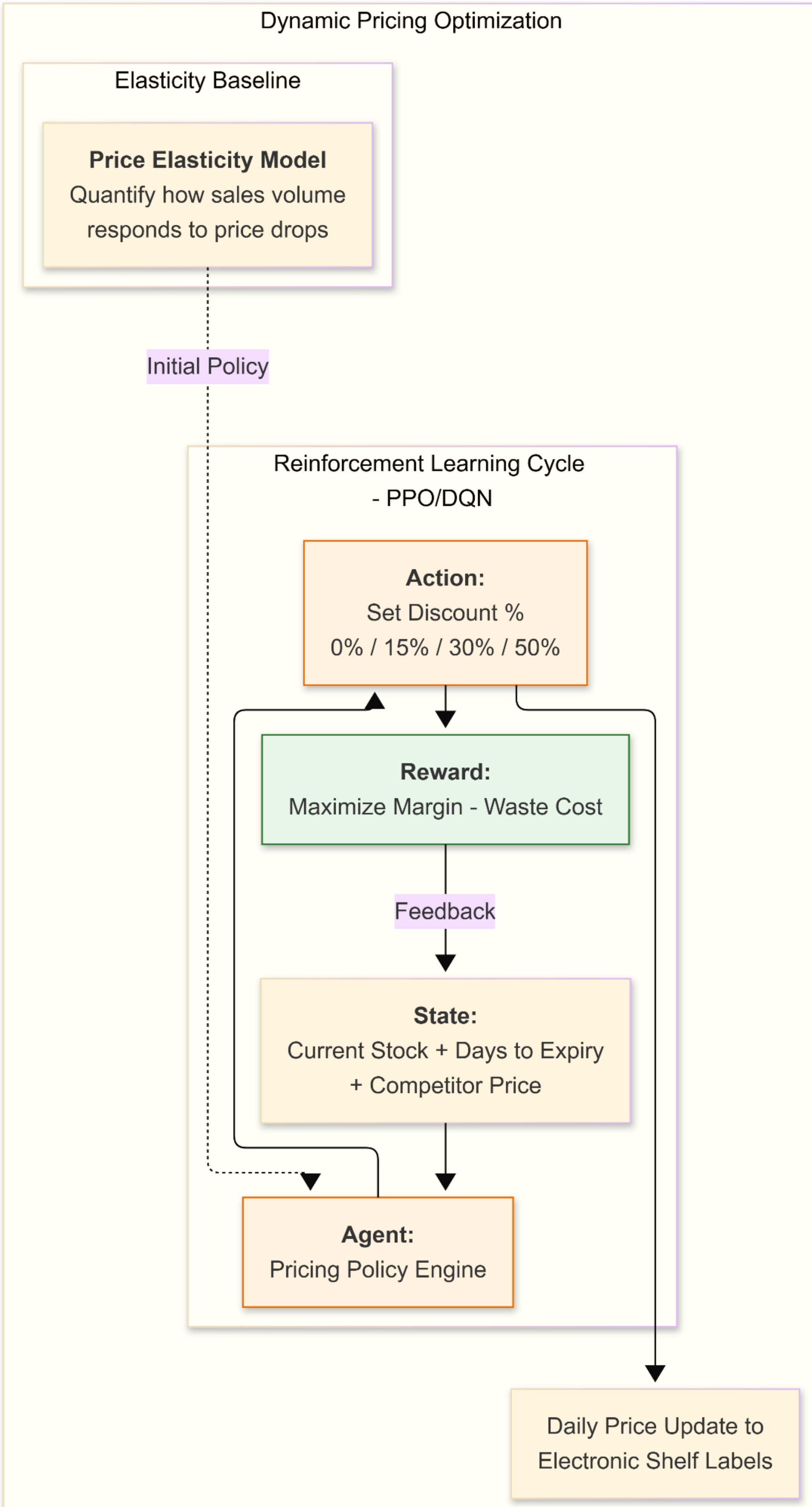
- **Layer 4 (Feature Store):** Provides the "State" variables (Stock, Expiry, Competitor Price).
- **Layer 5 (MLflow):** Serves the RL Agent via a REST API.
- **Layer 6 (Decision Engine):** The **Business Rules Engine** acts as a "Guardrail." If the RL agent suggests an 80% discount but corporate policy says "Max 50%," the Decision Engine overrides the agent.
- **Layer 7 (Write-back):** The approved price is pushed to **Symphony Gold**, which automatically updates the **Electronic Shelf Labels (ESL)**.

5. Operational Outcome: The "Smart Markdown"

Unlike a human who might forget to mark down the chicken until 6:00 PM, Model 3 might identify at 11:00 AM that the sales velocity is too slow for the current stock levels.

- **The AI Move:** It triggers a small **15% discount** early in the day.
- **The Result:** The chicken sells out by 4:00 PM at a higher total margin than if the store had waited until 6:00 PM to do a "desperation" 50% discount.

Summary: Model 3 ensures that Ahold Delhaize doesn't just "avoid waste"—it **optimizes the exit price** of every perishable item, turning potential losses into protected margins.


Dynamic Pricing Optimization
Elasticity Baseline
Price Elasticity Model
Quantify how sales volume responds to price drops
Initial Policy
Reinforcement Learning Cycle - PPO/DQN
Action:
Set Discount %
0% / 15% / 30% / 50%
Reward:
Maximize Margin - Waste Cost
Feedback
State:
Current Stock + Days to Expiry + Competitor Price
Agent:
Pricing Policy Engine
Daily Price Update to Electronic Shelf Labels

Summary of Algorithmic Synergy for Ahold Delhaize

1. **Model 1 (Forecasting)** feeds the **Model 2 (Waste)** engine. If the forecast is lower than current stock, the probability of waste increases.
2. **Model 2 (Waste)** serves as a "Negative Reward" trigger for **Model 3 (Pricing)**. If Model 2 predicts high waste, the Reinforcement Learning agent in Model 3 receives a massive penalty if it *doesn't* lower the price.
3. **Model 3 (Pricing)** then adjusts the price, which in turn changes the **Model 1 (Forecast)** because lower prices increase demand volume (Elasticity).

These two models represent the **Operational Execution** and **Strategic Procurement** logic. Model 4 focuses on moving goods efficiently through the network, while Model 5 ensures the source of those goods is reliable and sustainable.

MODEL 4: Replenishment Optimization (MEIO)

Goal: Optimize inventory across multiple tiers (Distribution Center to Store) while respecting physical and shelf-life constraints.

In the Ahold Delhaize 8-Layer Architecture, **Model 4 (M4: Replenishment Optimization)** is the "Tactical Orchestrator." While Model 1 predicts what will happen, M4 decides **what to do about it**.

M4 uses **Multi-Echelon Inventory Optimization (MEIO)** to balance stock across the entire supply chain—ensuring that the right amount of product is held at the Distribution Center (DC) and the right amount is pushed to the Store, preventing both empty shelves and full bins of spoiled food.

1. The Core Logic: What is "Multi-Echelon"?

In a traditional "Single-Echelon" system, the DC and the Store are optimized separately. This leads to "Bullwhip Effects" where small changes in store demand cause massive, unnecessary orders at the DC.

The MEIO Approach (M4):
Model 4 views the DC (SAP ECC/EWM) and the Store (Symphony Gold) as a **single organism**. It optimizes the "Stock Placement":

- **At the DC:** Hold inventory that is shared across many stores to handle "demand spikes" (Aggregated safety stock).
- **At the Store:** Hold only what is needed for immediate sales and visual abundance.

2. The Four Pillars of Model 4 Logic

A. Probabilistic Safety Stock Calculation

M4 consumes the **P10/P50/P90** outputs from Model 1.

- Instead of using a static "2 days of cover," M4 calculates safety stock based on **Lead-Time Variability** (from SAP TM) and **Forecast Error**.

- *Action:* If a route from a DC in the Netherlands to a store in Belgium is historically unreliable (high TM variance), M4 automatically increases the store-level safety stock for that specific route.

B. Shelf-Life Aware Replenishment

For perishables, M4 runs a **"Survival Logic"** calculation:

- **The Freshness Delta:** [Remaining Shelf Life] - [Lead Time to Store] - [Expected Days on Shelf].
- *Decision:* If the Freshness Delta is negative or near zero, M4 **blocks the replenishment**. It would rather have a controlled out-of-stock than ship a product that will expire before it can be sold.

C. The Constraint Solver (The Reality Filter)

This is where the math meets the physical world (EWM/TM functionality). M4 runs a **Linear Programming (LP) model** to solve for the following:

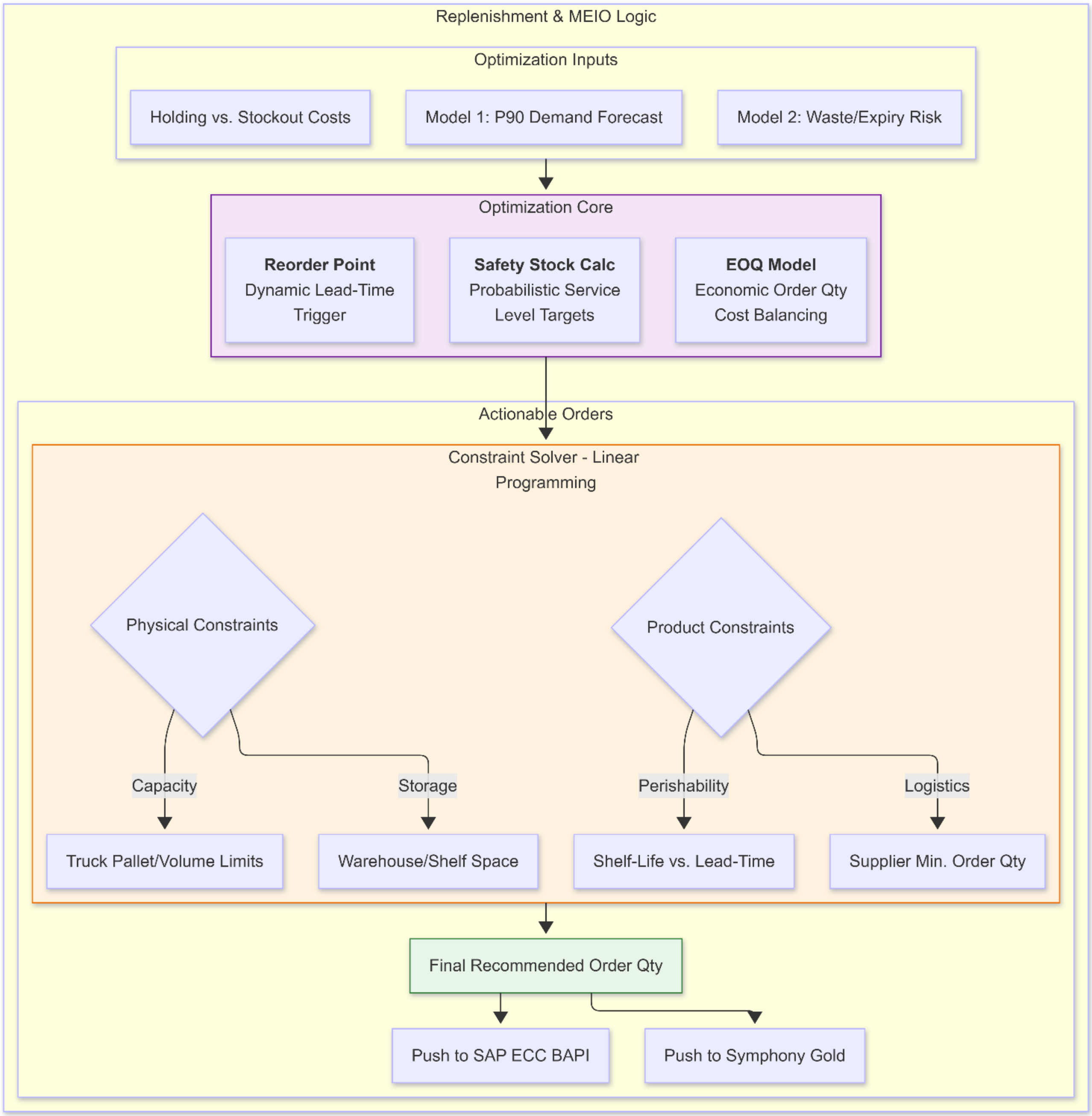
- **Cube/Weight Optimization (TM):** It "builds" a virtual truck. If the AI wants to send 10.5 pallets, but the truck holds 10, the Solver decides which 0.5 pallets are the least critical to margin and removes them.
- **Picking Capacity (EWM):** If the DC labor schedule (EWM Resource Mgmt) shows a 20% staff shortage, M4 reduces the total order volume to a level that the warehouse can physically pick, preventing a backlog.

D. Economic Order Quantity (EOQ) and Minimums

- M4 respects **Supplier MOQs** (Minimum Order Quantities) from SAP ECC.
- *Logic:* If a store needs 5 units, but the supplier only ships in cases of 12, M4 evaluates if the extra 7 units will likely result in waste (M2) or if they can be sold before expiry.

3. Business Value (The EBITDA Impact)

1. **Working Capital Reduction (€150M Goal):** By optimizing the "Echelons," Ahold Delhaize stops holding duplicate safety stock at both the DC and the store. This releases cash trapped in inventory.
2. **Waste Reduction:** By factoring in "Freshness Delta," M4 stops the flow of "dead on arrival" produce, directly lowering the scrap rate.
3. **Freight Optimization:** By filling trucks to their mathematical maximum (respecting TM weight/volume), logistics costs per unit are minimized.
4. **S/4HANA Sync:** The optimized inventory levels lead to more accurate "Inventory Valuation" in the **Universal Journal (ACDOCA)**, providing the CFO with high-integrity financial projections.



MODEL 5: AI-Driven Supplier Selection

Goal: Evaluate and rank suppliers using a multi-criteria scorecard to ensure supply chain resilience and ESG (Environmental, Social, and Governance) compliance.

In the Ahold Delhaize 8-Layer Architecture, **Model 5 (M5: AI-Driven Supplier Selection)** serves as the **Strategic Procurement Brain**. While Models 1 through 4 focus on *how much* to order and *when* to move it, Model 5 focuses on **from whom** to buy to ensure the long-term health, sustainability, and resilience of the Ahold Delhaize global supply chain.

By shifting from a "Price-Only" model to a **"Multi-Criteria Decision Making" (MCDM)** model, Ahold Delhaize can mathematically balance lower costs with lower carbon footprints and higher delivery reliability.

1. The Four Pillars of the Scoring Engine

Model 5 evaluates every supplier in the Ahold Delhaize network against four critical dimensions:

A. Supply Chain Resilience (Predictive Reliability)

- **Source Data:** SAP ECC (EKPO) and SAP TM.
- **Logic:** Uses **Bayesian Statistics** to move beyond "average lead time." It calculates the probability of a supplier failing during a specific event (e.g., "What is the risk of this supplier failing during the Christmas peak?").
- **KPIs:** Lead-time variance, order fill rate, and "Time to Recover" after a disruption.

B. ESG Compliance (Sustainability & Ethics)

- **Source Data:** S/4HANA ESG modules + External APIs (EcoVadis, MSCI).
- **Logic:** Tracks **Scope 3 Emissions**—the carbon produced by the supplier to create the product. It also analyzes social compliance (labor practices) from audit logs.
- **KPIs:** Carbon intensity per kg of product, Water stress index, Fair labor certification status.

C. Total Cost of Ownership (TCO)

- **Source Data:** SAP ECC (Valuation) and S/4HANA (Finance).
- **Logic:** AI looks beyond the unit price. It factors in "Hidden Costs": the cost of waste caused by poor quality (from Model 2), the cost of storage for bulk buys, and the logistics cost from SAP TM.
- **KPIs:** Landed cost, Waste-adjusted margin, Payment term efficiency.

D. Product Quality & Freshness

- **Source Data:** SAP QM (Quality Management) and EWM.
- **Logic:** Aggregates rejection rates and "Remaining Shelf Life at Receipt."
- **KPIs:** % of items rejected at the DC, average days of life remaining upon delivery.

2. The Algorithmic Heart: MCDM & Clustering

Model 5 uses two primary AI techniques to provide actionable rankings:

1. **Analytic Hierarchy Process (AHP):**
    - This is a mathematical technique that allows Ahold Delhaize leadership to set "Weights."
    - *Example:* During a crisis, "Resilience" might be weighted at 60%. During normal operations, "ESG" and "Cost" might be weighted at 40% each. The model re-ranks the supplier base instantly based on these corporate "dials."
2. **K-Means Clustering (Risk Profiles):**

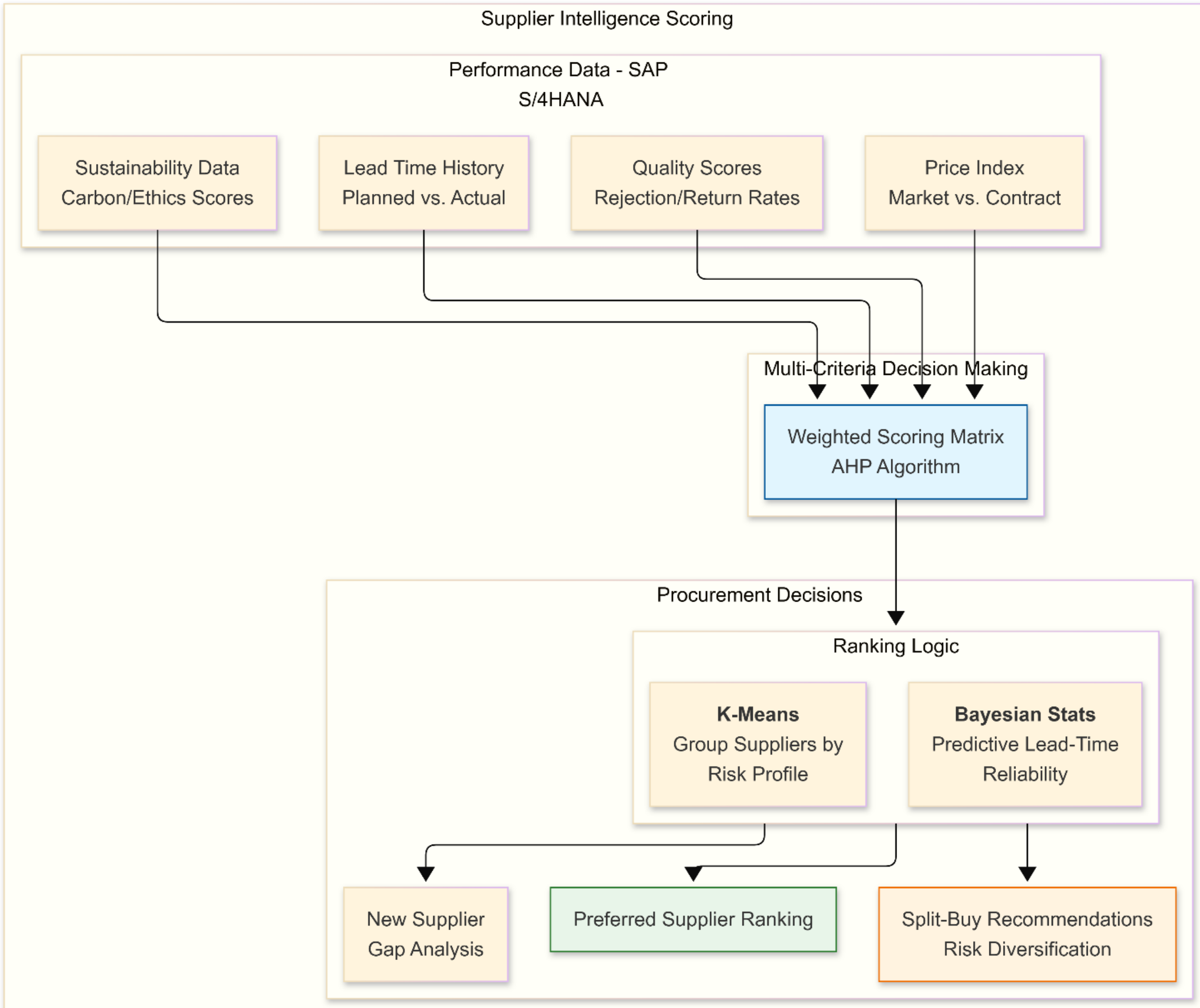
    - Model 5 groups suppliers into clusters:
        - *The Champions:* High ESG, High Reliability, Low TCO.
        - *The Risky Partners:* Low Cost, but high lead-time variance and poor ESG.
    - This allows procurement to identify "Single-Source Risks" where Ahold Delhaize is too dependent on a "Risky Partner."

3. Integration with the 8-Layer Architecture

- **Layer 4 (Feature Store):** Stores the "Supplier Reliability Index" and "Carbon-per-SKU" as features.
- **Layer 6 (Decision Engine):** When **Model 4 (Replenishment)** needs to place an order, the **Constraint Solver** checks Model 5. If the primary supplier is "High Risk," the Solver triggers a **"Split-Buy"**—ordering 50% from the primary and 50% from a highly resilient secondary supplier.
- **Layer 7 (Write-back):** The "Preferred Supplier" ranking is pushed back to **SAP S/4HANA**. This ensures that when a buyer opens a purchase requisition, the system automatically suggests the supplier that best balances margin with ESG goals.
- **Layer 8 (User Interface):** Category VPs use **SAP Analytics Cloud** to view the "EBITDA vs. ESG" tradeoff, allowing them to report to shareholders exactly how much carbon was removed from the supply chain through AI-driven selection.

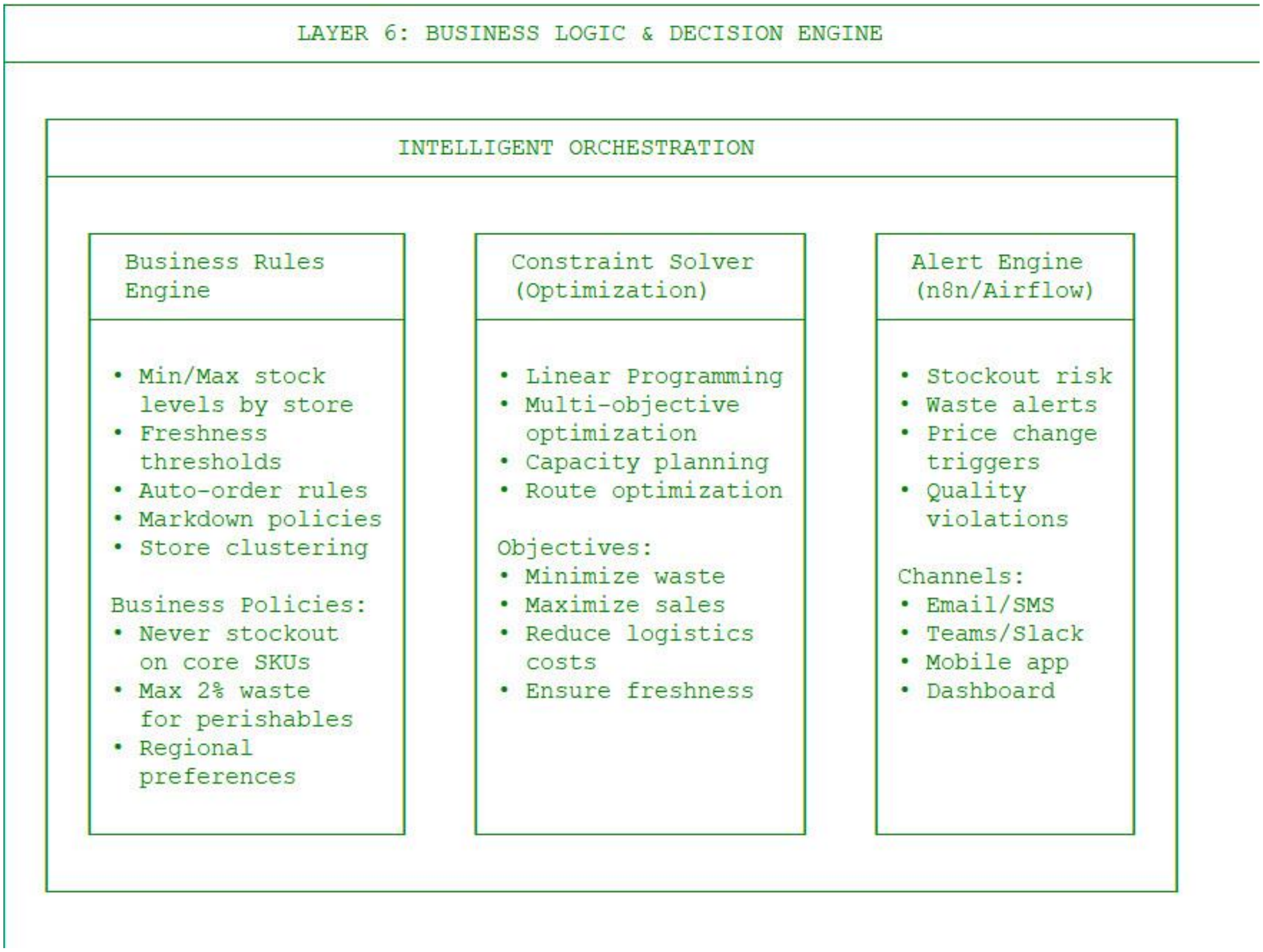
4. Business Value (The "Ahold Delhaize" Advantage)

1. **Resilience:** Prevents empty shelves during geopolitical or environmental disruptions by identifying "Back-up" suppliers *before* the crisis hits.
2. **ESG Leadership:** Directly supports Ahold Delhaize's commitment to the **Science Based Targets initiative (SBTi)** by mathematically favoring low-carbon suppliers.
3. **Waste Reduction:** By favoring suppliers with better "Freshness at Arrival" scores (from QM/EWM data), Model 5 reduces the waste potential even before the goods enter the Ahold Delhaize stores.
4. **Audit Defense:** Provides a high-integrity, automated audit trail for **EU Corporate Sustainability Reporting Directive (CSRD)** compliance.



Synergy between Model 4 and Model 5

1. **Lead-Time Feedback:** Model 5 (Supplier Selection) calculates the **Actual Lead Time Reliability**. This data is fed directly into Model 4's **Safety Stock** calculation. If a supplier becomes unreliable, Model 4 automatically increases the safety stock buffer for those SKUs.
2. **Quality-Waste Link:** Model 5 monitors **Quality Scores**. If a certain supplier's "Freshness at Arrival" drops, Model 2 (Waste Prediction) is alerted to adjust the expiry risk higher for those specific batches.
3. **Financial Alignment:** Both models feed into **SAP S/4HANA**. Model 4 optimizes Working Capital (Inventory on hand), while Model 5 optimizes Procurement Spend and Sustainability reporting for Ahold Delhaize's annual ESG goals.

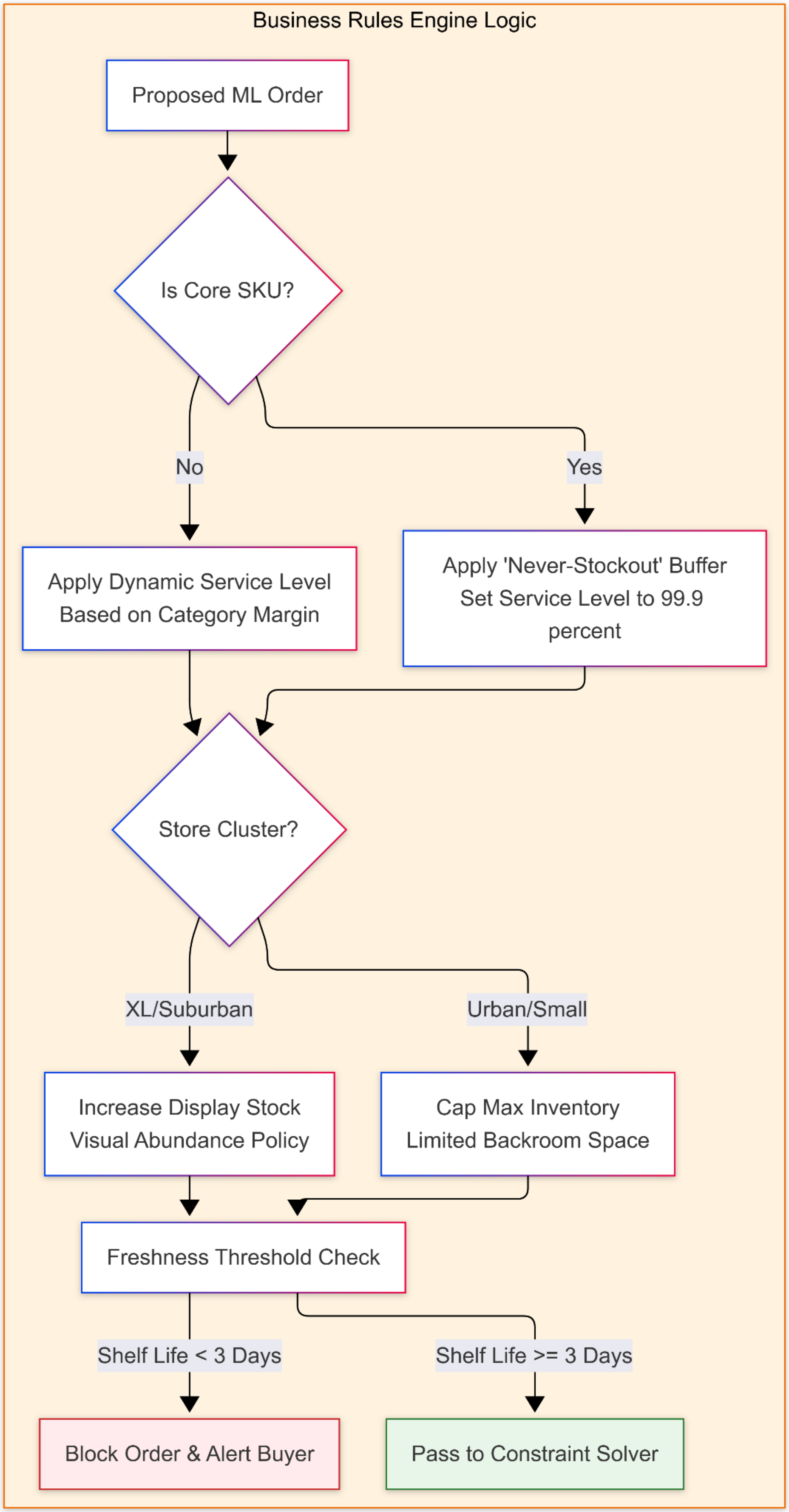


Layer 6 acts as the **Intelligent Orchestrator**. It translates the "mathematical ideals" from the AI Models (Layer 5) into "operational realities" that the ERP systems can execute.

Here is the detailed business logic for each stream within Layer 6.

1. Business Rules Engine (Policy Guardrails)

This stream ensures that AI recommendations align with Ahold Delhaize's corporate strategy, regional preferences (NL vs. CEE), and brand-specific "never-out" policies.

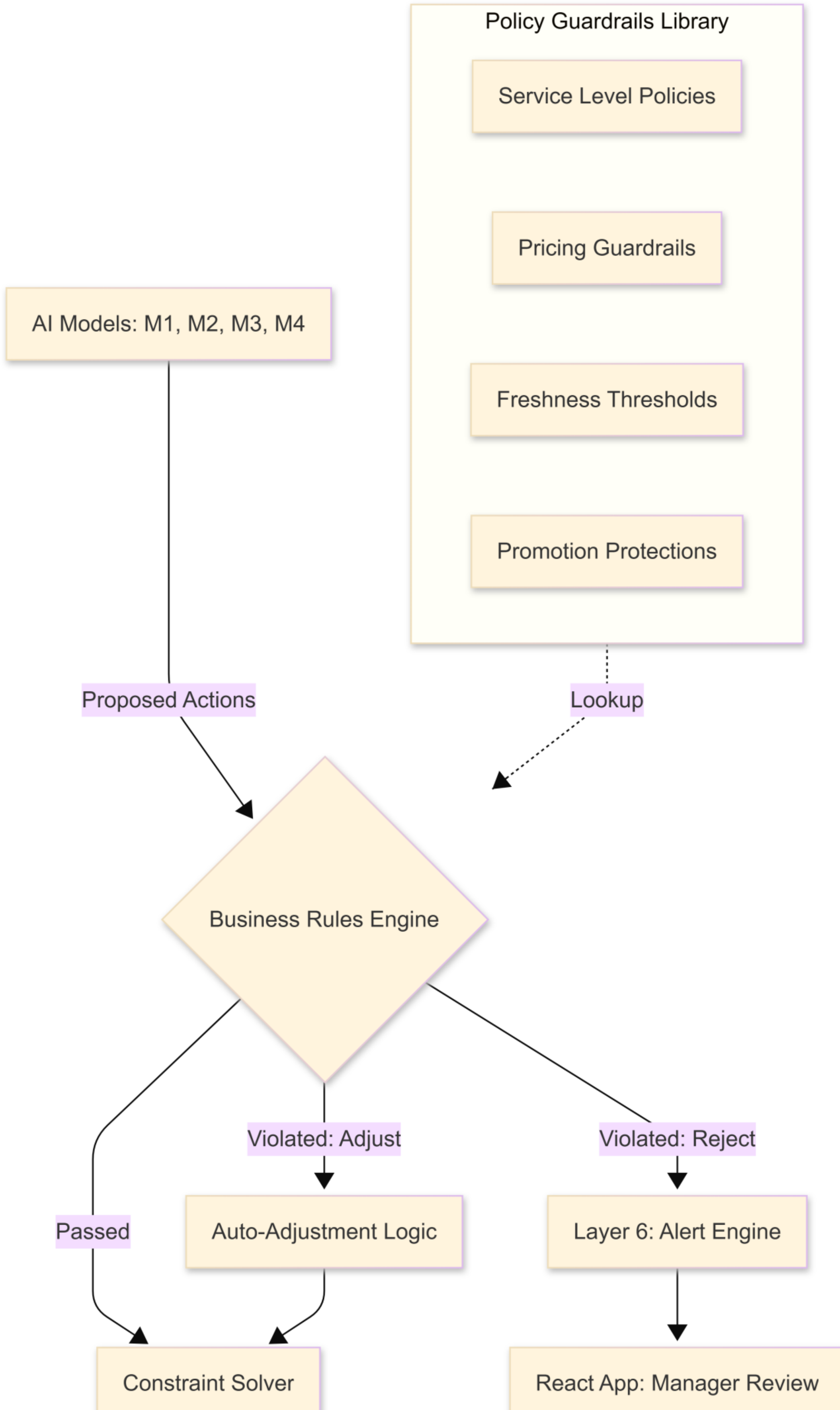

Business Rules Engine Logic
Proposed ML Order
Is Core SKU?
No
Yes
Apply Dynamic Service Level Based on Category Margin
Apply 'Never-Stockout' Buffer Set Service Level to 99.9 percent
Store Cluster?
XL/Suburban
Urban/Small
Increase Display Stock Visual Abundance Policy
Cap Max Inventory Limited Backroom Space
Freshness Threshold Check
Shelf Life < 3 Days
Shelf Life >= 3 Days
Block Order & Alert Buyer
Pass to Constraint Solver

In the Ahold Delhaize 8-Layer Architecture, the **Business Rules Engine (BRE)** serves as the **Strategic Filter**. While the AI models (Layer 5) are optimized for mathematical efficiency, the BRE ensures that those mathematical outputs align with **corporate strategy, legal regulations, and brand identity**.

It acts as the "Guardrail" that prevents the AI from making decisions that are technically optimal but business-irrational (e.g., discounting milk to zero or stocking out of bread during a national holiday).

1. The Logic Flow: From AI "Request" to Business "Approval"

The BRE sits between the AI Models (Layer 5) and the Physical Solver (Layer 6). It evaluates the "Proposed Action" against a library of hundreds of "Policy Rules."


Policy Guardrails Library
Service Level Policies
Pricing Guardrails
Freshness Thresholds
Promotion Protections
AI Models: M1, M2, M3, M4
Proposed Actions
Lookup
Business Rules Engine
Violated: Adjust
Violated: Reject
Passed
Auto-Adjustment Logic
Layer 6: Alert Engine
Constraint Solver
React App: Manager Review

2. Core Policy Guardrail Categories

For Ahold Delhaize, the BRE is categorized into four "Commandments":

A. Strategic Availability (The "Never-Out" List)

Ahold Delhaize’s brand trust relies on "Value Drivers" (Milk, Bread, Bananas, Diapers).

- **The Rule:** Any AI recommendation that suggests a "Service Level" below 99.5% for "Category A" SKUs is automatically rejected.
- **The Guardrail:** The BRE will "override" the AI to increase safety stock, even if the AI predicts a slight waste risk, to protect the customer’s basket completion.

B. Freshness & Safety (The Perishable Logic)

This ensures food safety and quality standards across the Albert Heijn, Delhaize, and CEE banners.

- **The Rule: Shelf-Life Buffer.** An order cannot be placed if the Remaining Shelf Life < (Lead Time + 2 Days).
- **The Guardrail:** If **Model 4** proposes an order for fresh salmon, the BRE checks the **SAP EWM** batch data. If the available batch in the DC is too old to survive the journey and stay on the shelf for 48 hours, the order is blocked.

C. Pricing & Margin Protection (Financial Guardrails)

This prevents **Model 3 (Reinforcement Learning)** from being too aggressive in clearing stock.

- **The Rule: Floor Price.** No markdown can exceed 50% without a Store Manager's digital approval. No markdown can be applied to "Premium" Private Label items (e.g., *AH Excellent*) during the first 24 hours of display.
- **The Guardrail:** If the AI suggests a 70% "Flash Sale" to clear strawberries, the BRE caps it at 50% and sends an alert to the **Alert Engine** for human intervention.

D. Promotion & Cannibalization Logic

Prevents the AI from ordering "Normal" stock when a "Promotional" equivalent is about to launch.

- **The Rule: Promo Blocking.** Block all "Regular" replenishment for SKU X if a "Bonus/Promo" version of SKU X is arriving within 48 hours.
- **The Guardrail:** This uses data from **Symphony Gold’s Promotion Management** module to ensure the store doesn't end up with double inventory.

3. Regional Contextualization (The "Federated" Feature)

Because Ahold Delhaize is a federation of brands, the BRE is not "one-size-fits-all." The engine uses a **Metadata-Driven approach**:

1. **Global Rules:** (e.g., Food safety standards) apply to all regions.
2. **Banner-Specific Rules: Albert Heijn (NL)** may have a "Visual Abundance" policy (keeping shelves 100% full for aesthetics), while **Mega Image (RO)** may prioritize "Waste Minimization" due to different margin structures.
3. **Local Holiday Logic:** The BRE automatically adjusts "Min/Max" stock levels based on a local calendar (e.g., *Sinterklaas* in NL, *Unification Day* in RO) which the base AI model might not have fully "learned" yet.

4. Technical Implementation (Databricks + JSON)

The BRE is implemented as a **High-Speed Decoupled Service** within the Databricks Lakehouse:

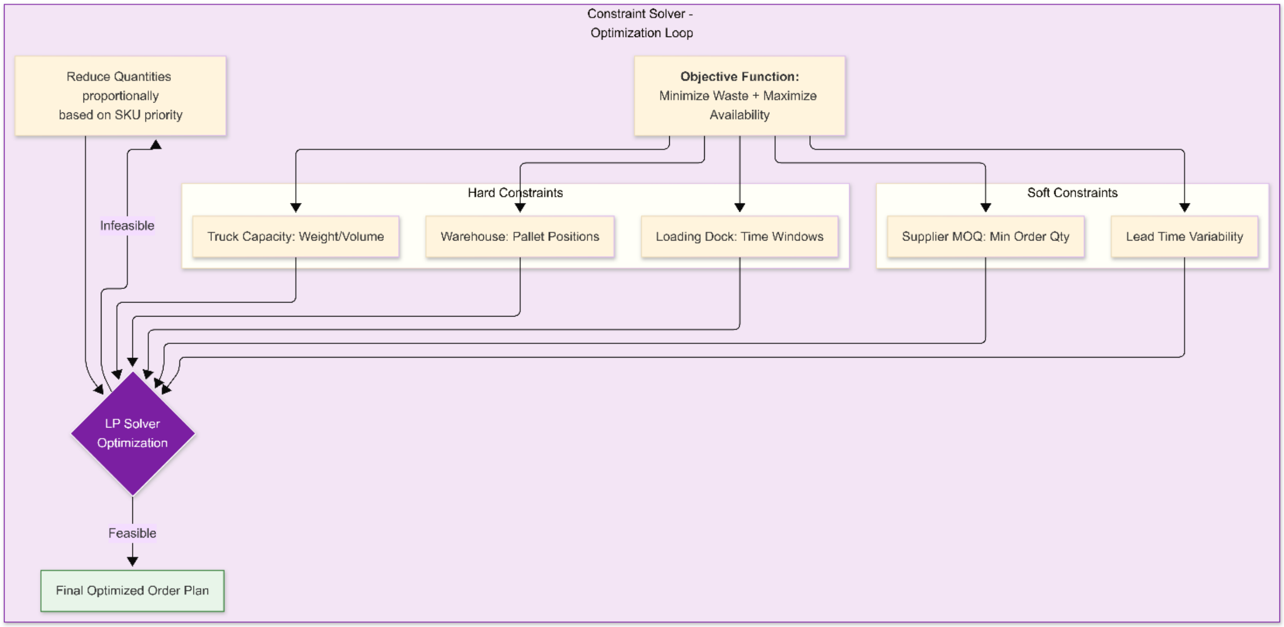
- **Configuration:** Rules are stored in **JSON/YAML files** or a low-code UI, allowing Business Analysts (not just Coders) to update a "Guardrail" in minutes.
- **Processing:** The engine uses **PySpark UDFs** (User Defined Functions) to evaluate millions of SKU-Store recommendations in parallel against the rule set.
- **Auditability:** Every time a rule overrides an AI recommendation, the BRE logs the **"Reason Code"** to the **Silver Layer**. This data is then used by the **MLOps team** to see where the AI is "fighting" with business policy, signaling a need for model retraining.

5. Summary of Business Value

- **Brand Protection:** Ensures the AI doesn't damage the brand through poor availability or excessive discounting.
- **Operational Control:** Gives Category VPs a "Kill Switch" or "Dials" to adjust the supply chain strategy instantly without rewriting code.
- **Regulatory Compliance:** Hard-codes EU food safety and pricing transparency laws into the automated ordering loop.

2. Constraint Solver (Physical Optimization)

This stream uses **Linear Programming** to solve the "Tetris" problem: How to fit the ideal order into physical trucks and warehouses while meeting supplier minimums.



In the Ahold Delhaize 8-Layer Architecture, the **Constraint Solver (Layer 6)** is the **"Reality Filter."**

While Model 1-4 calculate the "ideal" inventory levels based on math, and the Business Rules Engine (BRE) ensures those levels align with "policy," the Constraint Solver is responsible for ensuring the final order is **physically and logistically possible.** It reconciles the AI's "desires" with the hard limits of the **SAP EWM (Warehouse)** and **SAP TM (Transportation)** systems.

1. The Logic: "Mathematical Ideal" vs. "Physical Reality"

If the AI predicts a surge in demand for watermelons but the warehouse is out of forklift drivers, or the only available truck is a small refrigerated van, the Constraint Solver "trims" the order to fit the available resources.

Mathematical Engine

The solver uses **Mixed-Integer Linear Programming (MILP)** or **Heuristic Search Algorithms** (via Python libraries like Google OR-Tools, Pyomo, or PuLP) to solve the multi-objective optimization problem.

2. The Three Constraint Dimensions

The Solver ingests real-time CDC data from the **Logistics Digital Twin** (Layer 3) to evaluate three critical walls:

A. Volumetric & Payload Constraints (SAP TM Integration)

The solver performs "Virtual Load Building" to ensure the order fits the assigned vehicle.

- **The Data:** Ingests /SCMTMS/D_VEH_CAP (Vehicle Capacity) and /SCMTMS/D_TORROT (Route Data).
- **The Solver Logic:**
  - **3D Tetris:** It calculates the total cubic volume (m^3) and weight (kg) of the proposed order.
  - **Axle Weight:** It ensures the payload is balanced across the truck's axles.
  - **Temperature Separation:** If the truck has a "dual-temp" bulkhead, the solver ensures frozen items and ambient items are partitioned correctly.

B. Throughput & Resource Constraints (SAP EWM Integration)

The solver ensures the warehouse can physically handle the "picking" and "staging" of the order.

- **The Data:** Ingests /SCWM/AQUA (Available Quantity) and /SCWM/RSRC (Resource/Labor status).
- **The Solver Logic:**
  - **Labor Wall:** If EWM reports that 20% of pickers are absent, the solver "throttles" the total volume of orders to prevent a backlog at the loading dock.
  - **Staging Space:** It checks if there is enough physical square footage in the "Chilled Staging Area" to hold the pallets before the truck arrives.

C. Product-Specific Constraints (The "Freshness Clock")

Specifically for perishables, the solver calculates the **"Transit-to-Life"** ratio.

- **The Data:** Ingests Batch Expiry dates from EWM and Route Lead Times from TM.
- **The Solver Logic:**
  - **Freshness Threshold:** If a route to a remote store takes 8 hours and the batch in the warehouse only has 36 hours of life left, the solver **rejects** the order to prevent "Dead on Arrival" waste.
  - **MOQ Snapping:** It "snaps" AI decimal recommendations (e.g., 12.4 crates) to physical units (e.g., 12 full crates or 1 full pallet).

3. The Objective Function

The Solver doesn't just "cut" orders; it re-optimizes them. Its mathematical objective is:
Maximize:

$$\text{Total\,Margin} + \text{Freshness}$$

Minimize:

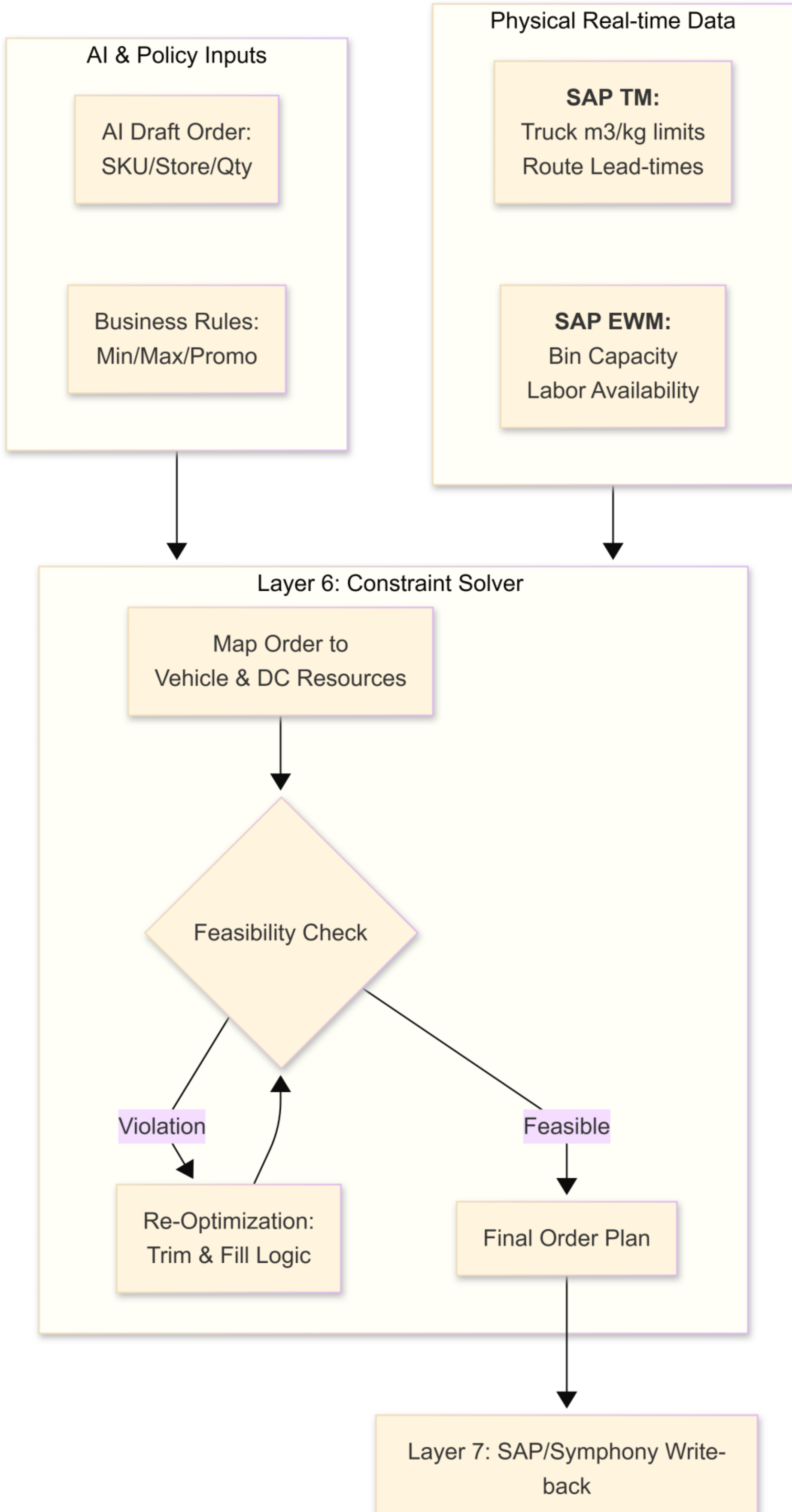
```
Logistics CostLogistics\,CostLogisticsCost
```

+

```
CO2 EmissionCO2\,EmissionCO2Emission
```

The Trade-off Logic:
If a truck is 95% full, the solver will search for "Fill-in" SKUs—low-priority but high-shelf-life items (like toilet paper or pasta)—to fill the remaining 5% and maximize the **Truck Fill Rate (TFR)**.

4. Detailed Execution Flow


AI & Policy Inputs
AI Draft Order: SKU/Store/Qty
Business Rules: Min/Max/Promo
Physical Real-time Data
SAP TM: Truck m3/kg limits Route Lead-times
SAP EWM: Bin Capacity Labor Availability
Layer 6: Constraint Solver
Map Order to Vehicle & DC Resources
Feasibility Check
Violation
Feasible
Re-Optimization: Trim & Fill Logic
Final Order Plan
Layer 7: SAP/Symphony Write-back

5. Operational Scenario: "The Saturday Night Squeeze"

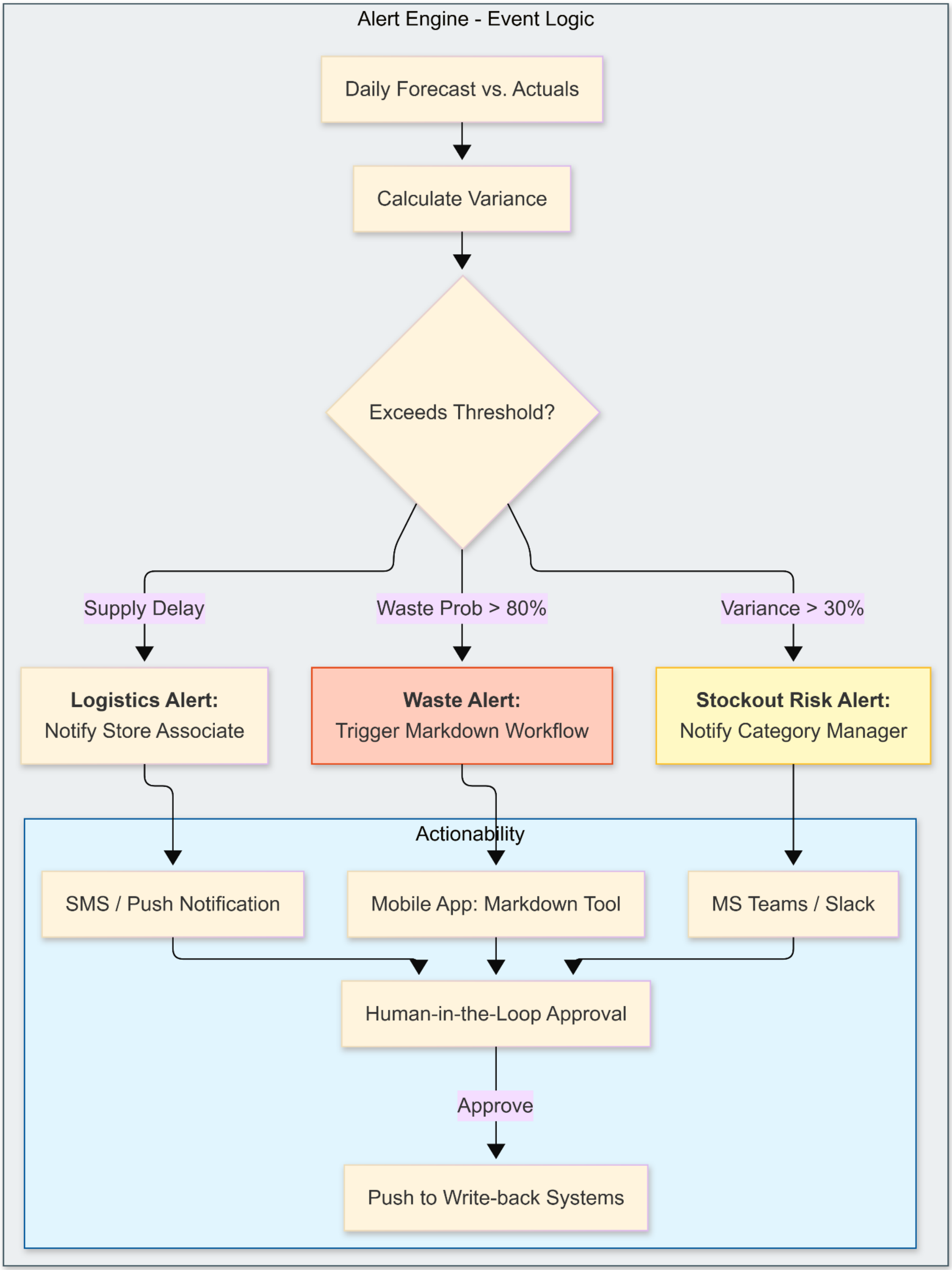
- **AI Proposal:** Model 1-4 proposes 5,000 units of fresh meat and produce for a major Delhaize store in Brussels to prepare for Sunday morning shoppers.
- **EWM Conflict:** EWM data shows the "Refrigerated Pick Face" is undergoing maintenance; only 3,000 units can be picked per hour.
- **TM Conflict:** The Saturday night truck is already 40% full with "Pre-ordered" bakery items.
- **Constraint Solver Action:**
    1. It identifies that 5,000 units won't fit the truck and can't be picked in time.
    2. It prioritizes "High-Waste Risk" items (berries/salads) over "Longer-Life" items (carrots/onions).
    3. It reduces the meat order to fit the weight limit.
    4. **Result:** It outputs a feasible order of 3,200 units that perfectly fills the truck and matches the warehouse labor capacity.

6. Summary of Business Value

1. **Reduction in "Refused Deliveries":** No more trucks arriving at stores with more pallets than the store's backroom can hold.
2. **Labor Efficiency:** Aligns the "Workload" (Orders) with the "Workforce" (EWM Labor), reducing overtime costs.
3. **Sustainability:** Maximizes truck fill rates, reducing the number of "Carbon-Empty" miles driven by the Ahold Delhaize fleet.
4. **Operational Trust:** Store managers trust the system because it never sends them an order they can't physically receive.

3. Alert Engine (Exception Management)

This stream monitors the gap between prediction and reality. It uses **n8n or Airflow** to trigger human intervention only when the AI cannot resolve a conflict autonomously.


Alert Engine - Event Logic
Daily Forecast vs. Actuals
Calculate Variance
Exceeds Threshold?
Supply Delay
Waste Prob > 80%
Variance > 30%
Logistics Alert: Notify Store Associate
Waste Alert: Trigger Markdown Workflow
Stockout Risk Alert: Notify Category Manager
Actionability
SMS / Push Notification
Mobile App: Markdown Tool
MS Teams / Slack
Human-in-the-Loop Approval
Approve
Push to Write-back Systems

In the Ahold Delhaize 8-Layer Architecture, the **Alert Engine (Layer 6)** is the system's **"Watchman."** While the AI and Solvers strive for 100% automation, the Alert Engine identifies the **exceptions**—the 15% of scenarios where the math fails, the data is corrupt, or a "black swan" event (like a sudden logistics strike or extreme weather) occurs.

It prevents "silent failures" by ensuring that the right human—whether a Store Manager in Brussels or a Category Buyer in Zaandam—is notified at the right time to take action.

1. The Core Logic: Exception Detection vs. Noise

The primary challenge in retail is **Alert Fatigue**. The Alert Engine uses **Dynamic Thresholding** to distinguish between normal operational "jitter" and a critical business risk.

The Detection Framework

The engine monitors the **Databricks Gold Layer** and **SAP Event Mesh** for four specific types of exceptions:

1. **Demand Anomaly:** Actual sales deviate from the Model 1 forecast by
2. `>X%> X\%>X%`

   . (e.g., A local festival causes a "run" on bottled water that the AI missed).

3. **Logistics Disruption:** A truck tracked in **SAP TM** is delayed beyond a "safety buffer," threatening a stockout of perishables.
4. **Freshness Crisis: Model 2** identifies a specific batch where the Probability of Waste has jumped from 20% to 80% within 4 hours.
5. **Integrity Failure:** The data feed from **Symphony Gold** or **SAP ECC** is delayed, meaning the AI is making decisions based on "stale" inventory.

2. Workflow Orchestration (n8n & Logic Apps)

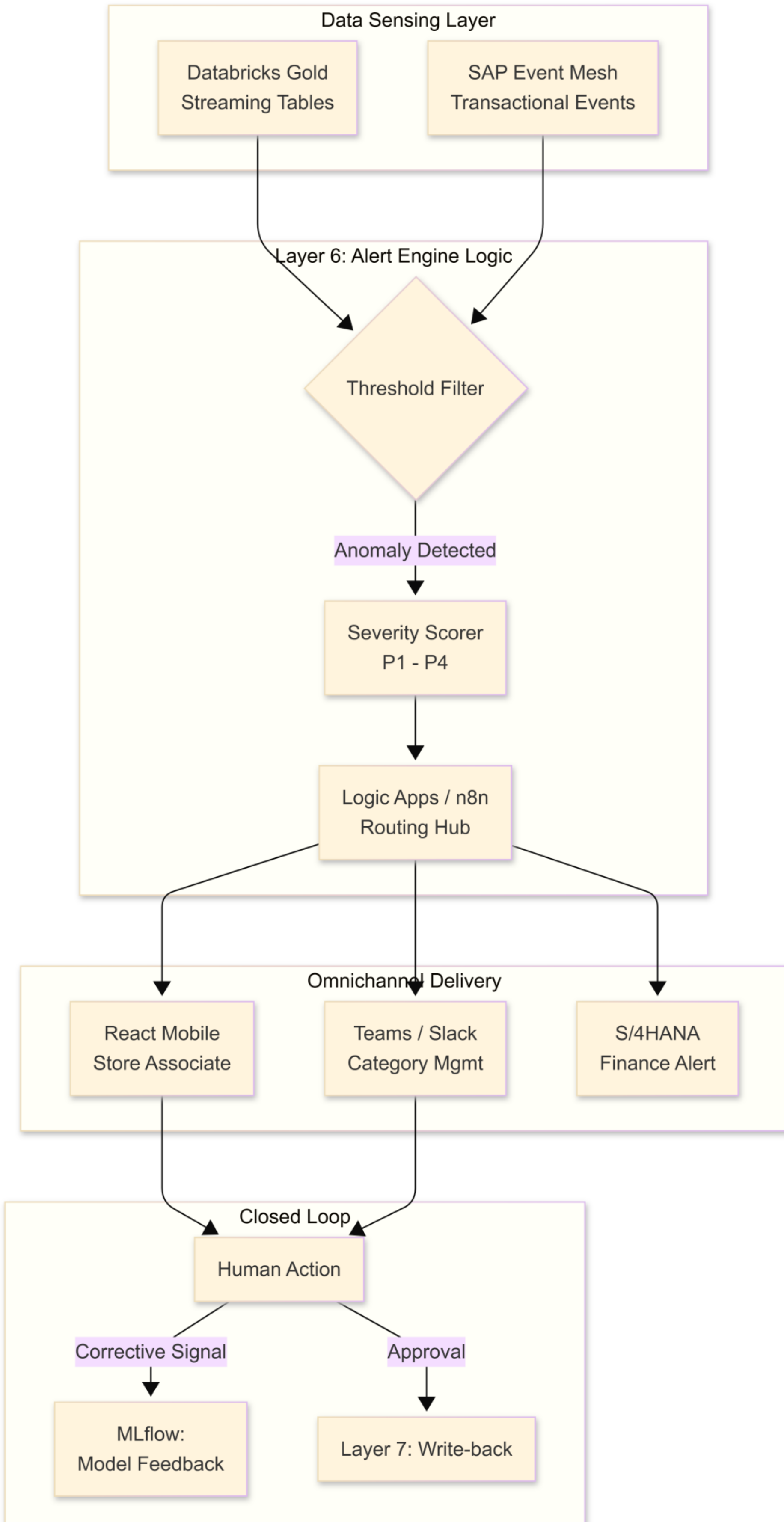
The Alert Engine isn't just a "sender of emails"; it is a **Workflow Orchestrator**. We use **n8n** (for flexible, low-code logic) or **Azure Logic Apps** (for enterprise-grade SAP connectivity).

The Step-by-Step Workflow:

- **Step 1: Event Capture:** An exception is detected in the Databricks Delta Lake.
- **Step 2: Severity Scoring:** The engine assigns a score (P1 - Critical to P4 - Informational).
- **Step 3: Routing:**
  - *P1 (Stockout in 2 hours):* Triggers an immediate **Push Notification** to the Store Manager's mobile React App via **Azure SignalR**.
  - *P2 (Supplier Lead-time Drift):* Creates a task in the **Category Buyer's** dashboard.
  - *P3 (Data Quality Issue):* Opens a ticket in **Azure DevOps** for the Data Engineering team.
- **Step 4: Human-in-the-Loop (HITL):** The user sees the alert, views the AI's "Reason Code," and clicks **Approve**, **Reject**, or **Modify**.

- **Step 5: Resolution Loop:** Once the human acts, the engine triggers the **Layer 7 Write-back** (e.g., updating a price in Symphony or a PO in SAP).

3. Detailed Alert Engine Components


Data Sensing Layer
Databricks Gold Streaming Tables
SAP Event Mesh Transactional Events
Layer 6: Alert Engine Logic
Threshold Filter
Anomaly Detected
Severity Scorer P1 - P4
Logic Apps / n8n Routing Hub
Omnichannel Delivery
React Mobile Store Associate
Teams / Slack Category Mgmt
S/4HANA Finance Alert
Closed Loop
Human Action
Corrective Signal
Approval
MLflow: Model Feedback
Layer 7: Write-back

4. Categorized Exception Scenarios

Exception Category	Trigger Example	AI Response	Alert Engine Action
Commercial	Competitor price drop detected via Web Scraping.	Model 3 suggests 20% price cut.	Notifies Category Manager for "Price Match" approval.
Operational	"Phantom Stock": POS shows sales, but Inventory stays high.	Model 1 predicts a stockout.	Sends "Inventory Verification" task to Store Associate's mobile app.
Logistics	SAP TM reports truck broke down.	Model 4 suggests re-routing from DC 2.	Notifies Warehouse Lead to prioritize "Emergency Pick" for DC 2.
Financial	S/4HANA shows waste costs exceeding budget.	N/A	Triggers "Strategic Review" alert for the CFO in SAP Analytics Cloud.

5. Strategic Value: "Human-in-the-Loop"

The Alert Engine is the bridge that builds **trust** in the AI.

- **Explainability:** Every alert includes the **"Why."** Instead of just saying "Discount this," the alert says: *"Tomato waste risk is 85% because sales slowed by 20% since the morning rain began."*
- **Continuous Learning:** When a Store Manager rejects an alert (e.g., "Don't discount, we have a private event tonight"), that rejection is fed back into **MLflow**. The AI learns the "Hidden Context" of that specific store, reducing future false alarms.

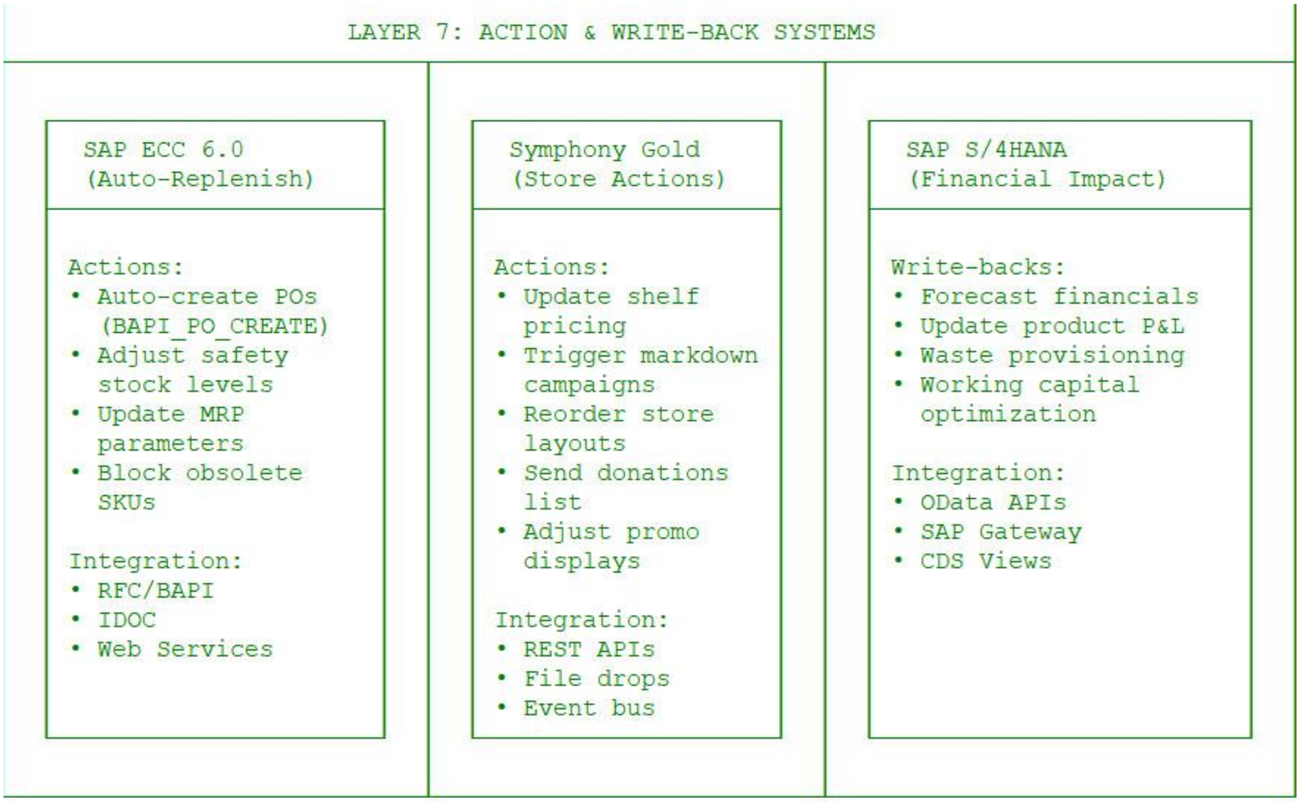
6. Summary for Ahold Delhaize

The Alert Engine ensures the inventory optimization remains **safe and accountable**. It allows the organization to scale AI across thousands of stores without needing thousands of analysts. It transforms store employees from "data entry clerks" into **"Decision Makers"** who only act when the system identifies a genuine risk to margin or availability.

Synergy of Layer 6 Orchestration

1. **The Filter (BRE):** It takes the "Model 1" forecast and says: *"I know the AI wants 500 units, but this is a 'Small Express' store, they only have room for 200."*
2. **The Realist (Solver):** It takes the filtered 200 units and says: *"The supplier only ships in multiples of 50. I will order 200, but I must remove 10 units of a different product to fit the truck volume."*

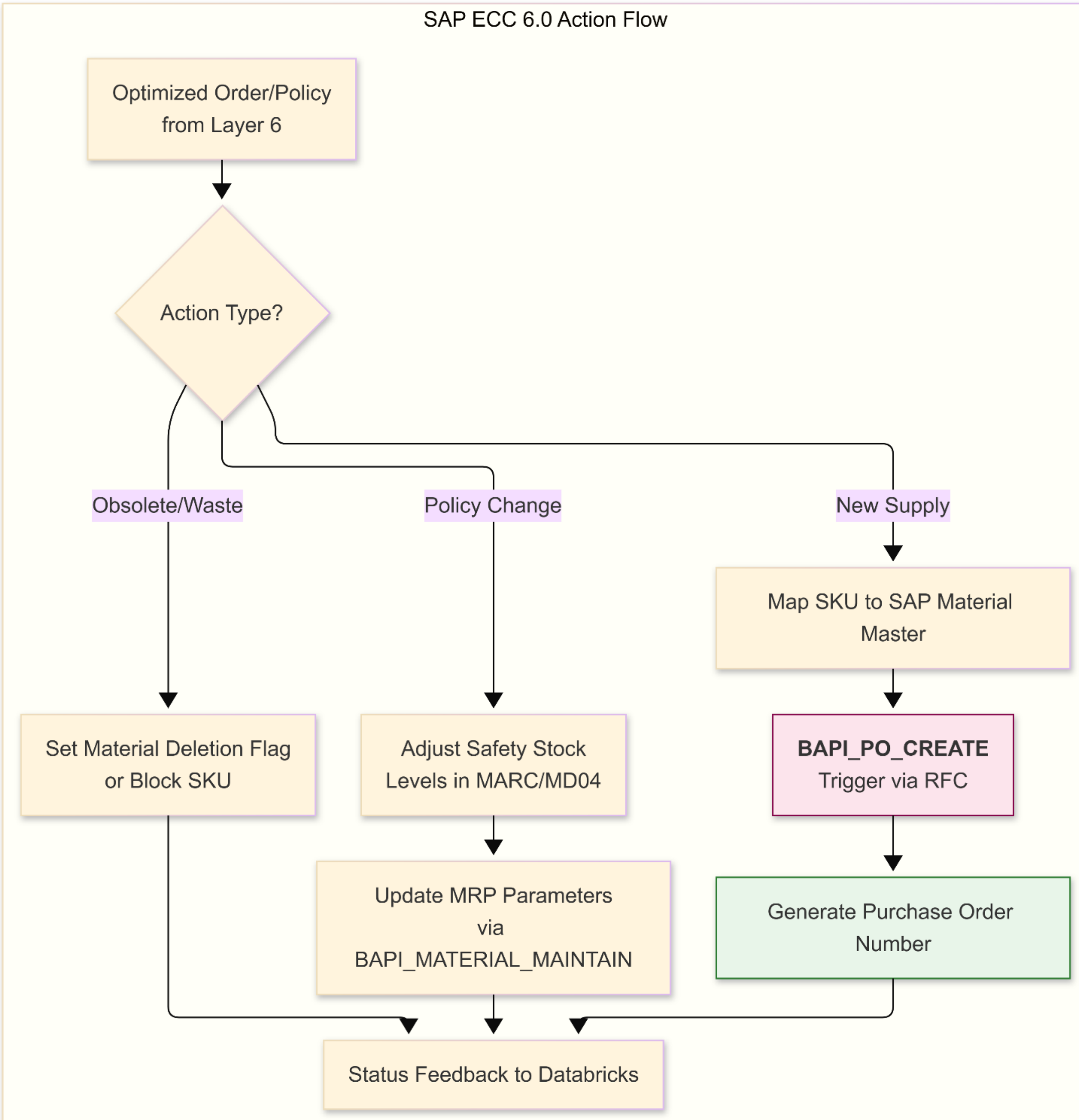
3. **The Watchman (Alerts):** If the truck is delayed by 4 hours, the Alert Engine calculates: *"20% of the produce on that truck will now expire before the weekend sale,"* and sends an immediate markdown alert to the store manager.



Layer 7 is the **Execution Layer**. It transforms the intelligent decisions from the Lakehouse into transactional reality within the three primary ERP systems.

1. SAP ECC 6.0 Flow: Automated Replenishment (Western Europe)

This stream handles the heavy lifting of procurement and supply chain adjustments for the legacy environment.

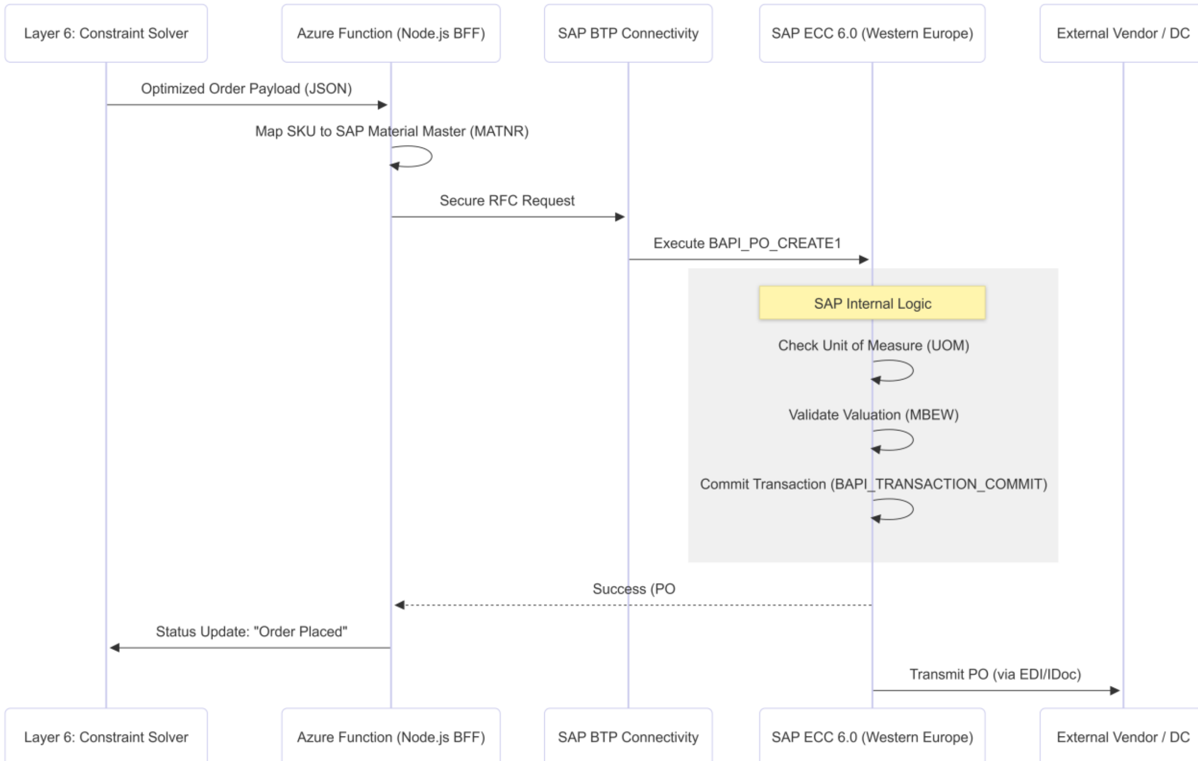


In the Ahold Delhaize 8-Layer Architecture, **Layer 7 (Action & Write-back)** for **SAP ECC 6.0** is the bridge where digital intelligence becomes physical supply chain execution.

For the Western Europe region (Albert Heijn, Delhaize), this flow automates the creation of procurement documents and the adjustment of master data, ensuring that the **Optimized Order Plan** generated in Databricks is executed without manual data entry in the legacy ERP.

1. The High-Level Execution Flow

The process follows a "Pull" logic: the AI "Pulls" an order requirement through the system, and the SAP ECC system "Executes" the transactional commitment.



2. Technical Components & Transaction Logic

The Automated Replenishment flow utilizes three primary **BAPI (Business Application Programming Interface)** sets to communicate with ECC 6.0:

A. Procurement: External Vendor Orders

- **BAPI used:** BAPI_PO_CREATE1
- **Logic:** For items supplied directly by external vendors to stores.
- **Action:** The AI provides the Vendor_ID, Material_ID, and Quantity. The BAPI creates a formal **Purchase Order (PO)**.
- **Automation Rule:** If the AI's confidence is
- >90%>90\%>90%

  , the PO is created as **"Released"** (Ready for Vendor). If confidence is lower, it is created as **"Parked"** for a buyer's review.

B. Stock Transfer: DC-to-Store (Internal)

- **BAPI used:** BAPI_PO_CREATE1 (with Stock Transfer Order type)

- **Logic:** For moving fresh produce from an Ahold Delhaize Distribution Center (DC) to a local Albert Heijn store.
- **Action:** Creates a **Stock Transfer Order (STO)**. This triggers the "Picking" process in **SAP EWM** (Layer 1) to prepare the pallet for the next available truck.

C. Dynamic Master Data Adjustment (MRP)

- **BAPI used:** BAPI_MATERIAL_MAINTAIN_DATA_WHSE
- **Logic:** AI identifies that a product's demand volatility has changed (e.g., a new trend in vegan products).
- **Action:** The system automatically updates the **Safety Stock (EISBE)** and **Reorder Point (MINBE)** levels in the MARC table for that specific store. This ensures the legacy MRP (Material Requirements Planning) in ECC stays aligned with the AI's future predictions.

3. Data Mapping: The "Translation" Step

Because the AI works in a modern Data Lake and ECC is a legacy system, the **Node.js BFF (Backend-for-Frontend)** performs critical mapping:

AI Variable (Databricks)	SAP ECC Field	Table/Structure
Global_Product_ID	MATNR (Material)	MARA
Store_ID	WERKS (Plant)	T001W
Calculated_Safety_Stock	EISBE	MARC
UoM_Conversion	MEINS to BASE_UOM	MARM
Supplier_ID	LIFNR (Vendor)	LFA1

4. Integration Security & Resilience

To connect **Azure/Databricks** to the **On-Premise SAP ECC**, the architecture uses:

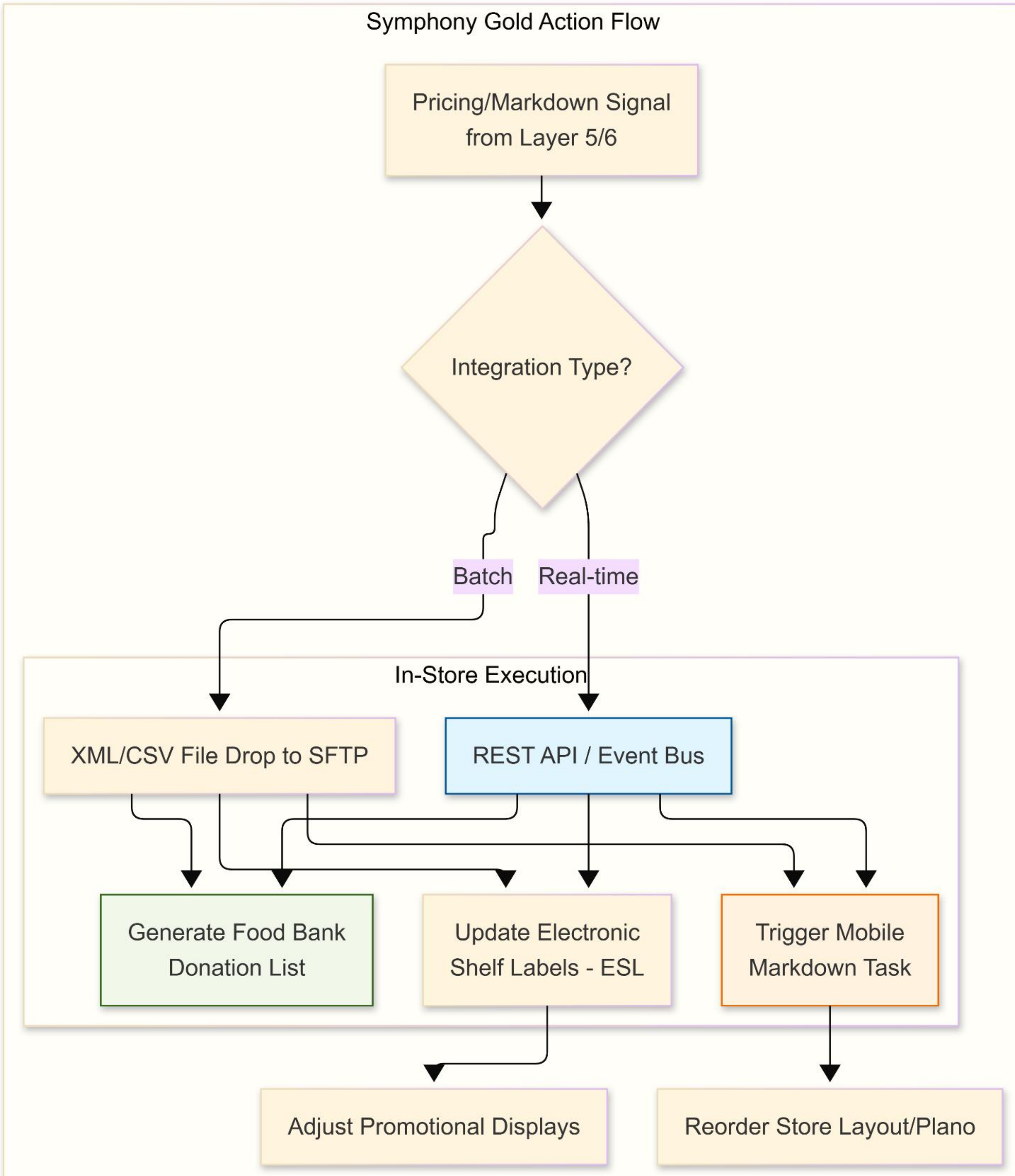
1. **SAP BTP Connectivity Service:** Creates a secure "Cloud Connector" tunnel. No need to open the SAP database to the public internet.
2. **State Management:** The Node.js BFF tracks every request. If SAP ECC is down for maintenance, the request is held in an **Azure Service Bus** queue and retried automatically when SAP is back online.
3. **Idempotency:** Each AI-generated order has a unique Correlation_ID. This prevents the system from accidentally creating duplicate Purchase Orders if a network glitch occurs during the BAPI call.

5. Business Impact for Ahold Delhaize (Western Europe)

1. **Elimination of "Order Lag":** In the legacy process, there was often a 4–8 hour delay between "Identifying a need" and "Placing the order." This flow reduces that to **seconds**.
2. **Labor Savings:** Store Managers in the Netherlands and Belgium save an estimated **70% of time** previously spent on manual ordering, allowing them to focus on customer service and store presentation.
3. **Accuracy:** By pushing safety stock levels directly into ECC MARC tables, the system ensures that even if the AI platform goes offline, the "Basic" SAP replenishment will still function using the AI's most recently optimized parameters.
4. **Closed-Loop Audit:** Every order created in ECC is linked back to the AI model version that recommended it, providing a **100% audit trail** for Group Finance (S/4HANA).

2. Symphony Gold Flow: Retail & Store Actions (CEE)

This stream focuses on the "Last Mile" of retail—shelf pricing, fresh food markdown, and store-level operations.

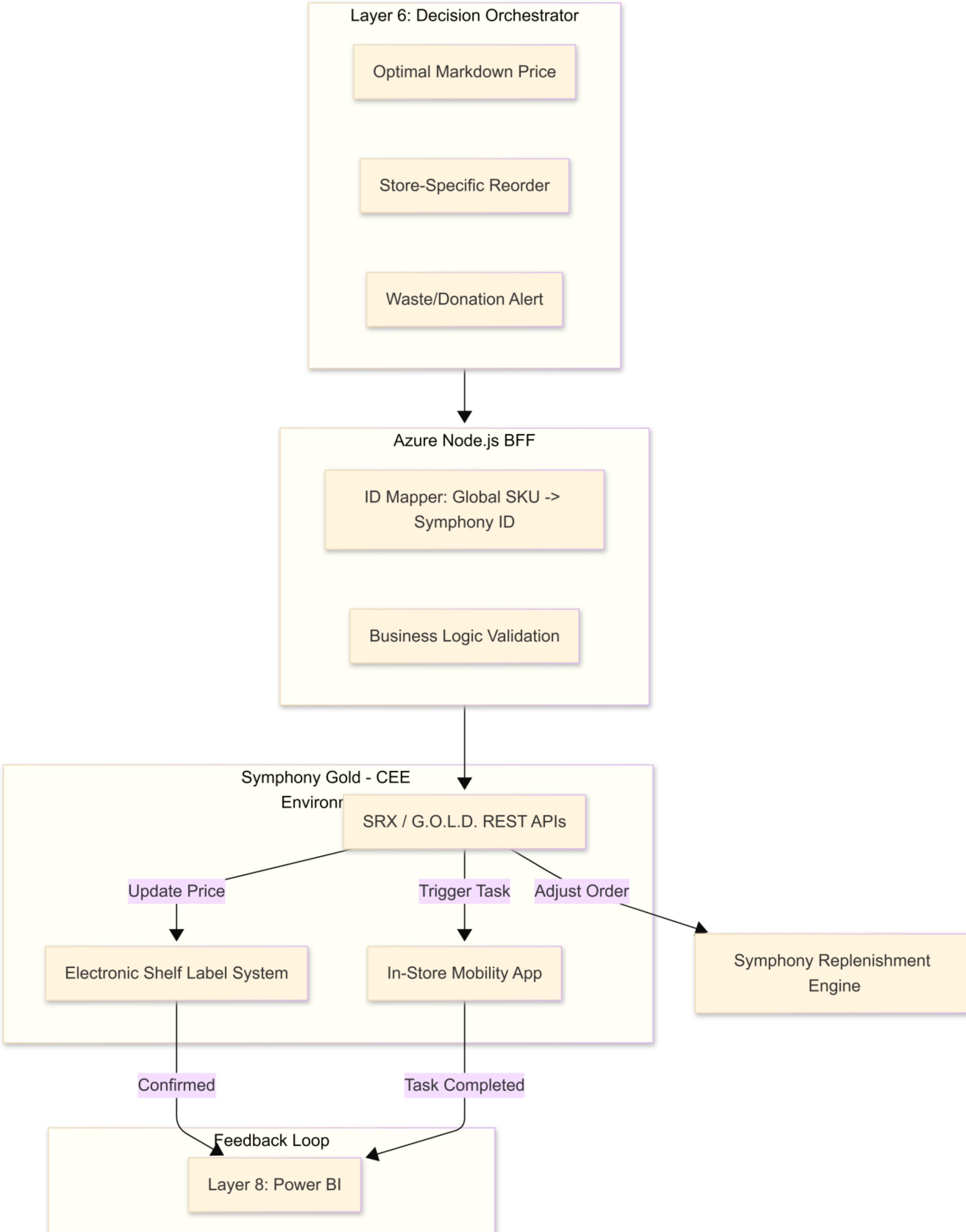


In the Ahold Delhaize 8-Layer Architecture, **Layer 7 (Action & Write-back)** for **Symphony Gold** represents the "Last Mile" of retail intelligence. While SAP ECC handles the massive back-end procurement in Western Europe, Symphony Gold is the operational heart for **Central & Eastern Europe (CEE)**, focusing on store-level agility, shelf-edge execution, and hyper-local fresh food management.

For banners like **Mega Image (Romania)** or **Albert (Czech Republic)**, this flow automates the transition from AI insights to physical store tasks.

1. The Retail Execution Flow

Unlike the "Procurement" focus of SAP, the Symphony Gold flow is **Action-Oriented**. It focuses on what the store associate sees and what the customer pays at the shelf.



2. Specific Functional Write-backs

The integration targets three core modules within the Symphony Gold (SymphonyAI) ecosystem:

A. Dynamic Pricing & ESL Updates (SRX Pricing)

- **The Action:** Model 3 (Reinforcement Learning) calculates that a batch of chicken in a Bucharest store must be discounted by 30% *right now* to avoid waste.
- **Execution:** The BFF triggers a **REST API** call to the Symphony Pricing module.
- **Physical Result:** The **Electronic Shelf Label (ESL)** at the aisle updates instantly. The POS (Point of Sale) is updated simultaneously to ensure price consistency at the checkout.

B. Perishable Task Management (Fresh Food Ops)

- **The Action:** Model 2 (Waste Prediction) identifies high-risk produce.
- **Execution:** Symphony Gold generates a "Prioritized Task" for the store associate.
- **Physical Result:** The associate receives a notification on their handheld device: *"Action Required: Move 5 boxes of Organic Bananas to the 'Quick Sale' bin and apply the AI-generated discount sticker."*

C. Store-Level Replenishment (Local "Pull")

- **The Action:** Model 4 (MEIO) determines that a store in Prague needs an emergency shipment of water due to a local event.
- **Execution:** The AI pushes a "Recommended Order" into the Symphony Gold Replenishment module.
- **Physical Result:** This "Recommendation" bypasses the standard nightly batch and is treated as a **High-Priority Store Pull**, notifying the Distribution Center for immediate picking.

3. Integration Patterns: API vs. Event Bus

Ahold Delhaize uses a hybrid integration strategy for the CEE region:

1. **REST API (Real-Time):** Used for **Pricing and Task Alerts**. These require sub-second latency to ensure a customer doesn't see one price on the shelf and another at the register.
2. **File Drops / SFTP (Batch):** Used for **Assortment Updates**. Large-scale changes to what products a store should carry are sent as XML files during low-traffic windows (2 AM - 4 AM).
3. **SAP Event Mesh (Asynchronous):** Symphony Gold acts as an "Event Subscriber." When a truck leaves a DC (tracked in SAP TM), the Event Mesh notifies Symphony Gold, which then updates the "Estimated Time of Arrival" for the store manager’s dashboard.

4. Data Mapping & Translation Logic

Symphony Gold uses a different nomenclature than SAP. The **Node.js BFF** performs the "Semantic Translation":

Global AI Term	Symphony Gold Equivalent	Context
SKU_ID	Product_Code / CIN	Global vs. Internal Symphony ID
Store_ID	Site_Code	Specific Retail Location

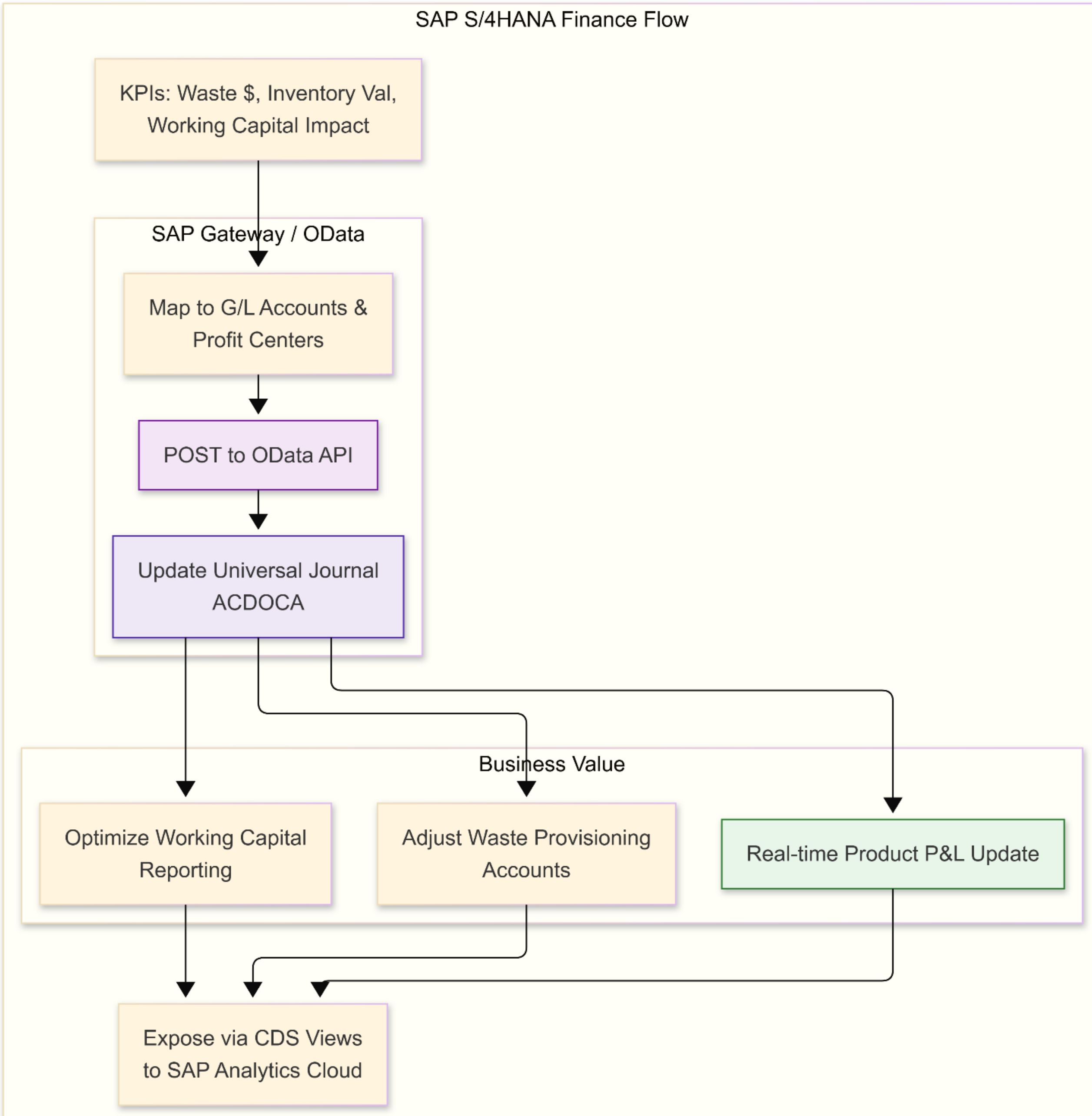
Markdown_Price	Promotion_Price_Index	Applied to the local price file
Safety_Stock	Minimum_Stock_Objective	Used by Symphony's local ordering engine

5. Business Value for CEE (Symphony Regions)

1. **In-Store Agility:** Unlike the centralized "Push" model of SAP, the Symphony flow enables a **"Pull" model**, where store-specific AI adjustments are made based on local hourly foot traffic.
2. **Waste Mitigation:** By pushing markdown triggers directly to the associate's mobile device, Symphony Gold ensures that the AI's "Waste Prediction" (Model 2) leads to immediate physical movement of goods.
3. **Cross-Banner Efficiency:** Even though Mega Image (Romania) and Albert (Czech Republic) use different Symphony instances, the **Databricks Layer 6** sends standardized instructions, ensuring consistent optimization across the whole CEE region.
4. **Auditability:** Every price change triggered by the AI is logged in Symphony's audit trail, providing 100% transparency for local tax and consumer protection regulations.

3. SAP S/4HANA Flow: Financial Synchronization (Group Level)

This stream ensures that the operational efficiencies (waste reduction, faster turnover) are reflected in the group's financial statements and P&L.

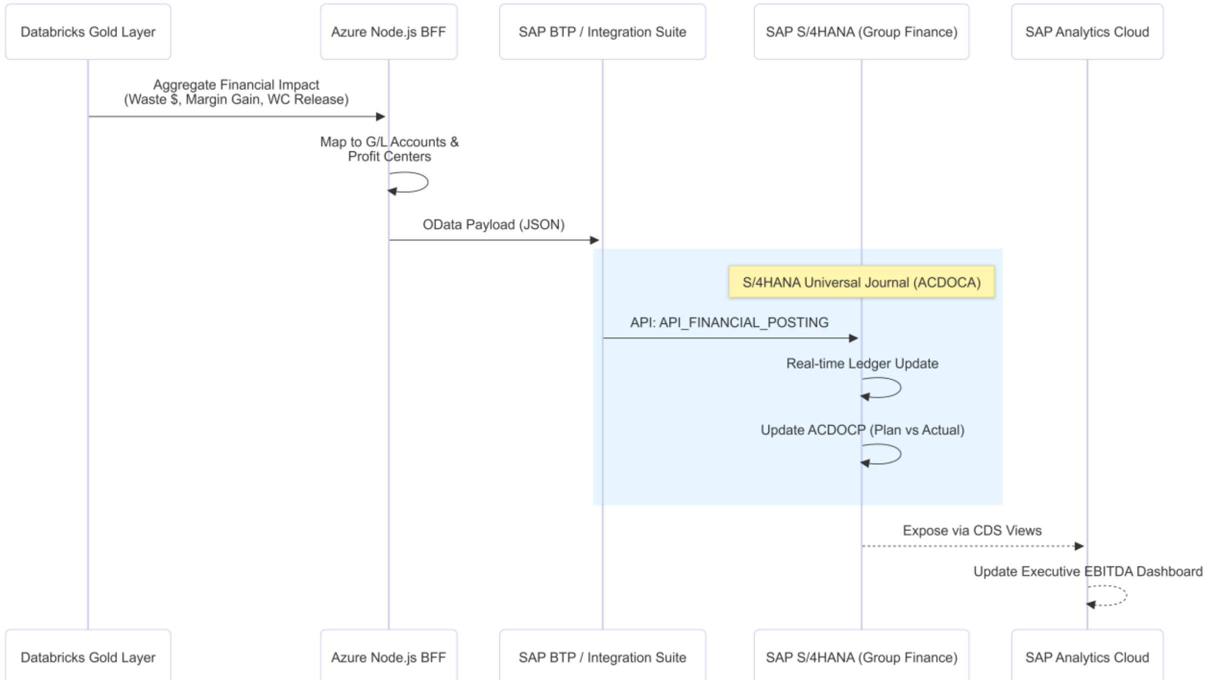


In the Ahold Delhaize 8-Layer Architecture, **Layer 7 (Action & Write-back)** for **SAP S/4HANA** represents the "Financial Truth."

While SAP ECC and Symphony Gold handle the physical movement of goods, S/4HANA acts as the **Central Financial Brain**. This flow ensures that every AI-driven operational decision—whether a markdown in Romania or a stock reduction in the Netherlands—is instantly translated into financial impact, providing the Group CFO with real-time visibility into the project's **€350M+ EBITDA target**.

1. The Financial Synchronization Flow

The flow moves from operational "events" to financial "postings" through a process of aggregation and valuation.



2. Key Technical Components

A. The Universal Journal (ACDOCA)

S/4HANA's "secret sauce" is the **Universal Journal**.

- **The Action:** The synchronization flow posts "Statistical Ledger" entries or "Journal Adjustments."
- **The Benefit:** It merges management accounting (CO) and financial accounting (FI). Ahold Delhaize can see waste costs broken down by **Material Group**, **Banner (Brand)**, and **Region** in a single table without the need for complex reconciliation.

B. OData & SAP BTP Integration

- **Integration Path:** The **Azure Node.js BFF** sends structured JSON to **SAP BTP (Business Technology Platform)**.
- **Safety Mechanism:** BTP performs "Pre-validation." It checks if the Profit Center exists and if the G/L account is open for the current period *before* attempting the post in S/4HANA. This prevents errors in the central ledger.

C. ACDOCP: The Planning Ledger

- **The Action:** The system writes back AI-calculated forecasts into the **ACDOCP (Planning Table)**.

- **The Benefit:** This allows for **Predictive P&L**. The CFO can see a "Financial Forecast" for the next 30 days based on the AI's demand predictions, rather than just looking at last month's history.

3. Financial Mapping Logic (The Translation)

The **Databricks Gold Layer** aggregates millions of micro-transactions (e.g., 50,000 individual banana sales) into "Financial Packets" before sending them to S/4HANA.

Operational Event (ECC/Symphony)	Financial Impact Category	S/4HANA Target (ACDOCA)
Inventory Optimization (M4)	Working Capital Release	Reduction in Inventory Value Asset
Markdown Approved (M3)	Margin Variance	Debit: Sales Promotion / Credit: Inventory
Waste Prevented (M2)	COGS Improvement	Reduction in Spoilage Provision Account
Supplier Reliability (M5)	Purchase Price Variance	Adjustment to Procurement Cost Center

4. Strategic ESG & Sustainability Write-back

A unique feature of this S/4HANA flow is the integration of **non-financial data**:

- **Carbon Ledger:** The flow posts "Carbon Savings" (derived from prevented waste) as a statistical value in the S/4HANA ledger.
- **CSRD Compliance:** Because the data is recorded in the "Financial Books," it carries the highest level of audit integrity required for the **EU Corporate Sustainability Reporting Directive (CSRD)**.

5. Business Value for Ahold Delhaize Group

1. **Real-Time ROI Tracking:** The CFO no longer waits for a quarterly report to see if the "Inventory Optimization Project" is working. They see the daily delta in the Universal Journal.
2. **Working Capital Optimization:** By automating the sync of inventory days (DIO), the group can optimize its cash position, potentially releasing the **€150M in cash** targeted in the business case.
3. **Unified Global Reporting:** Regardless of whether the operational data came from **Symphony (CEE)** or **ECC (WE)**, it arrives in S/4HANA in a perfectly standardized format.
4. **Audit Transparency:** Every financial entry is linked back to a **Databricks Job ID** and a **Model Version**, providing a "Clear Line of Sight" from a machine learning prediction to a financial result.

Cross-System Integration Logic

1. **SAP ECC 6.0 (The Muscle):** Uses **BAPIs and IDocs** to handle high-volume procurement. This is a "tight" integration where the Lakehouse acts as an external MRP engine.
2. **Symphony Gold (The Store):** Uses an **Event Bus or REST APIs** to communicate with store associates. It is more "agile," focusing on immediate actions like changing a price before a product expires.
3. **SAP S/4HANA (The Brain):** Uses **OData and CDS Views**. It doesn't care about individual crates of tomatoes; it cares about the *financial total* of all crates. It receives aggregated data to provide a "Single Version of the Truth" for Ahold Delhaize Group Finance.

LAYER 8: USER INTERFACES & ANALYTICS

DASHBOARDS & REPORTING

Power BI (Operational)	SAP Analytics Cloud (Strategic)	Custom React Apps (Store Managers)
Users: • Store managers • Category mgrs • Supply chain • Buyers	Users: • C-suite • Category VPs • Finance • Strategy	Users: • Store associates • Warehouse staff • Delivery drivers
Dashboards: • Daily inventory positions • Stockout alerts • Waste tracking • Forecast vs actuals • Replenishment queues	Analytics: • Waste trends • Forecast accuracy • Margin analysis • Supplier scorecard • Regional performance • ROI metrics	Features: • Mobile-first design • Barcode scanning • Real-time stock check • Markdown approval workflow • Photo upload (quality issues) • Task management
Refresh: Real-time	Refresh: Daily	Connectivity: Offline-capable

Layer 8 transforms complex data insights into actionable tools for three distinct user groups at Ahold Delhaize. Each stream uses a specific software stack optimized for its purpose: **High-speed operational monitoring**, **Strategic financial planning**, and **Mobile-first field execution**.

1. Power BI Stream: Operational Analytics

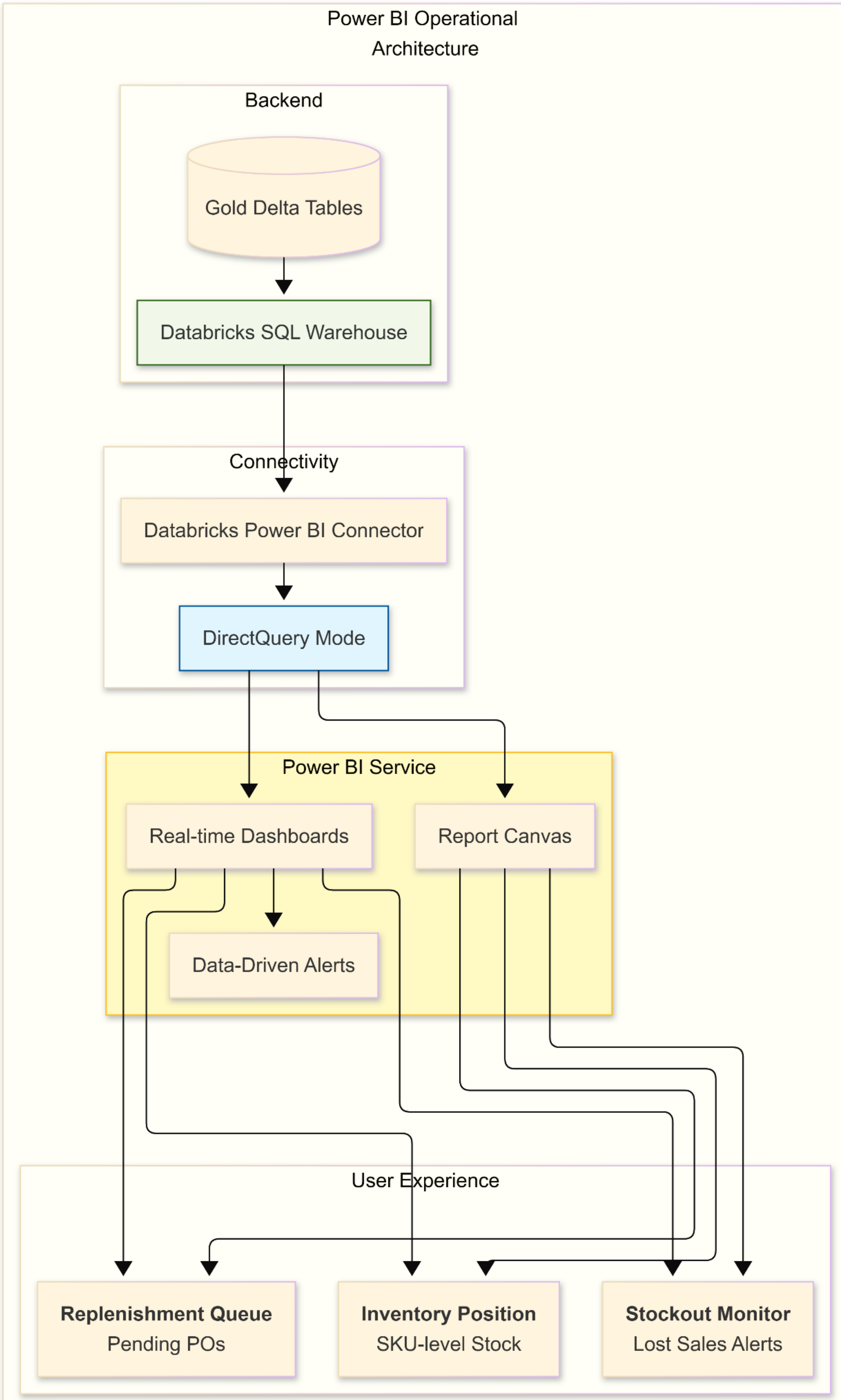
Focus: Real-time visibility for Store Managers and Category Buyers to manage daily stock levels and alerts.

In the Ahold Delhaize 8-Layer Architecture, the **Power BI Stream (Layer 8)** serves as the **Operational Command Center**. While SAP Analytics Cloud handles the "Big Picture" for the C-Suite, Power BI is the tool used by Store Managers, Category Buyers, and Supply Chain Analysts to manage the "Day-to-Day" pulse of inventory, waste, and availability.

This stream is designed for **high-velocity decision-making**, leveraging the "DirectQuery" capability of Databricks to provide near real-time visibility.

1. Technical Architecture & Component Flow

The Power BI stream bypasses traditional, slow "Import Mode" and instead queries the **Databricks SQL Warehouse** directly. This ensures that when a sale occurs in **Symphony Gold**, the dashboard reflects the stock change within minutes.


Power BI Operational Architecture
Backend
Gold Delta Tables
Databricks SQL Warehouse
Connectivity
Databricks Power BI Connector
DirectQuery Mode
Power BI Service
Real-time Dashboards
Report Canvas
Data-Driven Alerts
User Experience
Replenishment Queue
Pending POs
Inventory Position
SKU-level Stock
Stockout Monitor
Lost Sales Alerts

2. Core Operational Dashboards

Power BI is partitioned into three specific "Visibility Domains" for Ahold Delhaize:

A. The Inventory Health Dashboard

- **Target:** Store Managers.
- **KPIs:** On-Shelf Availability (OSA), Out-of-Stock (OOS) count, "Phantom Stock" alerts.
- **Function:** Visualizes the gap between the **Symphony Gold** system stock and the **Model 1** predicted demand.
- **Action:** If a store manager sees a "Red" status for Milk but the system says "10 units available," they know they have a "Phantom Stock" issue and need to perform a manual count.

B. The Freshness & Waste Monitor

- **Target:** Department Leads (Produce, Bakery, Meat).
- **KPIs:** Spoilage Rate (%), Days to Expiry, Markdown Success Rate.
- **Function:** Integrates **Model 2 (Waste Prediction)** logic. It lists the top 20 SKUs at risk of expiring in the next 48 hours.
- **Action:** Triggers the associate to verify the AI-suggested markdown in the **React Field App**.

C. The Replenishment Audit View

- **Target:** Category Buyers & Supply Chain Analysts.
- **KPIs:** Automation Rate (%), Manual Overrides, Forecast Error (MAPE).
- **Function:** Compares the AI's **Model 4 (MEIO)** reorder recommendation against what was actually sent to **SAP ECC**.
- **Action:** If a buyer sees that a specific supplier is consistently failing to meet the AI's requested lead times (using **Model 5** data), they can initiate a contract review in S/4HANA.

3. Key Technical Features for Ahold Delhaize

Feature	Implementation Detail	Business Benefit
DirectQuery	Queries the Delta Lake in real-time via Databricks SQL.	No data latency; managers see stock changes as they happen.
Row-Level Security (RLS)	Integrated with **Azure AD / Unity Catalog**.	An Albert Heijn manager only sees their store; a CEE lead sees only Symphony-managed regions.
Data Alerts	Native Power BI mobile alerts triggered by threshold breaches.	A Category Buyer gets a phone notification if a "Category A" SKU stockout exceeds 2%.
Drill-Through	Ability to click a "Region" and drill down to a "Specific Bin" in **SAP EWM**.	Enables root-cause analysis (e.g., "Why is this DC failing to pick fresh produce on time?").

4. The Data Refresh Strategy: "Pulse" vs. "Pulse"

- **Real-Time (DirectQuery):** Used for **Availability and Stockouts**. This is the most critical for the "Never-Out" core items.
- **Hourly Refresh:** Used for **Waste and Freshness**. Since spoilage doesn't change every second, an hourly update from the **Databricks Gold Layer** is sufficient and saves compute costs.
- **Daily Refresh:** Used for **Supplier Performance and Long-term Forecast Accuracy**. This is strategic data that Category Buyers use for weekly planning.

5. Business Value Summary

The Power BI Stream ensures that the **AI Models (Layer 5)** aren't a "black box." It provides the **transparency** needed to trust the automation. By visualizing the **€350M EBITDA impact** at the operational level, it allows regional managers to identify exactly which stores or suppliers are lagging, transforming raw data into a **proactive competitive advantage**.

2. SAP Analytics Cloud (SAC) Stream: Strategic Analytics

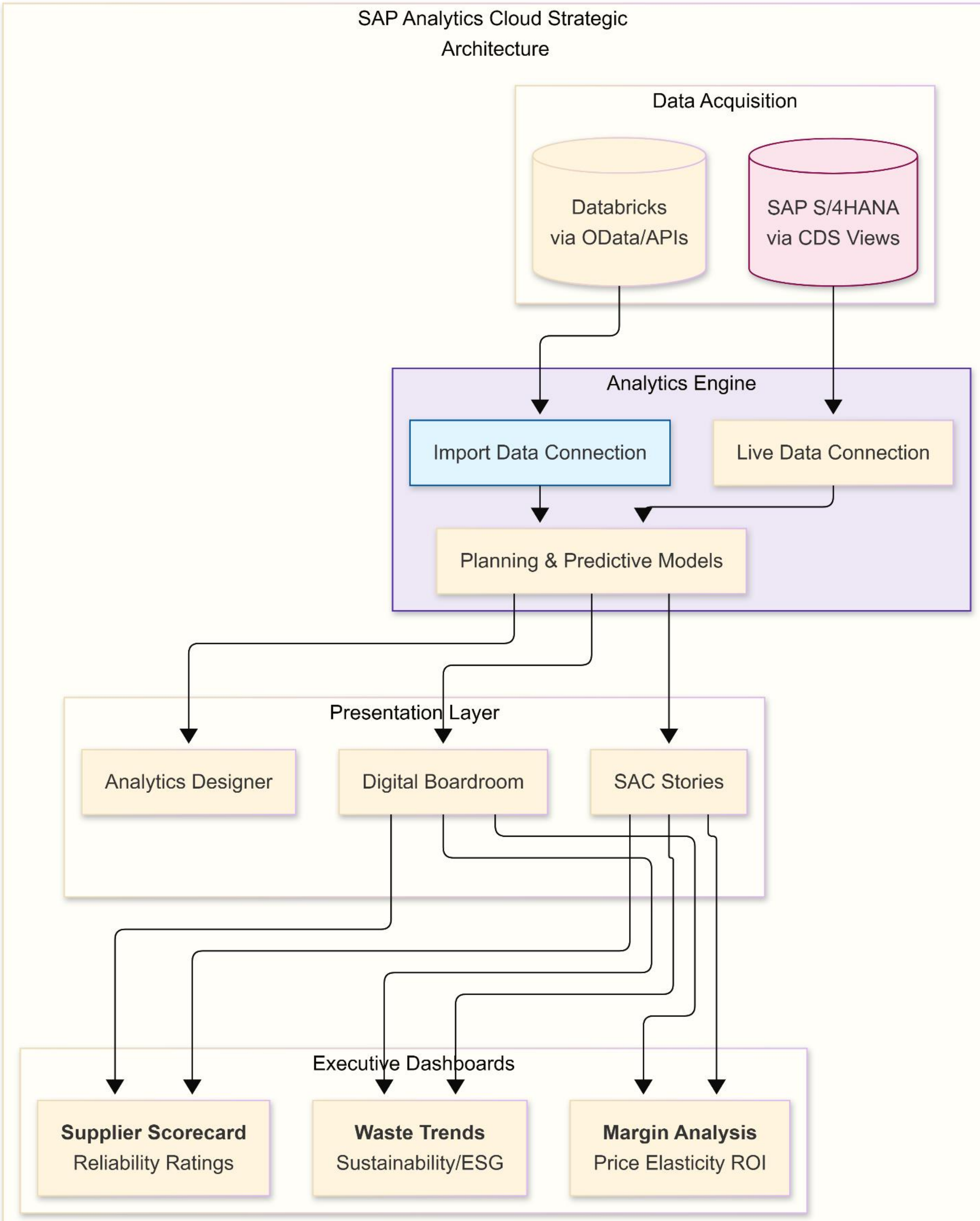
Focus: C-Suite and Finance visibility into group-level waste trends, supplier performance, and ROI metrics.

In the Ahold Delhaize 8-Layer Architecture, the **SAP Analytics Cloud (SAC) Stream (Layer 8)** serves as the **"Financial North Star."** While Power BI focuses on the "how" of daily operations, SAC is dedicated to the **"so what"** of financial outcomes and strategic steering.

It is the primary interface for the **C-Suite, CFO, and Category VPs**, providing a unified, auditable view of the **€350M+ EBITDA impact** across all banners (Albert Heijn, Delhaize, and CEE brands).

1. Technical Architecture: The "Live" Financial Hub

SAC is uniquely positioned to bridge the gap between the **Databricks Lakehouse** (ML insights) and **SAP S/4HANA** (Financial truth). It uses a hybrid data acquisition strategy.



2. Core Strategic Dashboards

SAC provides three "Executive Views" that align the AI project with corporate long-term goals:

A. The EBITDA Bridge (Financial Impact)

- **The View:** A "Waterfall Chart" starting from last year's profit.
- **The Logic:** It isolates the **incremental margin** gained specifically from the AI models.
  - *Segment 1:* Profit gained from reduced waste (Model 2).
  - *Segment 2:* Profit gained from increased availability/fewer stockouts (Model 1).
  - *Segment 3:* Cash released from optimized inventory levels (Model 4).
- **Target:** CFO and Group Strategy.

B. The Sustainability & ESG Ledger (The Carbon View)

- **The View:** Tracking "Prevented Spoilage" in terms of CO2 equivalents.
- **The Logic:** Links the physical tons of food saved (from Databricks) to Ahold Delhaize's **Scope 3 Emission** targets in S/4HANA.
- **Auditability:** Since this data is anchored to the S/4HANA financial books, it is ready for **EU CSRD (Corporate Sustainability Reporting Directive)** audits.

C. The Supplier Strategic Scorecard

- **The View:** A 4-quadrant matrix (Resilience vs. ESG vs. Cost vs. Quality).
- **The Logic:** Aggregates **Model 5 (Supplier Selection)** data.
- **Action:** If a Category VP sees that "Fruit Supplier X" has high margins but terrible "Freshness at Arrival" (causing downstream waste), they can initiate a strategic renegotiation or supplier switch.

3. Key Strategic Capabilities

Capability	Technical Detail	Business Value
What-If Simulation	Uses SAC Planning to model "Safety Stock" vs. "Waste" trade-offs.	*"If we increase Milk availability from 98% to 99.5%, what is the € impact on waste?"*
Smart Insights	Machine Learning-driven variance analysis.	Automatically explains why "Fresh Meat" margin dropped in Belgium (e.g., weather + transport delay).
Digital Boardroom	High-level interactive touch-screens for C-level meetings.	Enables real-time strategic pivot decisions during quarterly reviews.
Predictive Forecasting	Uses S/4HANA historical data + Databricks signals.	Provides a 12-month "Financial Outlook" for inventory valuation.

4. Integration Logic: Live vs. Acquired Data

- **Live Data (S/4HANA):** SAC connects via **CDS Views** to the Universal Journal. This means when a finance controller makes a manual adjustment in S/4HANA, the SAC Strategic Story updates **instantly**.
- **Acquired Data (Databricks):** High-volume ML features (like "Predicted Spoilage Probability") are imported into SAC daily. This data is merged with the financial data using a **Unified Dimension** (e.g., Global SKU ID or Region Code).

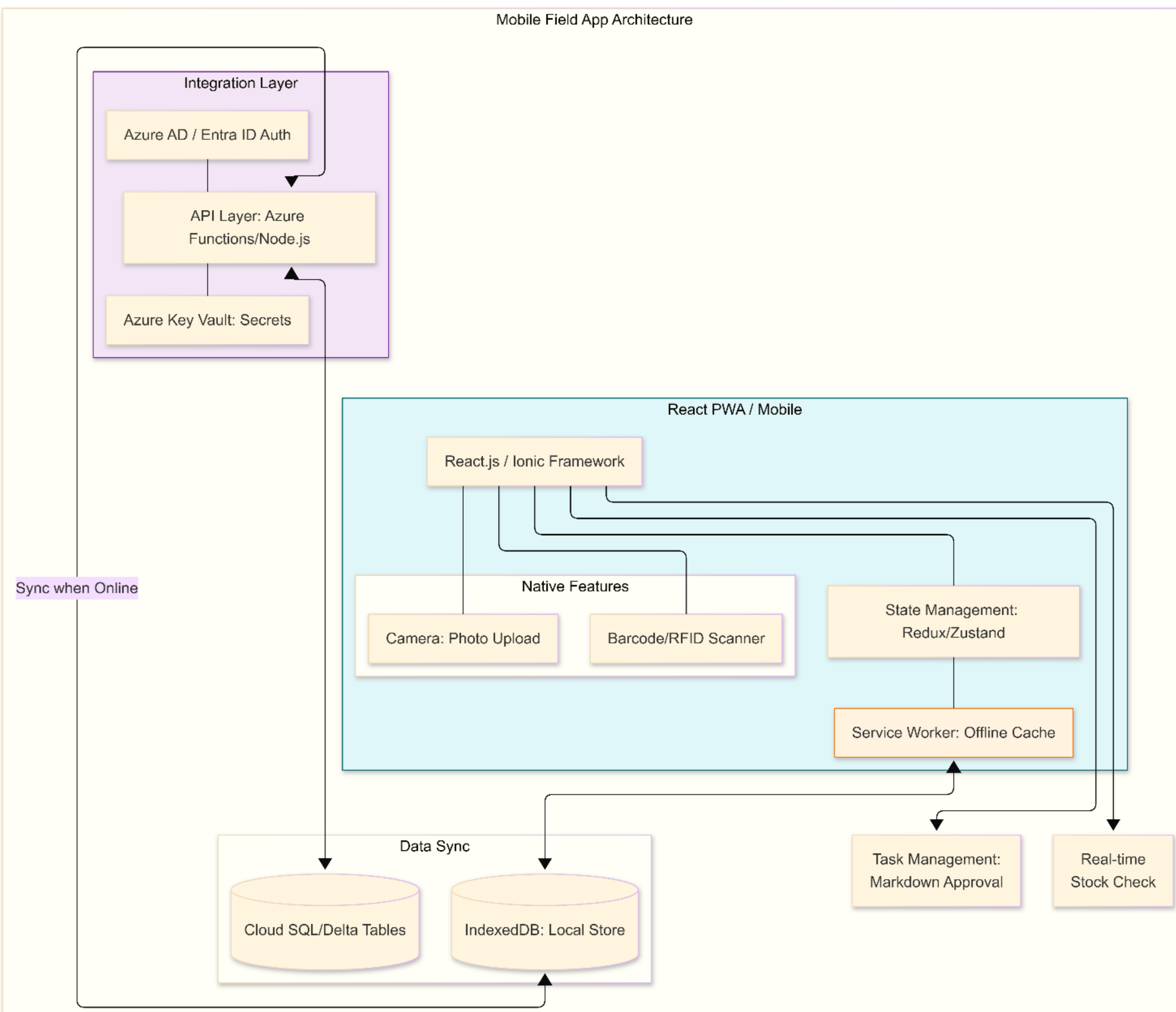
5. Summary of Business Value

The SAC stream ensures that the **AI project is never seen as just a "Technical Pilot."** By placing the AI results directly into the financial and strategic environment used by the C-Suite, it creates **Executive Accountability**.

It transforms Ahold Delhaize from a company that *reports* on last month's inventory to a company that **actively steers** its €350M+ EBITDA impact through a unified, predictive, and sustainability-linked financial interface.

3. Custom React App Stream: Field Operations

Focus: Mobile-first, offline-capable application for store associates and warehouse staff to perform physical tasks like markdown tagging and quality checks.



Summary of Software Components by Stream

Component	Power BI (Operational)	SAP Analytics Cloud (Strategic)	Custom React App (Field)
Frontend Framework	Power BI Web/Mobile	SAC Stories / Designer	React.js (Progressive Web App)
Data Engine	Databricks SQL Warehouse	SAP S/4HANA + Databricks	Node.js API + Azure Functions
Refresh Rate	Real-time (DirectQuery)	Daily / On-Demand	Event-driven / Real-time
Key Capability	Large-scale data slice/dice	Financial P&L Consolidation	Offline data entry & scanning
Security	RLS (Row Level Security)	SAP Role-Based Access	Azure AD + RBAC
Connectivity	Databricks Delta Sharing	OData & CDS Views	REST/GraphQL

As for Custom React App (Field) , while it appears as a **"Single Entry Point"** for the user (to simplify training and deployment), the underlying architecture is a **Multi-Tenant, Role-Based Modular Platform**.

Think of it as a **"Super App"** architecture where the features, data, and workflows visible to the user are dynamically injected based on their **Login Context** (Region, Store, and Job Role).

Here is how the architecture handles customer specific requirements:

1. Regional & ERP Contextualization (The "Digital Twin" Logic)

Because Western Europe (SAP) and CEE (Symphony) have different data structures, the app uses a **Regional Metadata Layer**.

- **Logic:** When an associate in the Netherlands (Albert Heijn) logs in, the app communicates with the **SAP ECC API wrapper**.
- **Logic:** When an associate in Czech logs in, the app switches its UI labels and backend routes to the **Symphony Gold API**.
- **Implementation:** The app detects the Store_ID and Region_Code from the user's **Azure AD (Entra ID)** profile upon login.

2. Role-Based Access Control (RBAC) & SKU Restrictions

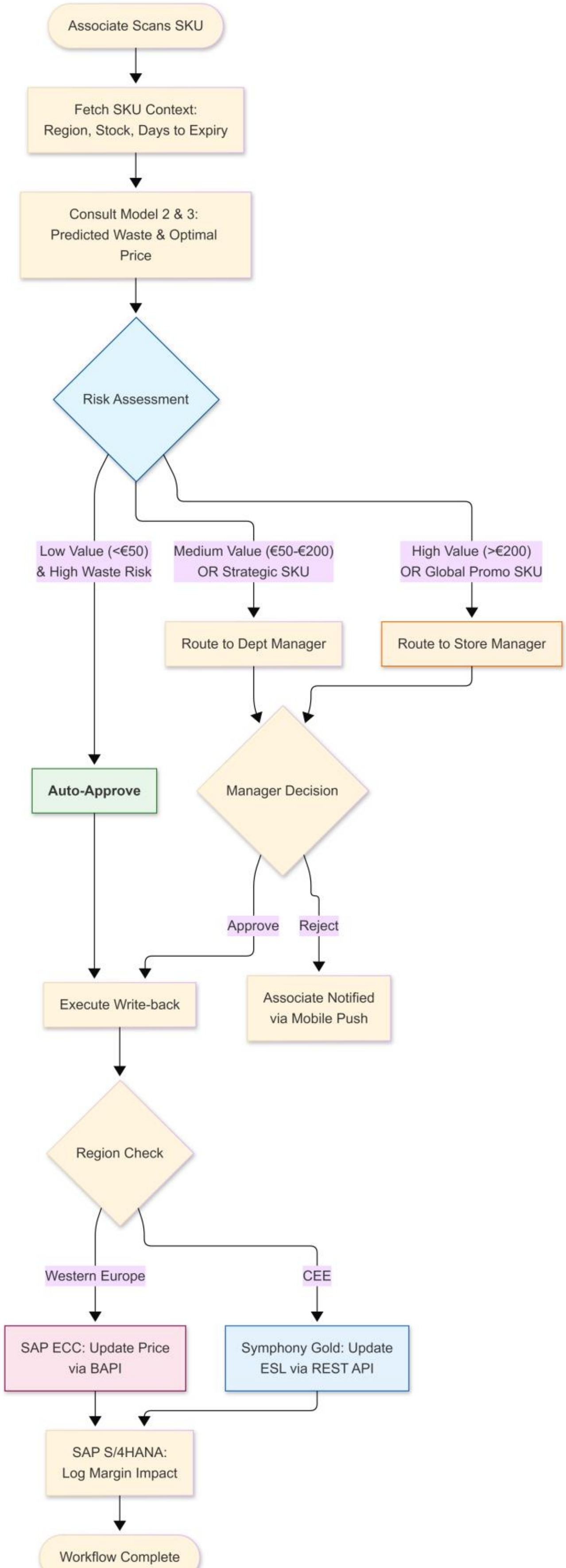
The app enforces a strict **"Need to Know/Need to Act"** policy. You wouldn't want a bakery associate accidentally approving a markdown for the meat department.

- **Store Associate:** View-only access to their specific department SKUs. Access to "Scanning" and "Reporting Quality Issue" modules.

- **Department Manager:** View access to all SKUs in their department. Can *initiate* markdown requests but not approve high-value ones.
- **Store Manager:** Approval dashboard for all "High-Variance" alerts. Cross-department visibility.

3. Workflow & Approval Architecture

This diagram shows how a **Markdown Approval Workflow** functions across different levels of responsibility. This flowchart describes the "Intelligent Routing" that determines who needs to approve a price change and why.


Associate Scans SKU
Fetch SKU Context:
Region, Stock, Days to Expiry
Consult Model 2 & 3:
Predicted Waste & Optimal Price
Risk Assessment
Low Value (<€50)
& High Waste Risk
Medium Value (€50-€200)
OR Strategic SKU
High Value (>€200)
OR Global Promo SKU
Route to Dept Manager
Route to Store Manager
Auto-Approve
Manager Decision
Approve
Reject
Execute Write-back
Associate Notified
via Mobile Push
Region Check
Western Europe
CEE
SAP ECC: Update Price
via BAPI
Symphony Gold: Update
ESL via REST API
SAP S/4HANA:
Log Margin Impact
Workflow Complete

4. Technical Component: The "Context-Aware" Middleware

To manage these restrictions, we introduce a **Backend-for-Frontend (BFF)** layer. This is a specialized API that sits between the React App and the Databricks/ERP layers.

Restriction Type	Implementation Method	Description
Regional Restriction	**Environment Variables & Headers**	The BFF routes traffic to the correct regional database (NL vs. CEE) based on the user's Store_ID.
SKU Scope	**Data Filtering (Row-Level Security)**	The API only returns SKUs that match the user's assigned Department_ID.
Business Role	**JWT Claims (Azure AD)**	The JSON Web Token (JWT) contains roles like role: "approver". If this claim is missing, the "Approve" button is physically removed from the UI.
Process Workflow	**State Machine (e.g., Temporal.io)**	Tracks the status of a request: Pending -> Under Review -> Approved -> Synced to ERP.

Why this is better than multiple apps:

1. **Unified Security:** One security audit for the whole platform.
2. **Shared Components:** The "Barcode Scanner" and "Photo Uploader" components are reused across all roles and regions.
3. **Global Visibility:** Ahold Delhaize HQ can see exactly how long it takes for a markdown to be approved in Romania vs. the Netherlands using a single set of telemetry.

Summary of the Workflow:

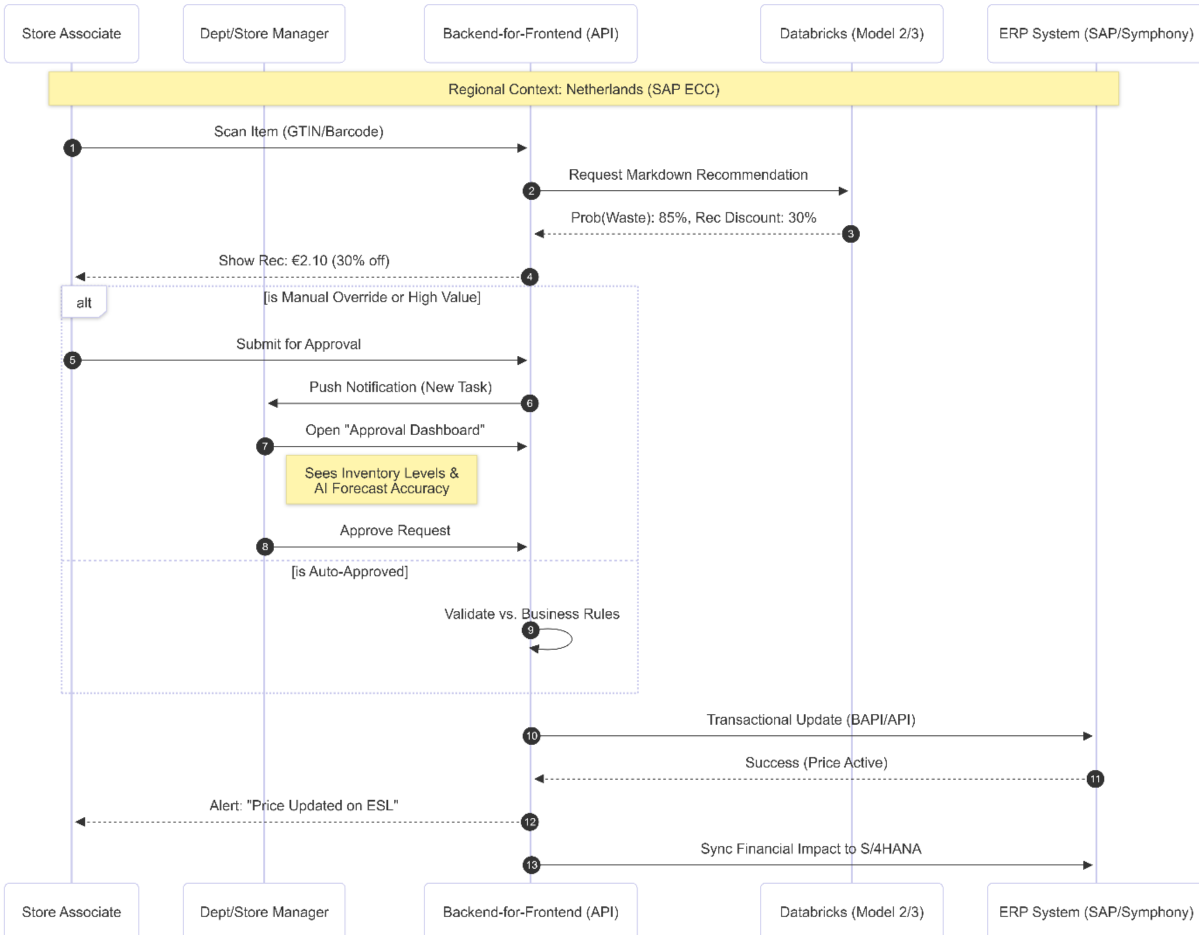
1. **Initiation:** Associate scans a product.
2. **Validation:** App checks SKU-role-region permissions.
3. **Intelligence:** Model 2 (Waste) suggests the discount depth.
4. **Approval:** If it crosses a financial threshold, it routes to a Manager.
5. **Execution:** Once approved, it triggers the BAPI (SAP) or REST API (Symphony) and updates the Electronic Shelf Label.

To design a **Markdown Approval Workflow** for a retail giant like Ahold Delhaize, the process must be resilient, audit-ready, and capable of handling regional differences (SAP vs. Symphony) while preventing "margin leakage" through over-discounting.

This workflow is integrated into the **Custom React App** but powered by the **Databricks Intelligence Layer** and the **BFF (Backend-for-Frontend)** middleware.

5. Multi-Role Sequence Diagram (The Human Process)

This sequence diagram illustrates how the **React App Shell** orchestrates the data between different users and the backend systems.



This detailed description breaks down the **Multi-Role Sequence Diagram** into functional phases. It illustrates how Ahold Delhaize employees (Store Associates and Managers) interact with the AI Intelligence Layer and the underlying ERP systems (SAP and Symphony Gold) to manage markdowns.

Overview of the Actors

1. **Store Associate (The Initiator):** The employee on the shop floor responsible for checking product freshness and scanning items.
2. **Dept/Store Manager (The Decider):** The authority responsible for margin protection and approving high-value or high-variance price changes.
3. **Backend-for-Frontend (BFF):** A specialized API layer that manages security, regional routing (SAP vs. Symphony), and communication with the AI models.

4. **AI Intelligence Layer (Databricks):** The "Brain" containing Model 2 (Waste Prediction) and Model 3 (Dynamic Pricing).
5. **ERP Systems (SAP ECC / Symphony Gold):** The execution systems that update price files and trigger Electronic Shelf Labels (ESL).

Phase 1: Identification & Discovery (The Associate)

The process begins on the store floor, typically during a "freshness check" or a "gap check."

- **Step 1: Physical Scan:** The Associate uses the **Custom React App** on a mobile handheld device to scan a product barcode (e.g., a bag of spinach expiring tomorrow).
- **Step 2: Contextual Lookup:** The App sends a request to the **BFF**. The BFF identifies the user's **Store ID** and **Region**.
- **Step 3: Data Fetch:** The BFF retrieves real-time inventory levels from the local ERP (SAP ECC or Symphony) and historical sales velocity for that specific store.

Phase 2: The Intelligence Trigger (The AI)

The human associate has identified the item; now the AI provides the "Advice."

- **Step 4: Inference Request:** The BFF sends the SKU data to **Databricks**.
- **Step 5: Predictive Analysis:**
    - **Model 2 (Waste Prediction)** calculates the probability that this specific SKU will go to waste based on remaining shelf life and predicted weekend weather.
    - **Model 3 (Dynamic Pricing)** calculates the "Price Elasticity"—how much we need to drop the price to ensure the item sells *before* it expires.
- **Step 6: Recommendation Delivery:** The AI returns a **Recommendation Package**: *"Predicted Waste Probability: 85%. Recommended Markdown: 30%. New Price: €1.40."*

Phase 3: Human Decision & Governance (The Manager)

This is where the business rules and job roles determine the workflow path.

- **Scenario A: Auto-Approval (Low Risk)**
    - If the item is low value (e.g., < €50 total batch value) and the AI confidence is high, the system bypasses the manager. The Associate sees a "Confirmed" message immediately.
- **Scenario B: Escalation (High Risk/Value)**
    - **Step 7: Submission:** For high-value items (e.g., expensive meat cuts or large quantities), the Associate clicks **"Request Approval."**
    - **Step 8: Notification:** The **BFF** sends a Push Notification to the **Manager's** device.
    - **Step 9: Managerial Review:** The Manager opens the app and sees a **Comparison Dashboard**: *Actual Price vs. Suggested Price vs. Waste Cost.*
    - **Step 10: Decision:** The Manager clicks **"Approve"** or **"Reject"** (with a reason code).

Phase 4: Transactional Execution (The ERP Sync)

Once the human (or the rule) has decided, the "Write-back" occurs.

- **Step 11: Regional Routing:** The BFF detects the region.
    - **Western Europe:** It triggers an **RFC/BAPI call** to **SAP ECC** to update the price condition table.
    - **CEE:** It triggers a **REST API** to **Symphony Gold** to update the retail price record.
- **Step 12: Visual Update:** The ERP pushes the update to the **Electronic Shelf Labels (ESL)**. The price on the shelf changes automatically in seconds.
- **Step 13: Confirmation:** The App notifies the Associate: *"Price updated successfully. Shelf label updated."*

Phase 5: Financial Impact & Learning (Closing the Loop)

The final phase ensures the group's books are balanced and the AI learns from the human's actions.

- **Step 14: S/4HANA Sync:** The BFF sends an asynchronous summary of the markdown to **SAP S/4HANA**. This records the "Price Variance" so Finance can track "Markdown Loss" vs. "Potential Waste Savings."
- **Step 15: Feedback to AI:** The decision (whether the manager agreed with the AI or changed the price) is logged back into **Delta Lake**. This "Human-in-the-loop" data is used to retrain the models, making future recommendations more accurate.

Business Value of this Sequence

1. **Efficiency:** Associates no longer guess the discount; they are guided by data.
2. **Margin Protection:** Managers only spend time on high-impact decisions (Exceptions), while the AI handles the "trivial" hundreds of daily fresh-food markdowns.
3. **Real-time Agility:** The time between "identifying a waste risk" and "changing the price on the shelf" is reduced from hours to seconds.
4. **Auditability:** Every step—who scanned it, what the AI recommended, and who approved it—is logged for Group Audit and ESG reporting.

Key Governance Restrictions in the Workflow

To maintain control over a fragmented environment, the following **Software Components** are built into the React App logic:

Feature	Implementation Logic	Purpose
Regional Geofencing	App validates user's IP/GPS or Store ID.	Prevents a manager in the Netherlands from accidentally approving a price change for a store in Romania.
Batch Expiry Lock	Model 2 checks "Batch ID."	Prevents the associate from marking down fresh stock when older stock of the same SKU is still on the shelf (FIFO enforcement).

Financial Guardrail	Hard-coded max discount (e.g., 75%).	Even the AI cannot suggest a price below a certain threshold without Category VP approval.
Audit Trail	Every click is logged to Delta Lake.	For compliance, Ahold Delhaize can audit why a specific markdown was approved (User ID, AI recommendation at that time, and Final Price).

6. Integration with ERP Write-back (Layer 7)

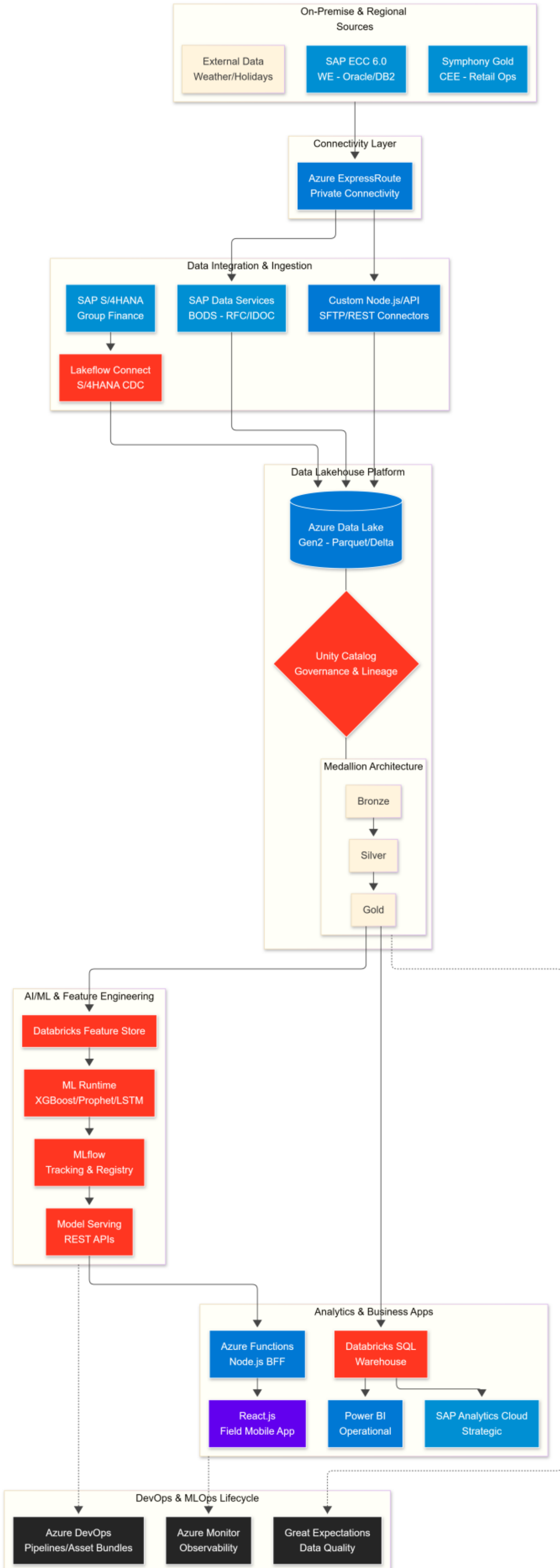
The final step of the workflow is specific to the underlying ERP:

1. **For Western Europe (SAP ECC 6.0):**
   - The App triggers the BAPI_PRICE_CONDITION or a custom IDoc to update the store-specific price condition table (A004/A005).
2. **For CEE (Symphony Gold):**
   - The App hits the **Symphony Retail AI Price Execution API**, which pushes the new price directly to the **Electronic Shelf Labels (ESL)** in that specific store aisle.
3. **For Group Finance (S/4HANA):**
   - An asynchronous message is sent to S/4HANA's **Universal Journal (ACDOCA)** to record the "Price Variance" as a marketing/waste expense for P&L transparency.

Technology Stack

Cloud: Azure (Ahold Delhaize standard) **Data Platform**: Databricks Lakehouse **Storage**: ADLS Gen2 **Integration**: BODS, Lakeflow Connect, APIs **ML/AI**: Prophet, LightGBM, XGBoost, TensorFlow **MLOps**: MLflow Registry **BI**: Power BI, SAP Analytics Cloud **Orchestration**: Databricks Workflows, n8n

This **Technology Architecture Diagram** visualizes the comprehensive stack for Ahold Delhaize, illustrating how the on-premise ERP systems connect via **Azure ExpressRoute** to the **Databricks Lakehouse** and how that intelligence is served back to the business.



On-Premise & Regional Sources
External Data Weather/Holidays
SAP ECC 6.0 WE - Oracle/DB2
Symphony Gold CEE - Retail Ops
Connectivity Layer
Azure ExpressRoute Private Connectivity
Data Integration & Ingestion
SAP S/4HANA Group Finance
SAP Data Services BODS - RFC/IDOC
Custom Node.js/API SFTP/REST Connectors
Lakeflow Connect S/4HANA CDC
Data Lakehouse Platform
Azure Data Lake Gen2 - Parquet/Delta
Unity Catalog Governance & Lineage
Medallion Architecture
Bronze
Silver
Gold
AI/ML & Feature Engineering
Databricks Feature Store
ML Runtime XGBoost/Prophet/LSTM
MLflow Tracking & Registry
Model Serving REST APIs
Analytics & Business Apps
Azure Functions Node.js BFF
Databricks SQL Warehouse
React.js Field Mobile App
Power BI Operational
SAP Analytics Cloud Strategic
DevOps & MLOps Lifecycle
Azure DevOps Pipelines/Asset Bundles
Azure Monitor Observability
Great Expectations Data Quality

Technical Component Breakdown

1. Ingestion Strategy (The Pipes)

- **SAP ECC (BODS):** Leverages **SAP Data Services** as the "Trusted Proxy" for the legacy environment. It handles the extraction of complex SAP Cluster tables using RFC/BAPI protocols and lands them as Parquet files in **ADLS Gen2**.
- **Symphony Gold (Custom API):** A **Node.js** layer (running on Azure) polls Symphony APIs or listens to file drops (SFTP), performing initial JSON-to-Parquet conversion.
- **S/4HANA (Lakeflow Connect):** Native Databricks CDC (Change Data Capture) streams financial changes from the **Universal Journal (ACDOCA)** directly into the Lakehouse with sub-minute latency.

2. Data Lakehouse (The Storage)

- **ADLS Gen2:** The physical storage layer for all structured and semi-structured retail data.
- **Unity Catalog:** The "Golden Thread" for **Security & Governance**. It enforces GDPR compliance across regions (NL vs. RO) and provides a unified lineage from the React App all the way back to the SAP source table.
- **Delta Lake:** Enables ACID transactions, allowing the system to handle real-time POS sales streams without corrupting the historical data used for model training.

3. AI & ML Lifecycle (The Intelligence)

- **Feature Store:** Centralizes features like "Rolling 7-day sales" or "Store Footfall." This ensures that the **Model Serving** endpoint (REST) uses the same math as the **Training** cluster.
- **MLflow:** Acts as the experiment lab. Every version of the "Perishable Waste Model" is logged here, allowing for easy rollback if forecast accuracy drops.
- **Model Serving:** Deploys trained models as **REST Endpoints** with serverless autoscaling, providing sub-second responses to the field mobile apps.

4. Delivery Layer (The Value)

- **Databricks SQL Warehouse:** A high-performance compute engine that allows **Power BI** and **SAP Analytics Cloud** to query the Delta Lake using standard SQL with low latency.
- **Azure Functions (BFF):** Acts as the secure bridge between the mobile React App and the internal SAP/Databricks APIs. It manages **Azure AD (Entra ID)** authentication and regional routing.
- **React.js Field App:** The final UI for store associates, optimized for mobile handheld devices with barcode scanning and push notifications for urgent markdowns.

5. MLOps & Quality (The Reliability)

- **Azure DevOps:** Automates the deployment of **Databricks Asset Bundles**, ensuring that code changes are tested via **pytest** in a staging environment before hitting production.

- **Great Expectations:** Automatically validates data quality at the Silver layer (e.g., checking that "Expiry Date" isn't in the past), preventing "Garbage In, Garbage Out" in the ML models.

In the Ahold Delhaize architecture, the **Data Engine** within the Custom React App (Field) serves as the "Operational Glue." Using **Node.js** and **Azure Functions** provides a serverless, event-driven middle layer that can handle the massive concurrency of thousands of store associates while bridging the gap between modern web apps and legacy ERPs.

Here is the detailed technical breakdown of how these components function together.

1. The Architecture Pattern: Backend-for-Frontend (BFF)

Instead of the React app communicating directly with SAP or Databricks (which would be slow and insecure), it communicates with a **Node.js API** hosted as an **Azure Function App**.

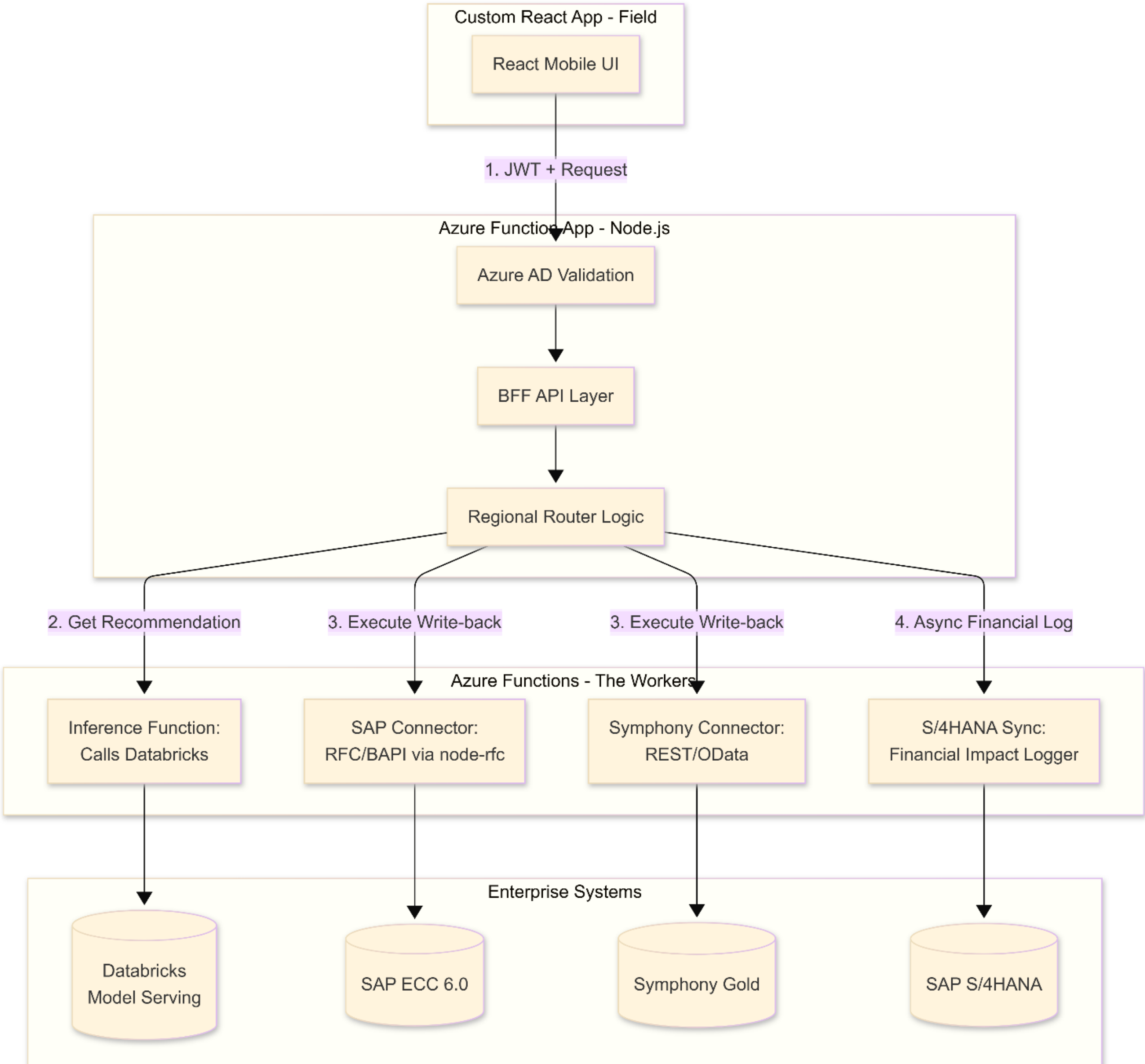
Why Node.js?

- **Asynchronous I/O:** Perfect for "waiting" on multiple legacy systems (SAP, Symphony) simultaneously without blocking the user interface.
- **JSON Native:** Modern ERP APIs and Databricks outputs use JSON, making data transformation minimal and fast.

Why Azure Functions?

- **Serverless Scaling:** During morning "freshness checks," usage spikes across 1,000 stores. Azure Functions scale instantly and cost $0 when nobody is using the app at night.
- **Micro-Services:** You can have one function for "SAP-Lookup" and another for "Symphony-Action," preventing a failure in one region from affecting the other.

2. Detailed Data Engine Flow (The "Action" Logic)



3. Component Deep-Dive

A. The Regional Router (Logic in Node.js)

The Node.js logic checks the User's store_id (extracted from the Azure AD token).

- **If Store = NL/BE:** It routes requests to **Function F2** (SAP ECC).
- **If Store = CZ/RO:** It routes requests to **Function F3** (Symphony Gold).
- **Benefit:** The React frontend code remains exactly the same for all regions; the complexity is hidden in the Data Engine.

B. The SAP Connector (Function F2 - node-rfc)

SAP ECC 6.0 often requires RFC (Remote Function Call) or BAPI protocols. Node.js uses the node-rfc library to communicate with the on-premise SAP Gateway.

- **Task:** It takes a JSON request from the app: { "sku": "123", "new_price": 1.50 }.
- **Action:** It converts this into an SAP-compliant BAPI structure (e.g., BAPI_PRICE_CONDITION_CHANGE) and executes the update.

C. The Inference Function (Function F1 - Databricks Integration)

This function acts as a proxy to the **Databricks Model Serving** (REST Endpoint).

- **Task:** It passes the current stock level and expiration date to the ML Model.
- **Action:** It receives the "Probabilistic Forecast" (P10/P50/P90) and translates the raw math into a UI-friendly recommendation: *"System suggests a 25% discount to clear stock by 6 PM."*

D. The Financial Logger (Function F4 - S/4HANA)

To keep the UI fast, this function is **Asynchronous (Fire-and-Forget)**.

- **Task:** Once a markdown is approved, this function sends a message to **SAP S/4HANA** using an **OData Service**.
- **Action:** It updates the "Provision for Waste" or "Marketing Expense" accounts in the Universal Journal (ACDOCA). The store associate doesn't have to wait for this to finish; it happens in the background.

4. Technical Implementation Details

Feature	Technology	Implementation Detail
Authentication	**MSAL / Entra ID**	React app gets a JWT; Azure Functions validate the signature and roles (Manager vs Associate).
Caching	**Azure Redis Cache**	Stores common SKU master data (names, images) for 1 hour to reduce hits on legacy SAP ECC.
Circuit Breaker	**Ocelot or Polly**	If SAP ECC is down for maintenance, the Node.js API returns a "Offline Mode" flag so the app can store actions locally.
Connectivity	**Azure VNET / Private Link**	Ensures data between Azure Functions and on-premise SAP/Symphony never touches the public internet.
Monitoring	**Azure Application Insights**	Tracks the "Time to Approve" a markdown across all regions for the C-Suite dashboard.

5. Summary of the "Data Engine" Role

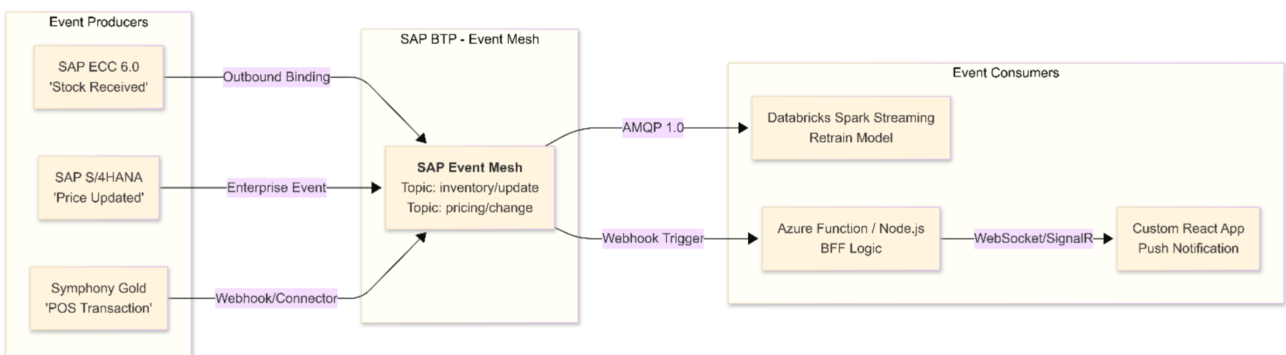
The Node.js + Azure Function setup allows Ahold Delhaize to **modernize the user experience without replacing the legacy ERPs**. It treats the ERPs as "dumb databases" and puts all the "intelligence" (regional routing, AI integration, and workflow logic) into this flexible, scalable cloud layer.

Integrating **SAP Event Mesh** (part of SAP BTP) into this architecture is the professional way to achieve a truly "Event-Driven" ecosystem. It acts as the **central nervous system** that connects the asynchronous legacy world of SAP ECC and Symphony Gold with the real-time needs of the React App and Databricks.

In this context, SAP Event Mesh replaces "polling" (asking the database every 5 minutes if there is new data) with "pushing" (notifying the system the millisecond a change occurs).

1. The Event-Driven Architecture (with SAP Event Mesh)

This diagram shows how SAP Event Mesh bridges the gap between the ERPs and the User Interface.



2. How it works for each ERP System

A. SAP ECC 6.0 (Western Europe)

Legacy ECC doesn't support events natively. You would use the **SAP Event Enablement Add-on**.

- **The Event:** A Goods Receipt (MIGO) is posted.
- **The Action:** The Add-on captures the database trigger and publishes a "Stock_Update" event to **SAP Event Mesh**.
- **The Result:** The React App instantly updates the "Stock on Hand" counter for the store associate without them needing to refresh the screen.

B. SAP S/4HANA (Group Finance)

S/4HANA has native integration with SAP Event Mesh.

- **The Event:** A Category Manager updates the "Markdown Provision" in S/4HANA.
- **The Action:** S/4HANA publishes a **Business Event** to the Mesh.
- **The Result:** The **SAP Analytics Cloud (SAC)** dashboard updates its "Projected Waste Cost" KPI in real-time.

C. Symphony Gold (CEE)

Since Symphony is non-SAP, we use the **Node.js Data Engine (Azure Functions)** as a bridge.

- **The Event:** A POS sale occurs in a store in Romania.
- **The Action:** A small connector in Symphony sends a REST call to an Azure Function, which then publishes the event to the **SAP Event Mesh**.
- **The Result: Databricks** (Model 1) receives the sale data immediately via a **Spark Structured Streaming** connector and updates the demand forecast for the next hour.

3. Impact on Layer 8 (User Interfaces)

Using SAP Event Mesh changes the **Refresh Rate** from "Scheduled" to "Reactive":

UI Component	Without Event Mesh (Polling)	With SAP Event Mesh (Event-Driven)
Custom React App	Associate must pull-to-refresh to see if a markdown was approved.	**Push Notification:** The app vibrates and updates the price the moment the manager clicks "Approve."
Power BI	Data is 15–60 minutes old due to batch ETL.	**Streaming Datasets:** Dashboards show sales and stockouts as they happen (Live-tailing).
Markdown Workflow	The manager has to check a "Pending List."	The manager receives an **Instant Alert** on their watch/phone only when a high-priority exception occurs.

4. Technical Implementation Detail

To connect the **React App (Field)** to **SAP Event Mesh**, we typically use a "Push Service":

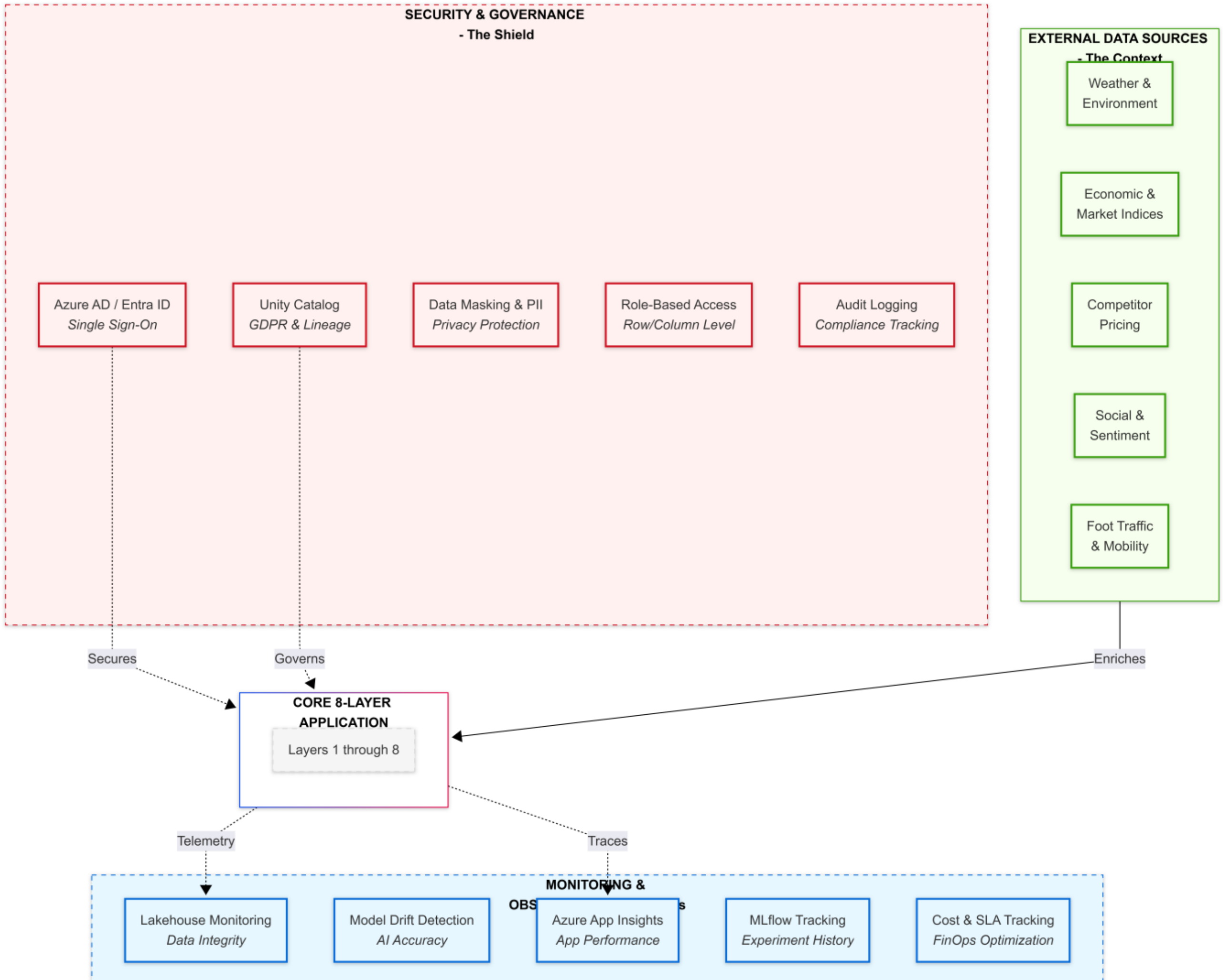
1. **SAP Event Mesh** receives the event from the ERP.
2. It triggers a **Webhook** pointing to an **Azure Function (Node.js)**.
3. The Azure Function uses **Azure SignalR Service** (or WebSockets) to "push" the message to the specific React App instance currently logged in by the Store Associate.

Why this is the "Gold Standard" for Ahold Delhaize:

- **Decoupling:** If SAP ECC goes down for maintenance, the Event Mesh buffers the messages. When ECC comes back up, the events are delivered. The React App doesn't "crash."
- **Scalability:** During peak holiday shopping (e.g., Christmas), the Event Mesh handles millions of POS events without putting a direct load on the React App’s API.

- **Multi-ERP Bridge:** It provides a single "message language" for both SAP and Symphony Gold regions, simplifying the logic in the Databricks AI layer.

Supporting Components: Cross-Cutting Architecture



Detailed Support Component Logic

1. Security & Governance (Protection Layer)

This ensures Ahold Delhaize stays compliant with EU regulations (GDPR) while operating a multi-region cloud.

- **Identity (Entra ID):** **Centralized Identity Management** and **Single Sign-On (SSO)**
- **Unity Catalog (The Heart of Governance):** This is the most critical supporting component. It provides:
  - **Data Lineage:** "Where did this tomato waste prediction come from?" (Traceable back to SAP ECC table MSEG).
  - **Fine-Grained Access:** Restricts Romanian sales data from being viewed by Western European users.

- **Data Masking:** Automatically hides customer loyalty IDs or supplier sensitive pricing from analysts who don't have "Clearance" roles.

2. Monitoring & Observability (Reliability Layer)

Ensures the system doesn't "fail silently."

- **Data Quality (Lakehouse Monitoring):** Uses **Great Expectations** logic. If the Symphony Gold POS feed suddenly shows "Zero Sales" for a whole region, the pipeline pauses and alerts the data engineers.
- **Model Drift:** AI models "age." If the Demand Forecast accuracy for fresh berries drops because of a new consumer trend, the **Drift Detector** triggers a re-training job in MLflow.
- **FinOps (Cost Tracking):** Monitors Databricks DBU usage and Azure egress costs to ensure the project stays within its €350M EBITDA impact budget.

3. External Data Sources (Enrichment Layer)

This adds "Human Context" that the ERP systems (SAP/Symphony) do not possess.

- **Environmental (Weather/Events):** Crucial for perishables. The system fetches forecasts every 6 hours.
- **Market Intelligence (Competitors/Economics):** If a competitor launches a "Buy 1 Get 1 Free" on strawberries, Ahold Delhaize's AI adjusts its own price elasticity model (Model 3) to remain competitive.
- **Mobility (Foot Traffic):** Uses SafeGraph or Google data to predict store busyness. Higher foot traffic predictions trigger higher "Safety Stock" recommendations in the MEIO solver (Model 4).

Synergy Example: The "Storm" Scenario

1. **External Data:** Weather API detects a major storm coming to Central Europe (CEE).
2. **Core App:** AI Model 1 predicts a spike in "pantry loading" (canned goods) and a drop in "fresh produce."
3. **Monitoring:** App Insights tracks the spike in API calls as the system re-calculates 10,000 SKUs.
4. **Security:** Unity Catalog ensures that this sudden surge in data processing is logged and audit-ready for the CEE regional leads.
5. **Action:** The system auto-adjusts the SAP EWM pick-lists to prioritize long-life items before the storm hits.

Key performance indicators

SUCCESS METRICS (KPIs)
Waste Reduction:
├── Target: Reduce perishable waste from 3.5% → 2.0% (€200M+ annual savings)
├── Metric: Daily waste \$ / Sales \$ by category & store
└── Dashboard: Real-time waste tracking with YoY comparison

Product Availability:
├── Target: Increase on-shelf availability from 94% → 98%
├── Metric: Stockout incidents per 1000 transactions
└── Dashboard: Stockout alerts by SKU, store, time-of-day

Forecast Accuracy:
├── Target: MAPE < 15% for fresh produce, < 10% for packaged goods
├── Metric: Weekly MAPE by category, brand, region
└── Dashboard: Forecast vs Actuals with confidence intervals

Working Capital:
├── Target: Reduce inventory days from 28 → 24 days (€150M cash release)
├── Metric: DIO (Days Inventory Outstanding) by category
└── Dashboard: Inventory aging, slow-moving SKU alerts

Operational Efficiency:
├── Target: Reduce manual ordering time by 70% (store manager productivity)
├── Metric: % of orders auto-generated vs manual
└── Dashboard: Automation rate by store, user satisfaction scores

Financial Impact:
├── Target: €350M+ annual EBITDA improvement (waste + availability + WC)
├── Metric: Incremental margin from AI optimization
└── Dashboard: ROI tracker, payback period analysis

To realize the **€350M+ annual EBITDA improvement**, Ahold Delhaize requires a **Unified Metric Store** approach. This ensures that a "Waste %" calculation in the Netherlands (SAP ECC) matches the calculation in Romania (Symphony Gold).

Below is the solution for the **KPI Calculation Engine** and the **Multilayer Dashboard Strategy**.

1. The KPI Calculation Engine (The "Metric Store")

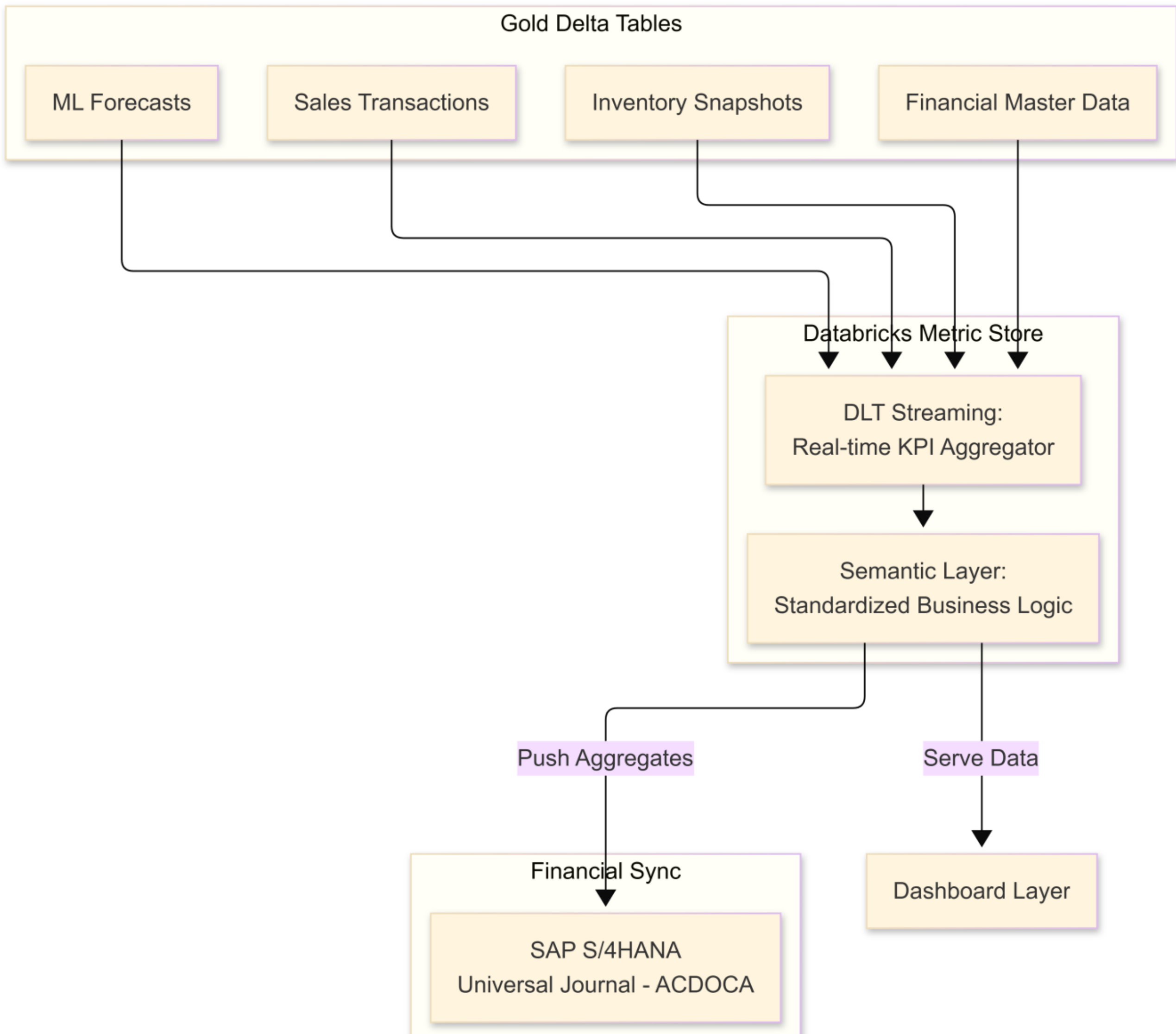
The engine resides within **Databricks SQL Warehouse**. It processes raw events into aggregated KPIs using **Delta Live Tables (DLT)** for reliability.

Technical Logic for Core Metrics

KPI	Calculation Logic (SQL/Python)	Data Sources
Waste Reduction	(SUM(Scrap_Value_at_Cost) / SUM(Gross_Sales_Value)) * 100	Symphony (Scrap Codes) + SAP (Material Valuation)
Product Availability	1 - (Count(SKU_Store_Days_Zero_Stock) / Total_SKU_Store_Combinations)	Symphony (Live Inventory) + POS (Zero-sales alerts)
Forecast Accuracy	1 - MAPE where MAPE = ABS(Actual - Forecast) / Actual	Gold.Demand_Forecast + Gold.Sales_Transactions
Working Capital (DIO)	(Avg_Inventory_Value / COGS_Daily) * 365	SAP MBEW (Cost) + Symphony Inventory Levels

Operational Efficiency	Count(Orders_Where_Modified = False) / Total_Orders	Layer 7 Write-back logs (Audit Trail)

Engine Architecture



2. Dashboard Strategy: Three Tiers of Visibility

Ahold Delhaize will use three distinct dashboard environments tailored to user roles.

Tier A: Strategic Executive Dashboard (SAP Analytics Cloud)

- **Target:** C-Suite, CFO, Chief Supply Chain Officer.
- **Focus:** Financial impact and ESG goals.
- **Key View:** "EBITDA Bridge" showing how much profit was gained specifically from AI-driven waste reduction vs. traditional sales growth.
- **Data Source:** SAP S/4HANA (Direct connection to Universal Journal).

Tier B: Operational Command Center (Power BI)

- **Target:** Category Managers, Regional Supply Chain Leads.
- **Focus:** Performance by banner (AH vs. Delhaize), category (Fruit vs. Meat), and region.
- **Key View:** "Forecast vs. Actuals" heatmaps. Allows a manager to see that "Fresh Berries in Belgium" has high forecast error, triggering a model review.
- **Data Source:** Databricks SQL Warehouse (DirectQuery for near real-time).

Tier C: Store Action Dashboard (Custom React App)

- **Target:** Store Managers, Department Leads.
- **Focus:** Immediate tasks.
- **Key View:** "Critical Expiry List" — a list of 10 items that must be marked down *today* to avoid total waste loss tomorrow.
- **Data Source:** Node.js BFF + Azure Redis Cache for instant responsiveness.

3. Comprehensive Solution Details

A. Waste & Freshness Tracking (Perishables Focus)

- **The Engine:** A specialized "Freshness Index" calculation that combines **EWM Goods Receipt Date** and **Symphony POS Expiry Logic**.

  The Dashboard: A Power BI "Waste Waterfall" showing: Planned Waste → Avoided via AI Markdown → Actual Waste → Donated to Food Bank.

B. Availability & Stockout Logic

- **The Engine:** Instead of just checking if stock is > 0, the engine uses **"Non-Zero Sales Intervals."** If a high-velocity item (Milk) has zero sales for 2 hours during peak time, the engine flags a "Phantom Stock" stockout even if the system says 5 units remain.
- **The Dashboard:** Real-time mobile alerts in the **React App** sending associates to specific aisles to verify stock.

C. Financial Impact & ROI Calculation

- **The Engine:** The **"Incremental Margin Calculator."** It compares the margin of a store using AI Optimization against a "control group" store (or historical baseline) to isolate the profit uplift.
- **The Dashboard:** In **SAP Analytics Cloud**, this shows the **Payback Period** of the project investment, updated weekly.

4. Implementation Readiness: The "Single Version of Truth"

To ensure these metrics are successful, the following **Governance Rule** is applied:

"No KPI is valid unless it is calculated within the Databricks Metric Store. Regional ERPs (SAP ECC/Symphony) provide the raw data, but they are not the source of truth for optimization performance."

Outcome:
By May 15, 2027 (Full Rollout), Ahold Delhaize leadership will have a **Real-time ROI Tracker** that proves the AI initiative has released **€150M in cash** (via Working Capital) and saved **€200M in physical food waste**, directly fulfilling the corporate sustainability and financial mandates.

Implementation Roadmap

Phase 1: Foundation (Months 1-3)

- Setup Databricks workspaces
- Establish connectivity
- Build DLT pipelines
- Migrate 2-3 years history

Phase 2: Pilot (Months 4-6)

- Focus: Netherlands fresh vegetables
- 50 pilot stores
- Baseline KPI measurement

Phase 3: Scale (Months 7-9)

- All perishables (dairy, bakery, meat)
- All Netherlands stores (~1000)
- 80% automation

Phase 4: Regional (Months 10-12)

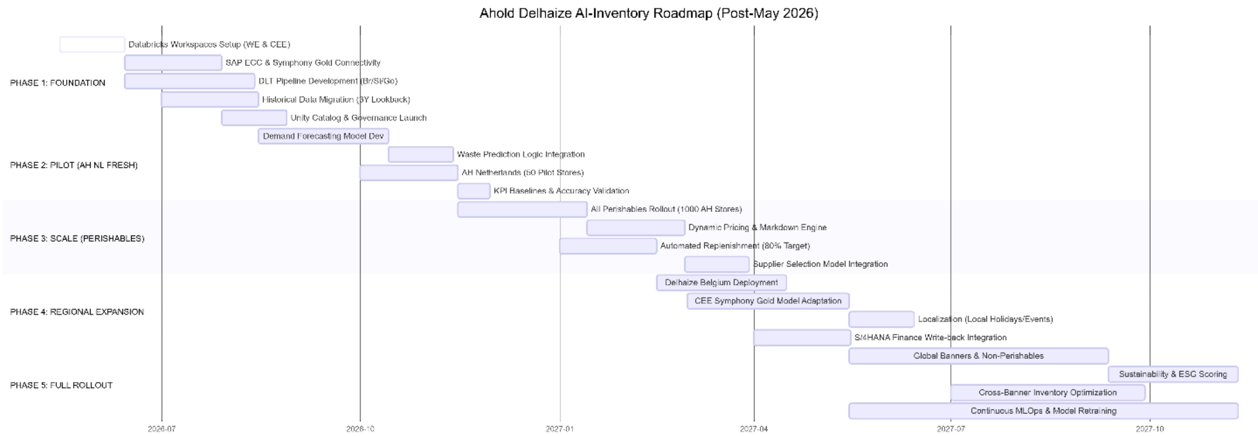
- Belgium expansion
- Central/Eastern Europe (Symphony)
- 85% automation

Phase 5: Full Rollout (Months 13-18)

- All banners, all regions
- Beyond perishables
- Continuous improvement

This **Gantt-style Roadmap** visualizes the 18-month journey from foundation to full-scale AI automation across the Ahold Delhaize enterprise.

This roadmap adjusts the implementation schedule to begin on **May 15, 2026**, aligning with the technical requirements of the hybrid SAP/Symphony landscape and the Databricks Lakehouse rollout.



Strategic Milestone Breakdown

Phase 1: The "Digital Bridge" (May – August 2026)

- **Focus:** Breaking the silos between Western Europe (SAP ECC) and CEE (Symphony Gold).
- **Key Deliverable:** A unified **Unity Catalog** on Databricks that allows a data scientist in Zaandam to see inventory patterns in Romania through a single governed interface.

Phase 2: Fresh Food Pilot (August – November 2026)

- **Focus:** The high-stakes Albert Heijn "Fresh" category.
- **Key Deliverable:** Moving from point-forecasts to **Probabilistic Demand Forecasting**. This allows store managers to understand not just *what* they will sell, but the *risk* of a stockout vs. the *risk* of waste.

Phase 3: National Automation (November 2026 – February 2027)

- **Focus:** Efficiency at scale.
- **Key Deliverable:** Achieving an **80% "No-Touch" Replenishment rate**. The system auto-triggers BAPIs in SAP ECC to create Purchase Orders without human intervention for standard, non-volatile items.

Phase 4: Cross-Border Harmonization (February – May 2027)

- **Focus:** Bridging the architectural gap between SAP and Symphony Gold regions.
- **Key Deliverable:** Adapting models to handle the different data structures of **Symphony Gold**. Integrating with **SAP S/4HANA** ensures that every avocado saved from waste is immediately reflected as improved working capital in the Group Financials.

Phase 5: Enterprise Intelligence (May – November 2027)

- **Focus:** Full group-wide optimization.
- **Key Deliverable:** Expansion to all categories (Dry Grocery, GM) and the introduction of **Sustainability Scoring**, where the AI prioritizes replenishment routes with the lowest carbon footprint.

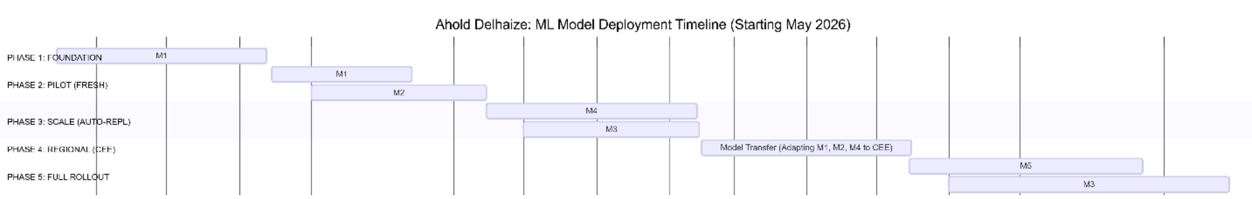
To ensure a successful deployment at Ahold Delhaize, the ML models are phased based on their **dependency on data maturity** and their **immediate impact on EBITDA (Waste vs. Availability)**.

The following division aligns the 5 models with the Roadmap starting **May 15, 2026**.

1. Model Implementation Phasing

Phase	Model	Focus & Logic
Phase 1: Foundation	**M1: Demand Forecasting (Baseline)**	**Prophet (Baseline):** Establish a statistical baseline using historical SAP/Symphony data to prove data pipeline reliability.
Phase 2: Pilot	**M1: Demand Forecasting (Advanced)** **M2: Waste Prediction**	**LightGBM/LSTM:** High-accuracy store-SKU-day forecasts. **Waste Prediction:** Specifically for fresh produce (tomatoes, berries) to identify "shrink" risk before it happens.
Phase 3: Scale	**M4: Replenishment Optimization** **M3: Dynamic Pricing (Intro)**	**MEIO:** Automating Safety Stock and EOQ directly into SAP ECC BAPIs. **Dynamic Pricing:** Using Price Elasticity to clear stock identified by Model 2 as "High Waste Risk."
Phase 4: Expansion	**M1, M2, M4 (CEE Adaptation)**	**Transfer Learning:** Adapting the Western Europe (SAP) models to the Central & Eastern Europe (Symphony Gold) consumer behavior and data formats.
Phase 5: Full Rollout	**M5: Supplier Selection** **M3: Dynamic Pricing (Advanced)**	**Supplier Selection:** Strategic scoring based on lead-time reliability and ESG. **RL Pricing:** Reinforcement learning for margin maximization across all categories.

2. Visual Roadmap: Model Deployment Timeline



3. Detailed Model Progression by Phase

Phase 1 & 2: The Forecasting Core (M1 & M2)

- **Why:** You cannot optimize inventory (M4) or price (M3) without a high-confidence demand signal.
- **Model 1 (M1):** Starts with **Prophet** for fast, interpretable trends. It then upgrades to **LightGBM** to handle "Cold Start" problems for new product launches in Symphony Gold.
- **Model 2 (M2):** Focuses on the "Perishable" use case. It outputs a probability of waste per batch, which acts as a "trigger" for the replenishment engine to slow down orders.

Phase 3: The Execution Engine (M4 & M3 Intro)

- **Why:** Once forecasts are accurate, we automate the "Buying" (Replenishment).
- **Model 4 (M4):** This is the **Multi-Echelon Optimization**. It connects the DC (SAP ECC) to the Store (Symphony Gold). It calculates the optimal "Safety Stock" to prevent stockouts during promotions.
- **Model 3 (M3 - Intro):** Starts with simple **Price Elasticity**. If M2 predicts a 70% chance of waste for strawberries in AH Netherlands, M3 suggests a markdown price to the Symphony Gold system.

Phase 4: The Integration Gap

- **Focus:** Bridging Western Europe and CEE.
- **Action:** We don't build new models here. Instead, we perform **Feature Alignment**. We ensure that "Material Group" in SAP ECC maps to "Sub-Class" in Symphony Gold so the models (M1, M2, M4) can run against Eastern European data without retraining from scratch.

Phase 5: Strategic Intelligence (M5 & M3 Advanced)

- **Why:** Shift from operational efficiency to strategic competitive advantage.
- **Model 5 (M5):** Uses a Multi-Criteria Decision Making (MCDM) algorithm. It scores suppliers not just on price, but on **Lead-Time Reliability** (critical for S/4HANA financial forecasting) and **Sustainability Scores** (Scope 3 emissions).
- **Model 3 (M3 - Advanced):** Moves to **Reinforcement Learning**. The model "learns" the best time to promote items to maximize total basket margin, rather than just clearing individual SKU stock.

The following Work Breakdown Structure (WBS) covers the 18-month implementation roadmap starting **May 15, 2026**.

The total estimated effort is approximately **4,750 Person-Days (PD)**. This reflects the complexity of integrating legacy SAP ECC and Symphony Gold with a modern Databricks AI stack for a multi-banner retail group.

Ahold Delhaize Inventory Optimization: WBS & Effort Estimation

WBS Code	Work Package / Deliverable	Estimated Effort (PD)	Key Roles Involved
1.0	**Foundation & Governance (Layer 3 & Cross-Cutting)**	**420**	**Architects, SecOps**
1.1	Unity Catalog setup & Cross-region RBAC configuration	80	Data Architect, SecOps
1.2	Azure ExpressRoute & VNet Private Link Integration	60	Cloud Network Engineer
1.3	Data Privacy (PII Masking) & GDPR Compliance Framework	100	Data Governance Lead
1.4	Databricks Workspace & DLT Environment Provisioning	60	Platform Engineer
1.5	Global Audit & Monitoring Dashboards (Azure Monitor)	120	DevOps Engineer
2.0	**Data Engineering & Integration (Layers 1, 2)**	**1,050**	**Data Engineers**
2.1	SAP ECC (BODS) ODP Delta Extractors (MM/SD/WM)	280	SAP Data Engineer
2.2	Symphony Gold REST/SFTP API Ingestion Pipelines	220	Data Engineer
2.3	SAP S/4HANA Lakeflow Connect CDC (ACDOCA)	120	SAP Data Engineer
2.4	**Logistics Stream:** SAP EWM/TM Table Replication	180	SAP Data Engineer
2.5	Silver Layer: Master Data Harmonization (SAP-SYM Mapping)	250	Senior Data Engineer
3.0	**AI Intelligence & Feature Store (Layers 4, 5)**	**1,000**	**Data Scientists**
3.1	Databricks Feature Store: Time-Series & Freshness Logic	250	Machine Learning Eng.
3.2	Model 1: Probabilistic Demand Forecasting (LSTM/LightGBM)	250	Data Scientist
3.3	Model 2: Perishable Waste Prediction Engine (XGBoost)	150	Data Scientist
3.4	Model 4: Multi-Echelon Inventory Optimization (MEIO)	250	Data Scientist (OR)
3.5	MLflow Lifecycle: Experiment Tracking & Model Registry	100	MLOps Engineer
4.0	**Orchestration & Decision Engine (Layer 6)**	**750**	**Logic Eng, Backend**
4.1	Business Rules Engine (BRE) & Policy Configuration	150	Business Analyst / Dev
4.2	**Multi-Objective Constraint Solver** (EWM/TM awareness)	400	Mathematical Optimizer
4.3	Exception Alert Engine (n8n/Logic Apps Integration)	200	Backend Developer
5.0	**App Development & Write-Back (Layers 7, 8)**	**1,130**	**Fullstack, SAP Dev**
5.1	Custom React Field App (Mobile UI/Scan/Workflow)	350	Frontend Developer
5.2	Node.js Backend-for-Frontend (BFF) on Azure Functions	220	Node.js Developer
5.3	SAP ECC (BAPI) & Symphony (REST) Write-back Connectors	280	SAP ABAP / API Dev

5.4	Power BI & SAP Analytics Cloud Strategic Dashboards	180	BI Developer
5.5	SAP Event Mesh Integration for Real-time ESL updates	100	Integration Developer
6.0	**Testing, Pilot & Rollout (Validation)**	**400**	**QA, PMO, Ops**
6.1	UAT, Data Validation (Great Expectations) & Stress Testing	150	QA Engineer
6.2	Pilot Rollout: 50 Stores (Fresh Produce - NL)	100	Project Manager
6.3	Regional Expansion Adaptation (CEE Symphony Context)	150	Local Product Owner
TOTAL	**Estimated Project Effort**	**4,750**	**~22 Full-Time Staff**

Work Package Deep-Dive & Deliverables

WP 2.5: Master Data Harmonization (250 PD)

- **Deliverable:** A unified "Global Silver SKU Table."
- **Challenge:** Mapping the 18-character MATNR in SAP ECC to the 13-character SKU_ID in Symphony Gold.
- **Outcome:** A single source of truth used by all AI models, allowing the same model to predict demand for "Albert Heijn Milk" and "Delhaize Milk."

WP 4.2: Multi-Objective Constraint Solver (400 PD)

- **Deliverable:** A Python-based LP (Linear Programming) Solver.
- **Logic:** This is the most complex component. It takes the AI forecast and "trims" it based on:
    - Warehouse Labor (from EWM).
    - Truck Payload (from TM).
    - Shelf Space (from Symphony).
- **Outcome:** Eliminates "Rejected Orders" caused by physical DC/Transport overloads.

WP 5.1: Custom React Field App (350 PD)

- **Deliverable:** A PWA (Progressive Web App) deployed to 15,000+ mobile devices.
- **Features:** Offline barcode scanning, real-time "Markdown Recommendation" display, and a manager approval workflow.
- **Outcome:** Direct empowerment of store floor associates to reduce waste in real-time.

Resource Strategy (The "22-Person" Team)

To complete this in 18 months, the following core team is proposed:

1. **Project Leadership:** 1 PM, 1 Solution Architect (Total 2).
2. **Data Squad:** 4 Data Engineers, 2 SAP Specialists (Total 6).
3. **AI Squad:** 3 Data Scientists, 1 Mathematical Optimizer, 1 MLOps Engineer (Total 5).
4. **Dev Squad:** 2 Frontend (React), 2 Backend (Node.js), 2 Integration (BTP/API) (Total 6).
5. **Quality/Governance:** 2 QA/Data Quality Specialists, 1 Governance/GDPR Lead (Total 3).

Execution Assumptions

- **Agile Methodology:** Bi-weekly sprints.
- **Environment:** Access to SAP ECC and Symphony Gold "Sandboxes" provided by Week 4.
- **Infrastructure:** Azure ExpressRoute already established by Ahold Delhaize Corporate IT.

Architectural Assumptions

The success of the Ahold Delhaize Inventory Optimization project relies on a set of foundational architectural assumptions. These assumptions define the boundaries of the solution and ensure that the multi-ERP, multi-region complexity is managed effectively.

1. Connectivity & Infrastructure Assumptions

- **Azure ExpressRoute Availability:** It is assumed that a stable, high-bandwidth Azure ExpressRoute connection exists between Ahold Delhaize's on-premise data centers (hosting SAP ECC and Symphony Gold) and the Azure Cloud environment.
- **Cloud-First Scaling:** The solution assumes that Databricks and Azure Functions will leverage **autoscaling**. Compute resources will expand during morning "Freshness Check" peaks and shrink during off-peak hours to optimize cost.
- **Regional Latency:** It is assumed that network latency between Central/Eastern Europe (Symphony Gold sites) and the primary Azure region is within acceptable limits (<100ms) for real-time ESL (Electronic Shelf Label) updates.

2. Integration & Data Assumptions

- **SAP ECC "ReadOnly" Extraction:** To protect the legacy ECC 6.0 production environment, all data extraction via SAP BODS will be performed against a **High-Availability (HA) Replica** or via ODP (Operational Data Provisioning) to prevent performance degradation of store operations.
- **Global Material ID (The "Golden SKU"):** It is assumed that a cross-reference mapping exists, or can be programmatically derived, between SAP Material Numbers (MATNR) and Symphony SKU IDs to allow for the creation of a unified **Silver Layer** in the Lakehouse.
- **Event Mesh Capability:** It is assumed that the SAP landscape is upgraded with the **Event Enablement Add-on**, allowing SAP ECC to publish events to the SAP Event Mesh.

3. AI & Modeling Assumptions

- **Data Sufficiency:** It is assumed that at least **24–36 months of historical sales and inventory data** are available from both SAP and Symphony to train the LSTM/LightGBM models to account for seasonality and holiday spikes.
- **Feature Parity:** The **Databricks Feature Store** will serve as the single source of truth for features. It is assumed that features calculated for "offline" training will be bit-identical to the features served "online" to the React App for inference.
- **Probabilistic vs. Deterministic:** The architecture assumes the business will transition to **Probabilistic Demand Signals (P10/P50/P90)** rather than single-point forecasts, requiring a shift in how store managers interpret "Safety Stock."

4. Logistics & Physical Constraints (EWM/TM)

- **Digital Twin Accuracy:** The **Constraint Solver (Layer 6)** assumes that SAP EWM and TM data (Bin capacity and Truck volume) is updated in the Lakehouse with near real-time latency (via CDC). If EWM data is stale, the solver will default to conservative "Safe" capacity limits.
- **Constraint Hierarchies:** In the event of a conflict, **Physical Constraints** (Truck Volume/Weight) take precedence over **AI Demand**. The system will "trim" the lowest-margin/longest-life items from the order first to ensure the truck can ship.

5. Security & Governance Assumptions

- **Unified Identity:** It is assumed that **Microsoft Entra ID (Azure AD)** is the global identity provider across all banners (Albert Heijn, Delhaize, etc.), enabling the use of JWT claims for Role-Based Access Control (RBAC).
- **GDPR Compliance by Design:** All PII (Personally Identifiable Information), such as customer loyalty data from Symphony POS, will be **masked or pseudonymized** at the Bronze-to-Silver transition within the Databricks DLT pipeline.
- **Regional Data Residency:** While data is centralized in a Lakehouse, **Unity Catalog** will be configured to restrict data access based on the user's regional legal entity, adhering to local data sovereignty policies.

6. Business & Process Assumptions

- **Human-in-the-Loop:** For the first 6 months (Pilot Phase), the system assumes a **100% manual review** for high-value markdowns. Transition to "85% Automation" will only occur once the Model Accuracy (MAPE) achieves the agreed-upon threshold.
- **Hardware Compatibility:** It is assumed that the **Custom React App** will run on existing Honeywell/Zebra mobile handheld devices used in stores, requiring no new hardware procurement.
- **ESL Connectivity:** It is assumed that stores are equipped with **Electronic Shelf Labels (ESL)** capable of receiving price updates via the Symphony Gold API gateway.

Risk Management & Readiness Observation

Implementing an AI-driven inventory optimization across a fragmented landscape like Ahold Delhaize’s (SAP vs. Symphony) carries high stakes. A failure in the "brain" (AI) or the "nerves" (Integration) can lead to empty shelves or massive fresh-food waste.

Below is the **Risk Management & Readiness Observation**, categorized by technical and operational dimensions.

1. Technical & Integration Risks (The "Plumbing")

Risk Factor	Impact	Mitigation Strategy
Legacy SAP Performance	High-volume BODS extraction could "choke" the SAP ECC 6.0 production instance.	Use **ODP (Operational Data Provisioning)** to offload delta extraction to the application layer; perform bulk loads from a secondary HA replica.
Data Silo Mismatch	SAP (WE) and Symphony (CEE) have different SKU hierarchies, making group-level AI training difficult.	Dedicated **WBS 2.5 (Master Data Harmonization)** to create a "Golden SKU" mapping in the Databricks Silver Layer.
Connectivity Latency	Real-time write-backs to Symphony Gold ESLs in Romania/Czech Republic might fail over ExpressRoute.	Implement **Circuit Breaker patterns** in Node.js BFF; use local caching and asynchronous event-driven updates via SAP Event Mesh.

2. AI & Data Quality Risks (The "Brain")

Risk Factor	Impact	Mitigation Strategy
Model Drift	A sudden change in consumer behavior (e.g., inflation spike) makes historical training data irrelevant.	Implement **Databricks Model Monitoring** with automated retraining triggers in MLflow when MAPE (error) exceeds 15%.
The "Black Box" Effect	Store managers ignore AI recommendations because they don't understand or trust the "why."	**Explainable AI (XAI):** Ensure the React App displays "Reason Codes" (e.g., *"Recommended 30% discount due to upcoming 30°C weather + 2-day shelf life"*).
Garbage In, Garbage Out	Incorrect stock counts in EWM lead to the AI ordering stock that won't fit in the warehouse.	**Great Expectations (GX)** integration in DLT pipelines to halt data flow if EWM bin-capacity data shows anomalies.

3. Operational & Logistics Risks (The "Reality")

Risk Factor	Impact	Mitigation Strategy
Physical Constraint Breach	AI orders a full truck, but SAP TM shows only a small van is available for that route.	**Layer 6 Constraint Solver:** This must be a "Hard Constraint." If TM/EWM data is missing, the system defaults to a conservative 70% capacity fill rate.
Labor Shortage	AI optimizes orders, but the DC lacks forklift drivers (EWM) to pick the increased volume.	Integrate **Resource Availability** from EWM into the Solver logic to "throttle" orders based on picking capacity.
Markdown Margin Leakage	Excessive auto-markdowns requested by associates could erode profitability.	Strict **RBAC (Role-Based Access Control)** where discounts >25% require a Store Manager's digital thumbprint in the React App.

4. Readiness Assessment (The "Go/No-Go" Checklist)

Before the **May 15, 2026** start date, Ahold Delhaize must satisfy these three readiness pillars:

A. Infrastructure Readiness (Cloud & Network)

ExpressRoute Load Test: Can the tunnel handle the concurrent streaming of ECC, Symphony, and S/4HANA data?

Databricks Unity Catalog: Are the regional "Metastores" configured to prevent data leakage between banners (e.g., AH vs. Mega Image)?

B. Data Readiness (The "Historical Moat")

3-Year History: Is there a clean, accessible archive of sales and waste data for all perishables?

Master Data Cross-Walk: Has the initial mapping of SAP Material Groups to Symphony Sub-Classes been validated by the Category VPs?

C. Organizational Readiness (The "Human Element")

Champion Network: Have "AI-Super Users" been identified in the 50 pilot stores in the Netherlands?

S/4HANA Finance Alignment: Has the CFO agreed on the "Waste Savings" formula to ensure the €350M EBITDA impact is measurable?

5. The "Safety Valve" Strategy

To manage the transition, the architecture includes a **"Manual Override Mode"**:

- **Phase 1 (Pilot):** The AI operates in **Shadow Mode**. It generates recommendations, but no BAPIs are triggered in SAP ECC. Humans must manually replicate the order.
- **Phase 2 (Automation):** The system moves to **Exception-Only Mode**. Orders are auto-created unless the variance vs. the previous week is >20%, in which case it triggers a "Manager Review" in the React App.

Conclusion:
The greatest risk is not technical—it is **operational trust**. By integrating **EWM/TM logistics constraints** into the AI's decision-making process (Layer 6), Ahold Delhaize mitigates the risk of "theoretically perfect but physically impossible" inventory plans.

Key Corporate Change Attributes

For Ahold Delhaize, transitioning to an AI-driven inventory model is not just a technical upgrade; it is a fundamental shift in how the corporate culture operates across different regional brands (banners).

The following **Key Corporate Change Attributes** define the organizational evolution required to support the 8-Layer architecture.

1. From "Gut Feeling" to "Algorithmic Trust"

- **The Shift:** Historically, store managers in Belgium or Romania relied on local experience and intuition to adjust orders. The new model requires trusting **Probabilistic Demand Signals (P10/P50/P90)**.
- **Change Action:** Corporate leadership must champion "Algorithmic Trust." This involves moving managers away from manual order entry and toward **"Management by Exception."** They only intervene when the AI alerts them via the React App (Layer 8).

2. Unified Language (Data Democratization)

- **The Shift:** Previously, "Waste" or "Safety Stock" might have been defined differently in SAP ECC (Western Europe) than in Symphony Gold (CEE).
- **Change Action:** The implementation of the **Unity Catalog (Layer 3)** forces a "Single Version of the Truth." Corporate Change involves aligning all regional Category VPs on a **Unified Semantic Layer**, ensuring that a 5% waste reduction in the Netherlands is measured identically to one in Greece.

3. Cross-Functional "Logistics-Aware" Procurement

- **The Shift:** Procurement (Buying) and Logistics (Warehouse/Transport) often operate as separate cost centers.
- **Change Action:** Because the **Constraint Solver (Layer 6)** integrates EWM and TM data, the Buying team must now become "Logistics-Aware." They must accept that the AI may "trim" a high-demand order if the warehouse is at capacity or a truck is unavailable. This requires a shift in performance incentives from "Volume" to **"Balanced Throughput."**

4. Human-in-the-Loop Accountability

- **The Shift:** AI is often feared as a "job killer" or a "black box" that cannot be corrected.
- **Change Action:** The **Custom React App (Layer 8)** is designed for "Human-in-the-loop" interaction. Corporate change focuses on empowering store associates as **"Data Validators."** If an associate identifies a local event the AI missed (e.g., an unannounced local street festival), they use the app to provide a feedback signal that retrains the model (Layer 5).

5. Financial Transparency (Real-Time P&L)

- **The Shift:** Inventory valuation and waste impact were traditionally "month-end" surprises for Group Finance.
- **Change Action:** With **S/4HANA OData Write-backs (Layer 7)**, the financial impact of every avocado saved is visible in real-time. This requires a change in financial leadership from "Retroactive Accounting" to **"Predictive Financial Steering,"** using SAP Analytics Cloud to adjust quarterly guidance based on AI performance.

6. Agile Governance & Regional Autonomy

- **The Shift:** Ahold Delhaize has a decentralized "Multi-Banner" culture. A "one-size-fits-all" approach from the Zaandam HQ would meet resistance in CEE.
- **Change Action:** The architecture supports **Regional Localization**. While the core AI "brain" is centralized in Databricks, the **Business Rules Engine (Layer 6)** allows regional banners (e.g., Mega Image in Romania) to maintain their own local market strategies (e.g., different markdown policies or local holiday calendars).

Summary of Change Impact

Attribute	Old Corporate State	New AI-Driven State
Decision Driver	Experience & Intuition	Data-Driven Probability
KPI Focus	Sales Volume	Sales Margin & Waste Reduction
Manager Role	Order Entry & Tallying	Exception Management & Strategy
Data Access	Siloed Regional ERPs	Global Lakehouse (Unity Catalog)
Sustainability	Reactive ESG Reporting	AI-Optimized Waste Prevention
Financials	Month-End Reconciliation	Real-Time Universal Journal Sync

Conclusion for Leadership:
The successful rollout on **May 15, 2026**, depends on the **"AI Fluency"** of the organization. Corporate Change efforts must focus on training staff not just how to use the "React App," but how to interpret AI confidence intervals and understand the physical constraints (EWM/TM) that the system is now automatically respecting.

Business Constraints for Implementation

For Ahold Delhaize, implementing a centralized AI "Brain" across a decentralized federation of retail banners involves several hard business constraints. These constraints act as the "guardrails" within which the technical solution must operate.

The following **Business Constraints** must be respected throughout the 18-month implementation roadmap starting May 15, 2026.

1. Operational "Peak Season" Freeze

Retailers generate a disproportionate amount of profit during holiday periods.

- **The Constraint:** No major deployments, code releases, or system write-back activations (Layer 7) are permitted during the **Golden Quarter** (November 15 – January 15) or the **two weeks leading up to Easter**.
- **Impact on Roadmap:** The "Scale-up" phase (Phase 3) must be carefully timed to ensure the 1,000-store rollout doesn't collide with the 2026 Christmas season.

2. Regional Banner Autonomy (The "Federated" Model)

Ahold Delhaize operates as a federation, not a monarchy.

- **The Constraint:** Local banners (Albert Heijn, Delhaize, Mega Image) must maintain final authority over **pricing and local assortment**. The AI can *recommend* a markdown or an order quantity, but the regional business units must have the power to override it based on local market conditions.
- **Impact on Architecture:** The **Business Rules Engine (Layer 6)** must allow for regional "toggles" where specific banners can opt-in or opt-out of certain AI behaviors.

3. SAP "Clean Core" Strategy

- **Impact on Architecture:** This reinforces the use of **Databricks** for all heavy computation and **SAP BTP / Azure Functions** for the integration logic.

4. Labor Union & Work Council Compliance

In Western Europe (NL/BE), store operations are governed by strict labor agreements.

- **The Constraint:** The **Custom React App (Layer 8)** must not increase the net "administrative burden" on store associates. Any new task (like approving a markdown) must be offset by the removal of a legacy manual task (like manual shelf-checking).
- **Impact on Design:** User Experience (UX) is a business constraint. If the app is not "faster than paper," it cannot be deployed.

5. Financial ROI & EBITDA Targets

The project is funded based on a specific business case.

- **The Constraint:** The solution must demonstrate a **measurable reduction in "Fresh Waste %"** within the first 6 months of the Pilot (Phase 2). The total project must hit a self-funding status (NPV positive) within 24 months of the May 2026 start date.
- **Impact on Roadmap:** This necessitates the "Fresh Produce" focus in the pilot, as it offers the highest potential for immediate waste-cost savings.

6. GDPR & Data Sovereignty

- **The Constraint:** While sales data is centralized for AI training, **Personally Identifiable Information (PII)** from loyalty programs in Symphony Gold must not cross regional borders (e.g., Romania to Netherlands) in an unmasked state.
- **Impact on Architecture:** This mandates the use of **Unity Catalog (Layer 3)** for automated data masking and regional row-level security.

7. "Availability Over Waste" (The Core Retail Rule)

- **The Constraint:** Ahold Delhaize’s primary brand promise is product availability. The AI must never optimize for "Zero Waste" at the expense of **On-Shelf Availability (OSA)** for "Value-Driver" SKUs (e.g., milk, bread, bananas).

- **Impact on ML Models:** The **Demand Forecasting (Model 1)** must use high "Service Level" coefficients (P90) for core items, even if it results in slightly higher waste risk, to satisfy this business priority.

8. Vendor Minimums (MOQ) and Logistics Reality

- **The Constraint:** AI recommendations must respect **Supplier Minimum Order Quantities (MOQ)** and **Standard Pack Sizes**. Ordering "3.5 cases of apples" is not a business reality.
- **Impact on Layer 6:** The **Constraint Solver** must "snap" all AI suggestions to the nearest logical logistic unit (e.g., full crates or full pallets).

Summary of Constraint Impact on the May 15, 2026 Start:

Constraint Category	Limitation	Mandatory Action for Kick-off
Timeline	18-month limit	Ensure "Phase 1: Foundation" is finished by Aug 2026 to avoid the Xmas freeze.
Process	No new labor burden	Validate React App UX with AH NL Work Council by July 2026.
Technical	SAP Clean Core	Confirm all Write-back is via standard BAPIs, not custom table writes.
Strategy	Availability First	Set P90 Quantile as the default for all "Category A" SKUs in the models.

In short: These constraints ensure that the "AI Brain" doesn't become an "Academic Exercise" but remains a **Physically Possible** and **Financially Accountable** retail tool.

Impact on change management

The transition to an AI-driven inventory model at Ahold Delhaize represents a fundamental pivot from **reactive retail operations** to **proactive supply chain orchestration**.

Starting **May 15, 2026**, the organization will undergo a 18-month transformation across four critical dimensions. The following summary outlines the impact on the enterprise resulting from the move to a unified Databricks-driven "Intelligence Layer."

1. Processes: From Manual Entry to Management by Exception

The most significant impact is the shift in the "replenishment heartbeat."

- **Closed-Loop Replenishment:** The inventory process moves from a manual "review and order" cycle to a "closed-loop" system. 85% of replenishment orders will be auto-generated by AI, shifting the business process toward **Exception Management**.
- **Real-time Freshness Cycle:** For perishable goods, the process of marking down prices is no longer a localized "end-of-day" task. It becomes a dynamic, intraday process triggered by Model 2 (Waste Prediction) and Model 3 (Dynamic Pricing), ensuring product turnover before spoilage.

- **Integrated Logistics Awareness:** The "Order-to-Delivery" process now factors in physical warehouse (EWM) and transport (TM) constraints *before* the order is placed, eliminating the process friction caused by rejected orders or half-empty trucks.

2. Policies & Regulations: Unified Governance & "Clean Core"

The project enforces a new era of group-wide standards and compliance.

- **SAP "Clean Core" Mandate:** A strict policy against custom modifications in legacy SAP ECC 6.0 is enforced. This shifts all innovation to "Side-by-Side" extensions (Azure/Databricks).
- **Global Data Sovereignty:** Using **Unity Catalog**, a new data governance policy is introduced. While data is centralized for AI training, strict "Regional Data Residency" policies ensure that sensitive PII and banner-specific pricing data remain compliant with local EU regulations and GDPR.
- **ESG & Waste Transparency:** Automated waste-tracking policies are integrated into the financial books (S/4HANA). This enables the organization to meet new ESG reporting regulations with high-integrity, audit-ready data regarding food waste and carbon footprints.

3. People: The Shift to "Algorithm Trust" & Data Literacy

The human element is the most sensitive dimension of the change management plan.

- **Trust-Building & Transparency:** Store Managers and Associates must move from "gut-feeling" ordering to trusting **Probabilistic Forecasts**. This requires a comprehensive training program focused on "Explainable AI" (XAI), where the system explains *why* a certain markdown is recommended.
- **New Role Profiles:** The role of the "Store Associate" evolves from clerical inventory counting to **Field Data Validator**. Associates will use the React App to provide "ground-truth" feedback to the AI (e.g., reporting a local event that spikes demand), making them active participants in the ML lifecycle.
- **Incentive Realignment:** Key Performance Indicators (KPIs) for staff will shift. Instead of being measured solely on sales volume, performance will now be tied to **Waste Reduction %** and **On-Shelf Availability (OSA)**, aligning personal goals with the corporate EBITDA targets.

4. Technology: Decoupling Intelligence from the Legacy Core

The technology landscape transitions from fragmented silos to a modern "Hub-and-Spoke" architecture.

- **Modular Architecture:** By moving the "brain" to Databricks and the "nerves" to Azure Functions/SAP Event Mesh, the organization effectively **decouples intelligence from the ERP**. This allows Ahold Delhaize to update AI models without touching the stable, legacy SAP ECC 6.0 core.
- **Event-Driven Responsiveness:** The shift from "Batch Processing" (once a night) to **Event-Driven Refresh Rates** means the IT organization must support a 24/7 real-time data flow.

This requires a higher level of DevOps and MLOps maturity to monitor pipeline health and model drift.

- **Mobile-First Field Empowerment:** Legacy green-screen terminals and paper-based workflows are replaced by a **Single-Entry "Super App"** (React.js). This modernizes the digital workplace for thousands of employees, increasing technical engagement and reducing onboarding time for new staff.

Summary of Impact Matrix

Dimension	Change Impact	Organizational Result
Processes	Manual → Predictive	**30% reduction in manual ordering effort.**
Policies	Siloed → Unified	**100% auditability of waste & pricing decisions.**
People	Clerical → Analytical	**Higher employee engagement via modern tools.**
Technology	Monolithic → Decoupled	**Faster time-to-market for future AI innovations.**

Conclusion: This transformation allows Ahold Delhaize to act as a **Unified Global Retailer** while respecting regional nuances, ultimately resulting in the targeted **€350M+ annual EBITDA improvement** through optimized margins and significant reduction in perishable waste.

Appendix

Data Flow Example

Scenario: Optimize fresh produce for Albert Heijn (Netherlands)

1. Ingestion (2 AM) → SAP ECC sales + Symphony POS + Weather API
2. DLT Processing (2:15-3:00 AM) → Bronze→Silver→Gold
3. ML Inference (3:00-3:30 AM) → Demand forecast + Waste prediction
4. Business Rules (3:30-4:00 AM) → Apply constraints + policies
5. Action (4:00-5:00 AM) → Auto-create POs (85%) + Alerts (15%)
6. Monitoring (Continuous) → Track accuracy, update models

This architecture handles the complexity of:

- **3 different ERP systems** (ECC, Symphony, S/4HANA)
- **Regional variations** (Western vs Central/Eastern Europe)
- **Real-time + batch processing**
- **Automated actions with governance**
- **Full auditability for compliance**

Perfect for scaling AI-driven inventory optimization across a complex, multi-country retail operation!

SCENARIO: Optimize fresh produce inventory for Albert Heijn stores in Netherlands

1. DATA INGESTION (Daily at 2 AM CET)

├─ SAP ECC 6.0: Extract yesterday's sales (VBRK/VBRP), stock movements (MSEG)

├─ Symphony Gold: Pull POS transactions, current inventory levels

└─ External: Fetch weather forecast for next 7 days

2. DELTA LIVE TABLES PROCESSING (2:15 AM - 3:00 AM)

├─ Bronze: Land raw data in Delta format

├─ Silver: Cleanse, deduplicate, enrich with master data

└─ Gold: Create ML-ready features (moving averages, seasonality, promotions)

3. ML INFERENCE (3:00 AM - 3:30 AM)

├─ Demand Forecasting Model: Generate P10/P50/P90 demand for next 7 days

├─ Waste Prediction Model: Flag high-risk SKUs (tomatoes, leafy greens)

└─ Replenishment Optimization: Calculate optimal order quantities

4. BUSINESS RULES APPLICATION (3:30 AM - 4:00 AM)

├─ Apply min/max stock levels

├─ Adjust for ongoing promotions

├─ Consider truck capacity constraints

└─ Respect supplier minimum order quantities

5. ACTION EXECUTION (4:00 AM - 5:00 AM)

├─ SAP ECC: Auto-create purchase orders via BAPI for 85% of SKUs

├─ Symphony: Update recommended order quantities for store managers

└─ n8n Workflow: Send alerts for manual review items (15% edge cases)

6. MONITORING & FEEDBACK (Continuous)

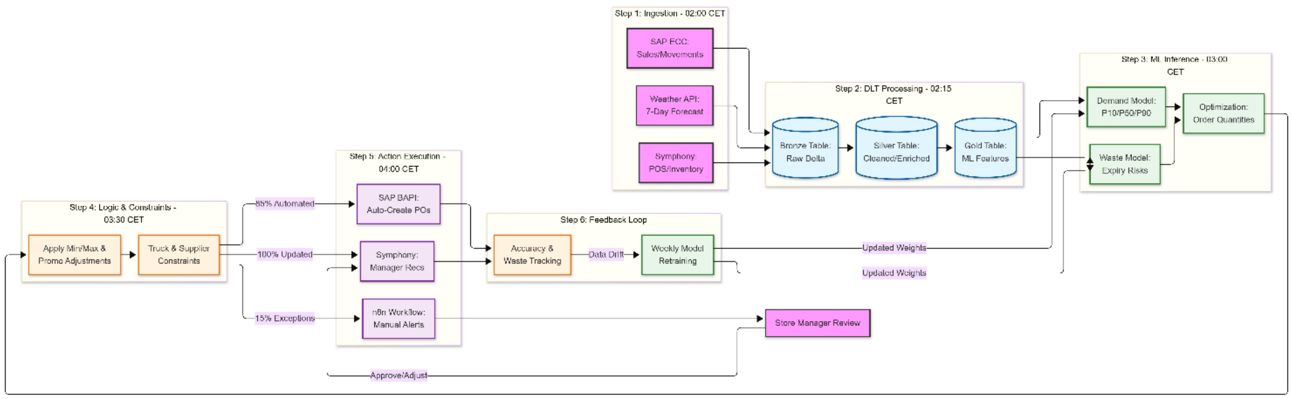
├─ Track actual vs predicted demand

├─ Monitor waste levels

├─ Calculate forecast accuracy

└─ Update models weekly based on performance

This **Data Flow Diagram (DFD)** illustrates the specific path data takes during the 24-hour cycle for Albert Heijn's fresh produce optimization. It emphasizes the transition from raw ERP data to AI-driven decisions and finally back to operational execution.



Data Flow Breakdown

1. **Ingestion Phase (02:00):** High-velocity POS data from **Symphony Gold** and inventory movements from **SAP ECC** are extracted simultaneously. Weather data is pulled to adjust for "barbecue weather" or rain, which drastically affects Albert Heijn's fresh produce sales.
2. **Medallion Processing (02:15):**
   - **Bronze:** Raw landing zone.
   - **Silver:** Joins sales with Material Master data (e.g., matching a barcode to its perishability profile).
   - **Gold:** Features like "rolling 7-day average sales" and "days until Sunday" are pre-calculated for the models.
3. **The Intelligence Core (03:00):**
   - The **Demand Model** provides a probabilistic range (P10–P90) rather than a single number to account for uncertainty.
   - The **Waste Model** identifies SKUs with high "shrink" risk (e.g., strawberries during a heatwave).

4. **Operational Constraints (03:30):** The raw AI output is "tempered" by reality. If the AI wants 100 crates of lettuce but the truck only holds 80, the **Constraint Solver** scales the order down. It also checks against Supplier Minimum Order Quantities (MOQ).
5. **Write-Back (04:00):**
    - **85% Success Path:** Standard replenishment is pushed directly into SAP ECC to create Purchase Orders.
    - **15% Exception Path:** High-value or high-risk orders are routed through **n8n** to a Store Manager's mobile app for a "human-in-the-loop" thumb-up/down.
6. **The Continuous Loop:** Actual sales at the end of the day are compared against the 3:00 AM forecast. If the **MAPE (Mean Absolute Percentage Error)** exceeds 15%, the system triggers a retraining notification in the **MLflow** registry.